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Aircraft handlers and related methods

Abstract

Aircraft handlers include a plurality of undercarriage support assemblies. Each undercarriage support assembly includes a body, a translation assembly operably attached to the body, and a support member operably coupled to and extending from the body. The support member comprises a support region configured to engage and operably support at least a portion of an undercarriage of an aircraft. The translation assembly is configured to support the undercarriage support assembly on a ground surface and permit the undercarriage support assembly to translate along the ground surface. The aircraft handlers further include one or more drive assemblies configured to move the aircraft handler along the ground surface, in which each drive assembly comprises the translation assembly of an undercarriage support assembly. Methods comprise supporting the undercarriage of the aircraft with the aircraft handler and moving the aircraft along the ground surface with the aircraft handler.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2980270	12/1960	Elliott	414/537	B64F 1/22
3119502	12/1963	Paul	414/427	B60P 3/11
3211425	12/1964	Greulich	414/592	B64F 1/22
4895319	12/1989	Bardsen	244/50	B64F 1/22
4911603	12/1989	Pollner	414/428	B64F 1/227
4911604	12/1989	Pollner	414/428	B64F 1/227
5562388	12/1995	Le Gall	414/427	B64F 1/22
5860785	12/1998	Eberspacher	244/50	B64F 1/227
2010/0254791	12/2009	Hwu et al.	N/A	N/A
2012/0215393	12/2011	Schiedegger	701/23	B60P 3/11
2017/0001511	12/2016	Kulkarni	N/A	B60L 50/40
2020/0298999	12/2019	Azaña Robles	N/A	B60T 8/266

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0739813	12/1995	EP	N/A
520051	12/1939	GB	N/A
2011030140	12/2010	WO	N/A

OTHER PUBLICATIONS

European Patent Office, Examination Report in related application EP 21 723 549.8, Jan. 17, 2025. cited by applicant

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Background/Summary

RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Patent Application No. 63/010,230, which is titled AIRCRAFT HANDLERS AND RELATED METHODS and was filed on Apr. 15, 2020. The entire disclosure of this patent application is hereby incorporated by reference.

FIELD

(1) The present disclosure relates to aircraft handlers and related methods.

BACKGROUND

(2) Often, aircraft need to be relocated while on the ground, such as before, after, or between airborne operations. Conventionally, ground relocation of aircraft is achieved by taxiing or by towing the aircraft with an aircraft tug. However, navigating aircraft within spatially confined or congested environments with these techniques can be difficult due to the large amount of ground space that is required for reorienting, turning, or rotating the aircraft with taxiing or towing. Additionally, it is sometimes necessary during ground relocation to move aircraft through or to within a confined volume, such as a stowage space, a shipping container, or an elevator. Generally,

when one or more dimensions of an aircraft exceed that of the confined volume, the aircraft must be disassembled or reconfigured to be moved through or fit within the confined volume, which can be time consuming and impose otherwise unnecessary requirements on the design of the aircraft. Thus, a need exists for devices and methods that improve aircraft ground mobility, such as relocation of aircraft within spatially confined spaces and movement of aircraft through or to within confined volumes.

SUMMARY

(3) Aircraft handlers and related methods are disclosed herein. The aircraft handlers are mobile, and are configured to engage and support, to thereby move, an aircraft relative to a ground surface. The aircraft handlers discussed herein include a plurality of undercarriage support assemblies. Each undercarriage support assembly includes a body, a translation assembly operably coupled to the body, and a support member operably coupled to and extending from the body. The support member comprises at least one support region configured to engage and operably support at least a respective portion of an undercarriage of an aircraft. The translation assembly is configured to support the respective undercarriage support assembly on a ground surface and permit the respective undercarriage support assembly to translate along the ground surface. The aircraft handlers further include one or more drive assemblies configured to move the aircraft handler along the ground surface, and each drive assembly comprises the translation assembly of an undercarriage support assembly. The aircraft handlers are thus configured to move (e.g., translate, rotate relative to one or more axes, etc.) an aircraft along the ground surface. In some examples, the aircraft handlers are controlled remotely by an operator, or operate autonomously or semi-autonomously. In some examples, the aircraft handlers are configured to support the aircraft in a level aircraft orientation and among a plurality of tilted aircraft orientations. The methods comprise supporting the undercarriage of the aircraft with the aircraft handler and moving the aircraft along the ground surface with the aircraft handler.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic plan view representing examples of aircraft handlers, according to the present disclosure.
- (2) FIG. 2 is a schematic plan view representing more examples of aircraft handlers, according to the present disclosure.
- (3) FIG. 3 is a schematic plan view of the example aircraft handlers of FIG. 2 in various configurations.
- (4) FIG. 4 is a schematic plan view representing additional examples of aircraft handlers according to the present disclosure.
- (5) FIG. 5 is a schematic side view representing the example aircraft handlers of FIG. 4 in various configurations.
- (6) FIG. 6 is an isometric view illustrating an example aircraft handler supporting an aircraft, according to the present disclosure.
- (7) FIG. 7 is another isometric view illustrating the example aircraft handler of FIG. 6 supporting an aircraft.
- (8) FIG. 8 is an isometric view of the example aircraft handler of FIG. 6.
- (9) FIG. 9 is another isometric view of the example aircraft handler of FIG. 6.
- (10) FIG. 10 is another isometric view of the example aircraft handler of FIG. 6.
- (11) FIG. 11 is an isometric view showing another example aircraft handler supporting an aircraft, according to the present disclosure.
- (12) FIG. 12 is another isometric view of the example aircraft handler of FIG. 11 supporting an

aircraft.

(13) FIG. 13 is an isometric view of the example aircraft handler of FIG. 11 according to the present disclosure.

(14) FIG. 14 is a side view of the example aircraft handler of FIG. 11, with the support arms of the aircraft handler illustrated in a level orientation.

(15) FIG. 15 is a side view illustrating the example aircraft handler of FIG. 11, with the support arms illustrated in a tilted orientation.

(16) FIG. 16 is a side view illustrating the example aircraft handler of FIG. 11, with the support arms illustrated in a stowed orientation.

(17) FIG. 17 is a flowchart that schematically represents methods, according to the present disclosure.

DESCRIPTION

(18) Aircraft handlers and related methods are disclosed herein. FIGS. 1-17 provide examples of aircraft handlers 10, and related methods 500 for utilizing aircraft handlers 10, according to the present disclosure. Elements that serve a similar, or at least a substantially similar, purpose are labeled with like numbers in each of FIGS. 1-17, and these elements may not be discussed in detail herein with reference to each of FIGS. 1-17. Similarly, all elements may not be labeled in each of FIGS. 1-17, but reference numerals associated therewith may be utilized herein for consistency. Elements, components, and/or features that are discussed herein with reference to one or more of FIGS. 1-17 may be included in and/or utilized with any of FIGS. 1-17 without departing from the scope of the present disclosure.

(19) FIG. 1 schematically represents examples of aircraft handlers according to the present disclosure, which are generally indicated with the reference number 10. The disclosure and drawings also present specific examples of aircraft handlers 10, which are designated with other reference numbers. For example, FIGS. 2-3 schematically represent examples of aircraft handlers 10 that are indicated at, and referred to herein as, aircraft handlers 300. FIGS. 4-5 schematically represent examples of aircraft handlers 10 that are indicated at, and referred to herein, as aircraft handlers 400. FIGS. 6-10 illustrate a more specific example of aircraft handlers 300 indicated at, and referred to herein as, aircraft handler 350, and FIGS. 11-16 illustrate a more specific example of aircraft handlers 400 indicated at, and referred to herein as, aircraft handler 450 according to the present disclosure. Generally, in FIGS. 1-5, elements that are likely to be included in a given (i.e., particular) example of an aircraft handler 10 are illustrated in solid lines, while elements, orientations, and/or configurations that are optional to a given example of an aircraft handler 10 are illustrated in broken lines. However, elements that are illustrated in solid lines in FIGS. 1-5 are not essential to all examples of aircraft handlers 10 of the present disclosure, and an element shown in solid lines in FIGS. 1-5 may be omitted from a particular example of an aircraft handler 10 without departing from the scope of the present disclosure. In schematic FIGS. 1-5, dot-dash and dot-dot-dash lines are utilized to illustrate various optional configurations of aircraft handlers 10. Dot-dash lines also may be utilized to indicate axes.

(20) The following discussion concentrates on the schematic representation of aircraft handler 10 in FIG. 1; however, where appropriate and to facilitate understanding of aircraft handler 10, reference to the illustrative, non-exclusive examples of aircraft handlers 300 in FIGS. 2-3 and 6-10 and of aircraft handlers 400 in FIGS. 4-5 and 11-16 are made. Example aircraft handlers 300 and 400 are non-exclusive and do not limit aircraft handler 10 to the illustrated embodiments of FIGS. 2-16. That is, aircraft handler 10 according to the present disclosure may incorporate any number of the various aspects, configurations, characteristics, properties of aircraft handlers 10 that are illustrated and discussed herein with reference to FIG. 1 and/or the embodiments of FIGS. 2-16, as well as variations thereof, without requiring the inclusion of all such aspects, configurations, properties, etc. For the purpose of brevity, each discussed component, part, portion, aspect, region, etc. or variants thereof may not be discussed, illustrated, and/or labeled with respect to aircraft handlers

300 and **400**; however, it is within the scope of the present disclosure that the discussed features, variants, etc. of aircraft handler **10** may be utilized with aircraft handlers **300** and **400**. Likewise, the specific illustrated and discussed aspects of aircraft handlers **300** and **400** may be utilized with other aircraft handlers **10** according to the present disclosure.

(21) With initial reference to FIG. **1**, aircraft handlers **10** comprise a plurality of undercarriage support assemblies **200**. Each undercarriage support assembly **200** comprises a body **12** and a support member **202** operably coupled to and extending from body **12**. Each support member **202** comprises at least one support region **20** configured to engage and operably support at least a respective portion of an undercarriage **108** of an aircraft. Stated differently, aircraft handler **10** comprises a plurality of support members **202**, each being configured to support a respective portion of an undercarriage of an aircraft. In some examples, support members **202** collectively are configured to support the entire undercarriage of the aircraft spaced apart from the ground surface **16**. Stated differently, in some examples, aircraft handlers **10** are configured to suspend the entire undercarriage of the aircraft spaced apart from the ground surface **16** when support member **202** of each undercarriage support assembly **200** supports at least the respective portion of the undercarriage of the aircraft.

(22) Each undercarriage support assembly **200** further includes a translation assembly **204** operably coupled to body **12**. Translation assembly **204** is configured to support the respective undercarriage support assembly **200** on a ground surface **16** and permit the respective undercarriage support assembly **200** to translate along ground surface **16**. Stated differently, aircraft handlers **10** comprise a plurality of translation assemblies **204** that support aircraft handler **10** on ground surface **16** and permit the aircraft handler **10** to translate along the ground surface **16**.

(23) Aircraft handlers **10** further include one or more drive assemblies **212** configured to move the aircraft handler **10** along the ground surface **16**, in which each drive assembly **212** comprises the translation assembly **204** of an undercarriage support assembly **200**. Put in slightly different terms, at least one undercarriage support assembly **200** comprises a respective drive assembly **212** and is configured to move aircraft handler **10** along ground surface **16**. In some examples, at least two undercarriage support assemblies **200**, and optionally each undercarriage support assembly **200** comprises a respective drive assembly **212**. Stated differently, in some examples, aircraft handler **10** comprises a plurality of drive assemblies **212**.

(24) Aircraft handlers **10** are configured to move an aircraft along ground surface **16**. More specifically, the one or more drive assemblies **212** are configured to move aircraft handler **10** along ground surface **16** while each support member **202** operably supports at least the respective portion of the undercarriage of the aircraft, such as to move aircraft handler **10**, together with the aircraft, along ground surface **16**.

(25) Aircraft handler **10** also may be referred to as an aircraft relocation cart, an aircraft dolly, aircraft lift cart, an aircraft handling assembly, and/or an aircraft relocation assembly. Aircraft handler **10** may be configured to move any suitable type of aircraft along ground surface **16** such as fixed wing aircraft, rotorcraft, helicopters, commercial aircraft, military aircraft, and/or private aircraft. Aircraft handler **10** also may be configured to move along any suitable ground surface **16**, such as while supporting the undercarriage of the aircraft spaced apart from ground surface **16**. For example, aircraft handler **10** may be configured to move along a ground surface **16** that is included in a runway, on a road, in an aircraft hangar, on a nautical ship, on an aircraft carrier, within a confined volume, within the trailer of a driven vehicle, within a shipping container, and/or within an elevator. Aircraft handler **10** also may be configured to move along a ground surface **16** that is level, sloped, even, and/or uneven.

(26) Each support member **202** and/or support region(s) **20** thereof is configured to support any suitable region of the undercarriage of the aircraft. In some examples, the undercarriage includes a landing gear assembly and support region **20** is configured to support at least a portion of the landing gear assembly, such as one or more wheels of the landing gear assembly. Additionally or

alternatively, the undercarriage, such as the undercarriage of a helicopter, may include one or more landing skids, and support region **20** may be configured to support at least a portion of the one or more landing skids. As shown in FIG. **1**, in some examples, support member **202** comprises more than one support region **20**, such as at least two support regions **20**, at least three support regions **20**, at least four support regions **20**, at least six support regions **20**, and/or at most ten support regions **20**. In some such examples, each support region **20** is configured to support a discrete or separate portion of the undercarriage of the aircraft. For example, each support region **20** may support one or more separate wheels of a landing gear assembly.

(27) Each support region **20** comprises any suitable structure for supporting at least the respective portion of the undercarriage of the aircraft in any suitable manner. For example, each support region **20** may comprise a platform, a surface, a prong, a fork, a ledge, a saddle, a basin, a recess, a notch, a slot, a clamp, a receiver, a friction fit, and/or combinations thereof. Each support region **20** may be configured to support, compliment, and/or operably engage with at least a portion of one or more types of undercarriages and/or with a specific portion of an undercarriage of one or more types of aircraft.

(28) As mentioned, in some examples, support members **202** are configured to support and/or suspend a substantial portion of, or the entirety of, the undercarriage of the aircraft spaced above ground surface **16**. Stated differently, in some examples, aircraft handler **10** is configured to support or suspend the entire weight of the aircraft above ground surface **16**. In some such examples, each support member **202** is operably positioned, oriented, and/or configured to translate laterally beneath at least the respective portion of the undercarriage of the aircraft, such as to be positioned to lift at least the respective portion of the undercarriage of the aircraft from direct contact with the ground surface. In some examples, each support member **202** is configured to lift at least the respective portion of the undercarriage of the aircraft from direct contact with the ground surface.

(29) In some examples, aircraft handler **10** further includes one or more lift mechanisms **26** each being included in an undercarriage support assembly **200**. Stated differently, in some examples, one or more undercarriage support assemblies **200** each comprise a respective lift mechanism **26**. When included, lift mechanism **26** operably couples a support member **202** of the respective undercarriage support assembly **200** to body **12** of the respective undercarriage support assembly **200**. Lift mechanism **26** is configured to selectively raise and lower the respective support member **202** relative to the ground surface **16**. In some examples, each support member **202** defines a proximal end that is positioned nearest body **12**, and lift mechanism **26** operably couples the proximal end of support member **202** to body **12** and selectively raises and lowers the proximal end to selectively raise and lower support member **202**. As discussed in more detail herein, in some examples, each lift mechanism **26** is configured to support the respective support member **202** in, and selectively raise and lower the respective support member **202** between two or more of a translation position, a lowered position, and a plurality of raised positions. In the translation position, support member **202** is positioned to translate laterally beneath at least the respective portion of the undercarriage of the aircraft. In the lowered position, support member **202** is positioned to support the respective portion of the undercarriage of the aircraft. In the raised position, support member **202** is spaced further away from the ground surface than in the lowered position and/or the translation position and is positioned to support the respective portion of the undercarriage of the aircraft spaced further away from ground surface **16** than in the lowered position.

(30) In some examples, each lift mechanism **26** is configured to selectively raise the respective support member **202** from the translation position to the lowered position and/or to among the plurality of raised positions, such that the respective support member **202** engages at least the portion of the undercarriage of the aircraft, and lifts at least the respective portion of the undercarriage of the aircraft from direct contact with ground surface **16**. In some examples, each lift mechanism **26** is configured to selectively raise the respective support member **202** from the

lowered position to among the plurality of raised positions, such that support member **202** supports the respective portion of the undercarriage of the aircraft spaced further apart from the ground surface than in the lowered position.

(31) In some examples, aircraft handlers **10** are configured to support the aircraft in a level orientation, in which the aircraft may be oriented generally horizontally relative to ground surface **16**, and among a plurality of tilted aircraft orientations, in which the aircraft is tilted relative to the ground surface **16** or the level aircraft orientation. In some such examples, aircraft handler **10** is configured to selectively tilt, or transition the aircraft between the level aircraft orientation and the plurality of tilted aircraft orientations. In some examples, lift mechanism(s) **26** are configured to selectively transition the aircraft between the level aircraft orientation and the plurality of tilted aircraft orientations by selectively raising and lowering the respective support member(s) **202** relative to ground surface **16**. More specifically, in some examples, aircraft handlers **10** are configured to support the aircraft in the level aircraft orientation when the support member **202** of each undercarriage support assembly supports at least the respective portion of the undercarriage of the aircraft and each lift mechanism **26** positions the respective support member **202** in the lowered position. In some examples, aircraft handlers **10** are configured to support the aircraft among the plurality of tilted aircraft orientations when each support member **202** supports the respective portion of the undercarriage of the aircraft and at least one lift mechanism **26**, and optionally each lift mechanism **26**, supports the respective support member **202** among the plurality of raised positions.

(32) Lift mechanism **26** includes any suitable structure and/or any suitable mechanism for raising and lowering support member **202**, and/or tilting the aircraft. As examples, lift mechanism **26** may include a jack mechanism, a hydraulic jack mechanism, a pneumatic jack mechanism, a mechanical jack mechanism, an electromechanical jack mechanism, a screw jack mechanism, a telescopic jack mechanism, and/or a chain driven lift mechanism. In some examples, lift mechanism **26** includes a pillar that extends upwardly from the respective translation assembly **204** and/or upwardly relative to ground surface **16**, and lift mechanism **26** is configured to move support member **202** up and down along the pillar.

(33) With continued reference to FIG. **1**, the one or more drive assemblies **212** comprise any suitable set of structures and/or mechanisms for moving aircraft handler **10** along ground surface **16**. In some examples, at least one drive assembly **212**, and optionally each drive assembly **212**, comprises one or more driven wheels **60** that support the respective undercarriage support assembly **200** on ground surface **16**. When included, driven wheels **60** are configured to receive power from one or more power sources **92** and facilitate movement of aircraft handler **10** along ground surface **16**. In some examples, drive assemblies **212** are configured to steer aircraft handler **10**, such as while moving aircraft handler **10** along ground surface **16**. In particular, drive assemblies **212** may be configured to rotate, turn, and/or move aircraft handler **10** and/or the aircraft **22** in a forward or in a reverse direction. In some examples, aircraft handler **10** comprises a plurality of drive assemblies **212**, in which each drive assembly **212** is configured to operate independently of one another, such as to steer and/or rotate aircraft handler **10** and/or the aircraft **22**. Additionally or alternatively, drive assemblies **212** are configured to operate in unison. More specifically, in some examples, at least one, some, or each driven wheel **60** is configured to pivot to steer, turn, and/or rotate aircraft handler **10** and/or the aircraft. Similarly, in some examples, at least one driven wheel **60** is configured to selectively and operably rotate in clockwise and counterclockwise directions such as to move aircraft handler **10** and/or the aircraft **22** in the forward and reverse directions along ground surface **16**. With this in mind, in some examples, driven wheel(s) **60** are configured to operate in unison, and in some examples, one or more driven wheels **60** are configured to operate independently.

(34) In some examples, one or more translation assemblies **204** each comprise one or more non-driven wheels **62**. Stated differently, in some examples, translation assembly **204** of at least one

undercarriage support assembly **200** comprises one or more non-driven wheels **62**. Additionally or alternatively, one or more drive assemblies **212** each comprise one or more non-driven wheels **62**. When included, non-driven wheels **62** are configured to support a respective translation assembly **204** and/or drive assembly **212** on ground surface **16** and rotate to permit the respective undercarriage support assembly **200** to translate along ground surface **16**. When non-driven wheels **62** are included in a drive assembly **212**, non-driven wheels **62** may provide additional support, stability, load bearing capacity, and/or improved steering to the respective drive assembly **212**. In some examples, drive assembly **212** and the respective undercarriage support assembly **200** comprises a chassis for supporting, arranging, and/or orienting driven wheels **60** and/or non-driven wheels **62**. Examples of suitable non-driven wheel **62** include caster wheels, rollers, and/or ball transfers.

(35) When aircraft handler **10** comprises lift mechanism(s) **26**, drive assembly(s) **212** may be configured to move aircraft handler **10** along ground surface **16** when lift mechanisms **26** position or support support members **202** in the translation position, in the lowered position, and among the lifted positions. Likewise, in some examples, drive assembly(s) **212** are configured to move aircraft handler **10** along ground surface **16** when support members **202** support the aircraft in the level aircraft orientation and when support members **202** support the aircraft among the tilted aircraft orientations. In some examples, drive assembly(s) **212** are configured to facilitate lifting of the aircraft from ground surface **16**. In particular, in some examples, drive assemblies **212** are configured to move aircraft handler **10** to translate support members **202** laterally beneath at least the respective portion of the undercarriage of the aircraft. When aircraft handler **10** includes lift mechanism(s) **26**, each lift mechanism **26** may be included in an undercarriage support assembly **200** that includes or that does not include a drive assembly **212**.

(36) As discussed in more detail herein, an aircraft may occupy or define an envelope, which is defined herein as a rectangular prism that encompasses the aircraft. The bottom side of the base of the rectangular prism is defined by ground surface **16**, and each other side of the rectangular prism includes at least one point that is contacted by, abuts, or is defined by, a maximal extent of the aircraft in that particular direction. For example, in some fixed wing aircraft, two opposed lateral sides of the rectangular prism are contacted by, or defined by two wing tips of the aircraft, a top or upper side of the rectangular prism is defined by an empennage of the aircraft, and the front and rear sides of the aircraft are defined by the nose and empennage of the aircraft. In some examples, aircraft handler **10** is configured to fit fully within the envelope of the aircraft when the plurality of undercarriage support assemblies support the respective portions of the undercarriage of the aircraft.

(37) Also as discussed in more detail herein, in some examples, aircraft handler **10** is configured to rotate the aircraft about a yaw axis of the aircraft. In some examples, aircraft handler **10** is configured to rotate the aircraft within a cylindrical volume having a radius that is less than a threshold fraction of a distance between the yaw axis of the aircraft and a maximum extend of the aircraft from the yaw axis.

(38) With continued reference to FIG. **1**, aircraft handler **10** comprises any suitable number of undercarriage support assemblies **200**, with examples including at least two undercarriage support assemblies **200**, at least three undercarriage support assemblies **200**, at least four undercarriage support assemblies **200**, at least five undercarriage support assemblies **200**, at least six undercarriage support assemblies **200**, at least eight undercarriage support assemblies **200**, at most three undercarriage support assemblies **200**, at most four undercarriage support assemblies **200**, at most five undercarriage support assemblies **200**, at most six undercarriage support assemblies **200**, at most eight undercarriage support assemblies **200**, and/or at most ten undercarriage support assemblies **200**. Each undercarriage support assembly **200** may include one or more of the same components, features, and/or functions as one or more other undercarriage support assembly **200**. Additionally or alternatively, each undercarriage support assembly **200** may include one or more

components, features, and/or functions that are different from that of one or more other undercarriage support assemblies **200**. In some examples, each undercarriage support assembly **200** is configured to support and/or engage a specific portion of the undercarriage of the aircraft.

(39) As shown in FIG. **1**, in some examples, the plurality of undercarriage support assemblies **200** comprises a front undercarriage support assembly **206**, a rear undercarriage support assembly **208**, and/or one or more distal undercarriage support assemblies **210**. In some examples, each of front undercarriage support assembly **206** and rear undercarriage support assembly **208** comprise a respective lift mechanism **26**. In such examples, each of front undercarriage support assembly **206** and rear undercarriage support assembly **208** are configured to selectively raise and lower the respective portions of the undercarriage of the aircraft relative to ground surface **16**. In some such examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** are configured to selectively raise and lower the respective portions of the undercarriage of the aircraft relative to support member(s) **202** of distal undercarriage support assembly(s) **210**. In such examples, aircraft handler **10** selectively tilts the aircraft between the level aircraft orientation and among the tilted aircraft orientations when each support member **202** supports the respective portion of the undercarriage of the aircraft, and support members **202** of front undercarriage support assembly **206** and rear undercarriage support assembly **208** are selectively raised and lowered by lift mechanisms **26** such as relative to support member(s) **202** of distal undercarriage support assembly(s) **210**.

(40) In some examples, at least some of the undercarriage support assemblies **200** are interconnected with one another. In some such examples, each undercarriage support assembly **200** is operably interconnected with at least one other undercarriage support assembly **200**. Stated differently, in some examples, the plurality of undercarriage support assemblies **200** are operably interconnected with one another. For some examples in which the plurality of undercarriage support assemblies **200** are operably interconnected with one another, fewer than all of the undercarriage support assemblies **200** comprise a respective drive assembly **212**.

(41) Undercarriage support assemblies **200** may be interconnected one another in any suitable manner. In some examples, each distal undercarriage support assembly **210** is operably coupled to front undercarriage support assembly **206** or rear undercarriage support assembly **208**. In some examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** are interconnected with one another. More specifically, in some examples, aircraft handler **10** comprises a connecting member **216** that is configured to operably interconnect front undercarriage support assembly **206** and rear undercarriage support assembly **208**. In some such examples, connecting member **216** supports and/or positions front undercarriage support assembly **206** and rear undercarriage support assembly **208** spaced apart from one another along ground surface **16**. As discussed in more detail herein, in some examples, front undercarriage support assembly **206** is operably coupled to a distal undercarriage support assembly **210** by a support arm **18**. In some examples, rear undercarriage support assembly **208** is operably interconnected to a distal undercarriage support assembly **210** by a support arm **18**. In some examples, each support arm **18** supports and/or positions distal undercarriage support assembly **210** spaced apart along ground surface **16** from front undercarriage support assembly **206** and/or rear undercarriage support assembly **208**. In other examples, aircraft handler **10** comprises one or more connecting arms **214** that are configured to operably interconnect rear undercarriage support assembly **208** with a distal undercarriage support assembly **210**. In some such examples, connecting arm(s) **214** are configured to support distal undercarriage support assembly **210** spaced apart from rear undercarriage support assembly **208** along ground surface **16**.

(42) In some examples, connecting member **216** comprises one or more communication conduits **66** that are configured to transmit data, communication signals, and/or control signals, between front undercarriage support assembly **206** and rear undercarriage support assembly **208**.

Additionally or alternatively, in some examples, connecting member **216** comprises one or more

power supply conduits **68** that are configured to transmit power, such as electrical power, between front undercarriage support assembly **206** and rear undercarriage support assembly **208**. In some examples, communication conduit(s) **66** are in communication with controller **90**. In some examples, power supply conduits **68** are in communication with power source(s) **92**.

(43) Additionally or alternatively, in some examples, at least one undercarriage support assembly **200**, and optionally each undercarriage support assembly **200**, is configured to move independently of the other undercarriage support assemblies **200** along ground surface **16** and/or is mechanically disconnected from the other undercarriage support assemblies **200**. For some examples in which undercarriage support assemblies **200** are mechanically disconnected from one another, each undercarriage support assembly **200** comprises a respective drive assembly **212** to permit each undercarriage support assembly **200** to move along the ground surface independently of one another. In some such examples, each undercarriage support assembly **200** comprises one or more respective power sources **92** configured to supply power to various components of the respective undercarriage support assembly **200**. In some such examples, front undercarriage support assembly **206**, rear undercarriage support assembly **208**, and optionally each distal undercarriage support assembly **210**, comprises a respective controller **90** configured to control the respective drive assembly **212** and/or synchronize the operation of the undercarriage support assemblies **200**. In some such examples, controllers **90** are configured to communicate with one another wirelessly.

(44) With continued reference to FIG. **1**, aircraft handler **10** occupies a footprint on ground surface **16** that is defined herein as the total area on ground surface **16** occupied by undercarriage support assemblies **200** as well as the area of ground surface **16** between undercarriage support assemblies **200**. As a more specific example, for some examples in which aircraft handler **10** comprises four undercarriage support assemblies **200**, aircraft handler **10** occupies a generally rectangular footprint encompassing each undercarriage support assembly **200** and the area therebetween. In some examples, aircraft handler **10** comprises a deployed configuration, in which aircraft handler **10** is configured to support an aircraft, and a stowed configuration in which aircraft handler **10** is configured to be stowed. In some examples, aircraft handler **10** is configured to occupy a footprint in the stowed configuration that is smaller than the footprint that the aircraft handler **10** occupies in the deployed configuration. In some examples, two or more undercarriage support assemblies **200** are positioned further apart from one another in the deployed configuration than in the stowed configuration, such that the area along ground surface **16** between the at least two undercarriage support assemblies **200** is smaller in the stowed configuration. In this way, the stowed configuration permits aircraft handler **10** to be stored in a smaller storage area. In some examples, aircraft handler **10** is configured to move undercarriage support assemblies **200** relative to one another to transition aircraft handler **10** between the stowed and deployed configurations.

(45) In some examples, aircraft handler **10** is configured to selectively transition between the deployed configuration and the stowed configuration. For some examples in which aircraft handler **10** comprises support arm(s) **18**, each support arm **18** is configured to selectively permit the undercarriage support assemblies **200**, that the support arm **18** interconnects to, to translate relative to one another along ground surface **16** and permit aircraft handler **10** to transition between the stowed configuration and the deployed configuration. Similarly, for some examples in which aircraft handler **10** comprises connecting member **216**, connecting member **216** is configured to selectively permit front undercarriage support assembly **206** and rear undercarriage support assembly **208** to translate relative to one another along ground surface **16** and permit aircraft handler **10** to transition between the stowed configuration and the deployed configuration. Likewise, for some examples in which aircraft handler **10** comprises connecting arm(s) **214**, connecting arms **214** are configured to selectively permit rear undercarriage support assembly **208** and distal undercarriage support assembly **210** to translate relative one another along ground surface **16** and permit aircraft handler **10** to transition between the stowed configuration and the deployed configuration.

(46) As mentioned, in some examples, aircraft handler **10** comprises one or more power sources **92** that are configured to selectively supply power to one or more components, assemblies, and/or elements of aircraft handler **10** to facilitate or power the one or more functions, operations, and/or actions that the one or more components, assemblies, and/or elements of aircraft handler **10** are configured to perform. In some examples, each component, assembly, and/or element of aircraft handler **10** that receives power from power source(s) **92** may be referred to as an actuator and/or a powered component, powered assembly, and/or powered element. Power source(s) **92** comprise any suitable type, number, and/or combination of power sources **92** for powering the one or more powered elements, powered assemblies, and/or powered components of aircraft handler **10**. In some examples, power source(s) **92** are configured to selectively store power and selectively distribute power to one or more powered element(s), powered part(s), and/or powered component(s) of aircraft handler **10**.

(47) Power source(s) **92** are configured to supply any suitable type of power to the more powered element(s), powered part(s), and/or powered component(s) of aircraft handler **10** such as mechanical power, electromechanical power, pneumatic power, hydraulic power, electrical power, and/or combinations thereof. In some examples, one or more powered element(s), powered assembly(s), and/or powered component(s) of aircraft handler **10** are configured to receive electrical power. In such examples, power source(s) **92** may include one or more batteries for selectively storing electrical power and distributing electrical power. In some such examples, the one or more batteries are rechargeable and may be selectively recharged by an external power source. Additionally or alternatively, in some examples, power source(s) **92** are configured to receive one or more fuels, such as liquid petroleum based fuels and/or gaseous fuels such as propane and/or hydrogen, and convert the one or more fuels to useable power. As another example, power source(s) **92** may comprise an internal combustion engine for converting the one or more fuels to mechanical power, pneumatic power, and/or hydraulic power, which power source(s) **92** in turn selectively distribute to the one or more powered element(s), powered part(s), and/or powered component(s) of aircraft handler **10**.

(48) As more specific examples, each drive assembly **212** may be a powered assembly, and one or more components thereof, such as driven wheels **60**, are configured to receive power from power source(s) **92** to move, steer, and/or rotate aircraft handler **10** and/or the aircraft along ground surface **16** and/or to facilitate lifting of at least a portion of the undercarriage of the aircraft from ground surface **16**. In some examples, each lift mechanism **26** is a powered lift mechanism **26** and configured to receive power from power source(s) **92** to facilitate raising and lowering of the respective support member **202** and/or tilting of the aircraft.

(49) When included, power source(s) **92** are positioned along and/or within aircraft handler **10** in any suitable one or more locations. In some examples, power source(s) **92** are positioned along and/or within a body **12** of at least one undercarriage support assembly **200** and/or within connecting member **216**. Additionally or alternatively, in some examples, one or more power source(s) **92** are positioned proximate or within the component, element, and/or assembly that each of the one or more power source(s) **92** are configured to power.

(50) With continued reference to FIG. **1**, in some examples, aircraft handler **10** comprises a controller **90** that is programmed to control various actuators of aircraft handler **10** (e.g., the drive assembly(s) **212** and/or lift mechanism(s) **26**). When included, controller **90** is in communication (e.g., electrical communication and/or wired or wireless communication) with power source(s) **92** and one or more actuators of aircraft handler **10**. Controller **90** receives power (e.g., electrical power) from power source(s) **92** and selectively distributes the power provided by power source(s) **92** to the one or more actuators according to a control scheme. In particular, methods **500** discussed below in relation to FIG. **17** describe an example control scheme that may be utilized by controller **90** to regulate the amount of power supplied to the one or more actuators of aircraft handler **10**.

(51) Controller **90** may be any suitable device or devices that are configured to perform the

functions of the controller discussed herein. For example, the controller may include one or more of an electronic controller, a dedicated controller, a special-purpose controller, a personal computer, a special-purpose computer, a display device, a logic device, a memory device, and/or a memory device having non-transitory computer-readable media suitable for storing computer-executable instructions for implementing aspects of systems and/or methods according to the present disclosure.

(52) Additionally or alternatively, controller **90** may include one or more computer(s), which may include, or be configured to read, non-transitory computer-readable storage, or memory, media suitable for storing computer-executable instructions, or software, for implementing methods or steps of methods according to the present disclosure. Examples of such media include CD-ROMs, disks, hard drives, flash memory, etc. As used herein, storage, or memory, devices and media having computer-executable instructions as well as computer-implemented methods and other methods according to the present disclosure are considered to be within the scope of subject matter deemed patentable in accordance with Section 101 of Title 35 of the United States Code.

(53) More generally, controller **90** sends command signals (e.g., digital signals and/or electrical signals) to the one or more actuators of aircraft handler **10** to adjust operation thereof. In some examples, controller **90** is configured to perform various actions, such as to control the one or more actuators, based on input. In particular, controller **90** may comprise computer-readable instructions stored in non-transitory memory, in which the computer-readable instructions comprise instructions for controlling the one or more actuators of aircraft handler **10**. Controller **90** also may comprise a processing unit that is configured to execute the stored computer-readable instructions to control operation of the one or more actuators of aircraft handler **10**.

(54) In some examples, aircraft handler **10** is configured to be operated by an operator, such as one or more people, in which controller **90** is configured to receive input (such as directly or wirelessly) from the operator to control the one or more actuators of aircraft handler **10** and/or the operation of aircraft handler **10**. In some such examples, aircraft handler **10** is configured for remote control, and controller **90** is configured to receive input from the operator at a remote location, such as via wireless communication (e.g., digital) between one or more remote devices controlled by the operator and a receiver included in controller **90**. Additionally or alternatively, in some examples, aircraft handler **10** is autonomous or semi-autonomous and controller **90** is configured to control operation of the one or more actuators of aircraft handler **10** and/or the operation of aircraft handler **10** without input or with limited input from an operator. In some such examples, aircraft handler **10** comprises one or more sensors **98**, that are in digital communication (e.g., wired or wireless communication) with controller **90**. In particular, aircraft handler **10** may comprise one or more sensors **98** that are configured to collect sensor data respective to a position, speed, acceleration, rotation and/or orientation of aircraft handler **10**, one or more sensors **98** that are configured to collect sensor data respective to one or more entities within the environment of aircraft handler **10**, such as the location thereof, and/or one or more sensors **98** that may be associated with each actuator of aircraft handler **10** and configured to collect sensor data respective to the operation of the actuator. In such examples, controller **90** is configured to receive sensor data input from each sensor **98** and issue control outputs to the one or more actuators of aircraft handler **10** based on the sensor data and the control scheme. In this way, controller **90** is configured to utilize input from sensor(s) **98** to control the various actions that aircraft handler **10** is configured to perform without or with limited input from an operator. Examples of suitable sensors **98** that may be included in aircraft handler **10** include one or more displacement sensors (e.g., non-linear displacement sensors) that are configured to collect sensor data respective to the movement of lift mechanism **26** and/or a respective support member **202**. As another example, aircraft handler **10** may include one or more alignment sensors (such as laser-based sensors) positioned proximate each support region **20** that are configured to detect the position or alignment of the undercarriage of the aircraft relative to each support region **20**. As yet another example, aircraft handler **10** may comprise a

steering angle sensor that may be installed in one or more of the drive assemblies **212** that is configured to detect the steering angle of the respective drive assembly **212** and/or aircraft handler **10**. As yet more examples, aircraft handler **10** may comprise one or more sonar sensors, one or more infrared sensors, one or more laser sensors, and/or one or more GPS sensors, that are configured to detect entities within the environment of aircraft handler **10** and/or location of aircraft handler **10** within its environment and/or one or more speedometers and/or one or more accelerometers that are configured to detect movement of aircraft handler **10**, in which data from each of these sensor types may be utilized by controller **90** to navigate or control movement of aircraft handler **10** along ground surface **16** in any suitable environment.

(55) FIGS. **2** and **3** schematically represent examples of aircraft handlers **10** that are indicated at and referred to herein as aircraft handlers **300**. With initial reference to the examples of FIG. **2**, aircraft handlers **300** include a front undercarriage support assembly **206** that is configured to support a forward portion of an undercarriage **108** of an aircraft, and a rear undercarriage support assembly **208** that is configured to support an aft portion of an undercarriage of the aircraft. At least one of front undercarriage support assembly **206** and rear undercarriage support assembly **208** includes a respective drive assembly **212**. In some examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** each comprise a respective drive assembly **212**.

(56) In some examples, aircraft handler **300** further includes a distal undercarriage support assembly **210** and at least one connecting arm **214** that operably couples distal undercarriage support assembly **210** to rear undercarriage support assembly **208**. In some such examples, rear undercarriage support assembly **208** is configured to engage and support a first aft portion of the undercarriage of the aircraft and distal undercarriage support assembly **210** is configured to support a second aft portion of the undercarriage of the aircraft. In some examples, the undercarriage of the aircraft comprises a front wheel assembly, a first rear wheel assembly, and a second rear wheel assembly laterally spaced apart from the first rear wheel assembly. In some such examples, front undercarriage support assembly **206** is configured to support the front wheel assembly, rear undercarriage support assembly **208** is configured to support the first rear wheel assembly, and distal undercarriage support assembly **210** is configured to support the second rear wheel assembly.

(57) In some examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** each comprise a lift mechanism **26**. Each lift mechanism **26** is configured to selectively position the respective support member **202** in the lowered position, among the plurality of raised positions, and optionally in the translation position, as discussed herein. In some examples, aircraft handler **10** is configured to support the aircraft in the level aircraft orientation when support member **202** of front undercarriage support assembly **206** and support member **202** of rear undercarriage support assembly **208** are in the lowered position, and aircraft handler **300** is configured to support the aircraft among the plurality of tilted aircraft orientations when the support member **202** of front undercarriage support assembly **206** and support member **202** of rear undercarriage support assembly **208** are among the plurality of raised positions.

(58) In some examples, aircraft handlers **300** further comprise one or more ground support chasses **236** each being operably coupled to, or defining a portion of, body **12** of an undercarriage support assembly **200**. In some examples, ground support chassis **236** is configured to provide additional stability to aircraft handler **300** and/or the respective undercarriage support assembly **200**.

Additionally or alternatively, in some examples, ground support chassis **236** is configured to distribute the weight of the aircraft over a larger area of ground surface **16** when aircraft handler **300** supports the aircraft. In some examples, front undercarriage support assembly **206** comprises a ground support chassis **236** of the one or more ground support chasses **236** that is referred to herein as a front ground support chassis **236**. Additionally or alternatively, in some examples, rear undercarriage support assembly **208** comprises a ground support chassis **236** of the one or more ground support chasses **236** that is referred to herein as a rear ground support chassis **236**.

(59) Each ground support chassis **236** comprises a pair of projections **238** that extend from body **12**

laterally spaced apart from one another and that form an open region **240** therebetween. In some examples, the pair of projections **238** are positioned such that support member **202** of the respective undercarriage support assembly **200** is positioned within open region **240**. Stated differently, in some examples, support member **202** is positioned between the pair of projections **238**. In some examples, a portion of translation assembly **204** and/or drive assembly **212** of the respective undercarriage support assembly **200** supports projections **238** on and/or above ground surface **16**. In some such examples, one or more driven wheels **60** and/or non-driven wheels **62** are positioned along a ground surface-facing region of each projection **238** and support projection **238** on and/or above ground surface. When ground support chassis **236** is included in front undercarriage support assembly **206**, each projection **238** is referred to herein as a front projection **238** and the pair of projections **238** is referred to herein as front pair of projections **238**. When ground support chassis **236** is included in rear undercarriage support assembly **208**, each projection **238** is referred to herein as a rear projection **238** and pair of projections **238** is referred to herein as rear pair of projections **238**.

(60) In some examples, ground support chassis **236** comprises a gate **242** that is operably coupled to and/or extends from a distal region of ground support chassis **236**, which is positioned furthest from support member **202** of the respective undercarriage support assembly **200**. As shown in FIG. 2, in some examples, gate **242** comprises a closed configuration **244**, in which gate **242** is operably coupled to and interconnects the pair of projections **238**. As shown in FIG. 3, in some examples, gate **242** comprises an open configuration **246**, in which gate **242** does not interconnect projections **238**, provides access to the front open region **240**. In some examples, gate **242** is only operably coupled to one projection **238** in open configuration **246**. Alternatively, gate **242** comprises two segments, each being operably coupled to a respective projection **238**, and configured to selectively interconnect with and disconnect from one another. In such examples, the two segments of gate **242** are disconnected from one another to provide access to the open region **240** in open configuration **246** and interconnected with one another in closed configuration **244**. In some examples, the respective undercarriage support assembly **200** is configured to translate laterally beneath the respective portion of the undercarriage of the aircraft when gate **242** is in open configuration **246** and is configured to support the respective portion of the undercarriage of the aircraft when gate **242** is in the closed configuration **244**.

(61) As shown in FIG. 2, either or both of front undercarriage support assembly **206** and rear undercarriage support assembly **208** may include a respective gate **242**. Gate **242** is configured to operably and selectively interconnect projections **328** in any suitable manner. In some examples, gate **242** is hingedly coupled to one projection **238** and the other projection **238** comprises a coupler that is configured to selectively couple to a corresponding coupler included in gate **242**. In some such examples, ground support chassis **236** and/or gate **242** comprises an actuator that is configured to selectively couple and release gate **242** from the other projection and/or an actuator that is configured selectively to pivot gate **242** about the hinged coupling to transition gate **242** between closed configuration **244** and open configuration **246**.

(62) With continued reference to FIG. 2, in some examples, support member **202** of front undercarriage support assembly **206** and/or rear undercarriage support assembly **208** comprises a fork assembly **248**. Each fork assembly **248** comprises a plurality of prongs **250** that project outwardly from the respective body **12** laterally spaced apart from one another. In the examples shown in FIG. 2, each fork assembly **248** comprises a pair of prongs **250** that are laterally spaced apart from one another. Prongs **250** collectively form support region **20** of the respective support member **202**, with each prong **250** forming a respective portion of support region **20**. Stated differently, in some examples, prongs **250** collectively are configured to support the respective portion of the undercarriage of the aircraft. For examples in which front undercarriage support assembly **206** and/or rear undercarriage support assembly **208** comprise lift mechanism **26** together with fork assembly **248**, lift mechanism **26** is configured to selectively raise and lower fork

assembly **248** relative to ground surface **16**.

(63) In some examples, the lateral separation between prongs **250** permits fork assembly **248** to translate laterally beneath the respective portion of the undercarriage of the aircraft, such as when lift mechanism **26** positions fork assembly **248** in the translation position. In some examples, prongs **250** engage the portion of the undercarriage of the aircraft when lift mechanism **26** raises fork assembly **248** from the translation position to the lowered position and/or to among the raised positions. As a more specific example, for some examples in which the respective portion of the undercarriage of the aircraft comprises a wheel or a wheel assembly, prongs **250** of fork assembly **248** are configured to be positioned on opposing sides of and below the wheel or wheel assembly in the translation position, and engage the opposing sides of the wheel or wheel assembly when fork assembly **248** is raised to the lowered position and/or among the raised positions.

(64) In some examples, fork assembly **248** is configured to adjust the lateral spacing between prongs **250**. In some examples, the lateral spacing between prongs **250** is adjusted to accommodate aircraft undercarriages having different dimensions and/or configurations. Additionally or alternatively, in some examples, fork assembly **248** is configured to adjust lateral spacing between prongs **250** to translate laterally beneath, and engage the respective portion of the undercarriage of the aircraft. In some more specific examples, fork assembly **248** is configured to widen the lateral spacing between prongs **250**, such as when fork assembly **248** is in the translation position and/or to configure fork assembly **248** to translate laterally beneath the respective portion of the undercarriage of the aircraft. In some examples, fork assembly **248** is configured to narrow the lateral spacing between prongs **250** once prongs **250** are positioned laterally beneath the respective portion of the undercarriage of the aircraft such that prongs **250** engage the portion of the undercarriage of the aircraft, and optionally lift the portion of the undercarriage of the aircraft from direct contact with ground surface **16**. In this way, fork assembly **248** also may be described as being configured to transition prongs **250** from the translation position to the lowered position, in which prongs **250** support the respective portion of the undercarriage spaced above the ground surface **16**. As shown in FIG. 2, in some examples, fork assembly **248** comprises a lateral translation actuator **249** that is configured to adjust the lateral spacing between prongs **250**, as discussed herein.

(65) In some examples, each fork assembly **248** further comprises a prong gate **252** that may be configured to reinforce fork assembly **248** and/or selectively retain the respective portion of the undercarriage of the aircraft on fork assembly **248**. In some examples, prong gate **252** is operably coupled to, or extends from a distal region of fork assembly **248** that is positioned furthest from body **12**. As shown in FIG. 2, in some examples, prong gate **252** comprises a prong gate closed configuration **254**, in which prong gate **252** is operably coupled to and interconnects the plurality of prongs **250**. As shown in FIG. 3, in some examples, prong gate **252** comprises a prong gate open configuration **256**, in which prong gate **252** does not interconnect prongs **250** and provides access to the lateral spacing between prongs **250**. In some examples, prong gate **252** is operably coupled to only one prong **250** of the plurality of prongs **250** in prong gate open configuration **256**. In some examples, gate **242** is only operably coupled to one projection **238** in open configuration **246**. Alternatively, prong gate **252** comprises two segments, each being operably coupled to a respective prong **250**, and configured to selectively interconnect with and disconnect from one another. In such examples, the two segments of prong gate **252** are disconnected from one another to provide access to the lateral spacing between prongs **250** in prong gate open configuration **256** and interconnected with one another in the prong gate closed configuration **254**.

(66) In some examples, fork assembly **248** is configured to translate laterally beneath the respective portion of the undercarriage of the aircraft when prong gate **252** is in prong gate open configuration **256**. More specifically, in some such examples, prong gate **252** provides access to the lateral separation between prongs **250** in prong gate open configuration **256** to permit fork assembly **248** to translate laterally beneath the respective portion of the undercarriage of the aircraft. In some

examples, prong gate **252** is configured to reinforce fork assembly **248** and/or selectively retain the respective portion of the undercarriage of the aircraft on fork assembly **248** when prong gate **252** is in prong gate closed configuration **254**. More specifically, in some examples, prong gate **252** provides a brace against lateral movement of the respective portion of the undercarriage of the aircraft from fork assembly **248** when prong gate **252** is in prong gate closed configuration **254**. In some examples, prong gate **252** is configured to selectively transition to prong gate open configuration **256** to permit fork assembly **248** to adjust the lateral spacing between prongs **250** as discussed herein.

(67) Prong gate **252** is configured to selectively and operably interconnect prongs **250** in any suitable manner, such as in a manner similar to that discussed herein for gate **242**. More specifically, for some examples in which fork assembly **248** comprises a pair of prongs, prong gate **252** is hingedly coupled to one prong **250** is configured selectively couple to and release from the other prong **250**. In some examples, prong gate **252** is configured to selectively pivot between prong gate closed configuration **254** and prong gate open configuration **256**. In some such examples, fork assembly **248** comprises one or more actuators for selectively pivoting prong gate **252** and/or for selectively coupling and releasing one prong **250** from the other prong **250**.

(68) As shown in FIG. 2, in some examples, each undercarriage support assembly **200** defines a respective coordinate system **290** that includes a vertical axis **292** which extends at least substantially transverse to, or perpendicular to, ground surface **16** when undercarriage support assembly **200** is positioned on ground surface **16**, a lateral axis **294** which extends perpendicular to vertical axis **292**, and a longitudinal axis **296** which extends perpendicular to both vertical axis **292** and lateral axis **294**. When aircraft handler **300** is supported on ground surface **16**, lateral axis **294** and longitudinal axis **296** extend at least substantially parallel to ground surface **16**. Lateral axis **294** extends at least substantially transverse to the direction which the respective support member **202** extends from body **12**, and a longitudinal axis **296** extends at least substantially parallel to the direction which the respective support member **202** extends from body **12**.

(69) The origin of coordinate system **290** is positioned at a center point **298** of an interface between body **12** and support member **202** when support member **202** is in the translation position or positioned nearest to ground surface **16**. For example, when support member **202** comprises a pair of prongs **250**, the origin of coordinate system **290** is positioned centrally between the pair of prongs **250** and along body **12** where prongs **250** extend from body **12** in the translation position. Coordinate systems **290** are illustrated as spaced apart from center point **298** on front undercarriage support assembly **206** and rear undercarriage support assembly **208** in FIG. 2 for sake of clarity. Thus, as discussed herein, when two components of aircraft handler **300** are described as being “laterally spaced apart from one another” the two components may be spaced apart along lateral axis **294**. Likewise, lift mechanism **26** may be described as being configured to move support member **202** along vertical axis **292**.

(70) With continued reference to FIGS. 2-3, in some examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** are spaced apart from one another along ground surface **16**. As mentioned, in some examples, aircraft handler **300** comprises connecting member **216** that is configured to operably interconnect front undercarriage support assembly **206** with rear undercarriage support assembly **208**. In some examples, connecting member **216** is configured to position front undercarriage support assembly **206** and rear undercarriage support assembly **208** relative to one another. More specifically, in some examples, connecting member **216** is configured to position front undercarriage support assembly **206** and rear undercarriage support assembly **208** with a desired lateral spacing therebetween. Additionally or alternatively, in some examples, connecting member **216** is configured to support front undercarriage support assembly **206** and rear undercarriage support assembly **208** aligned with, or offset from, one another.

(71) Front undercarriage support assembly **206** and rear undercarriage support assembly **208** are

described herein as being “aligned” with one another when lateral axis **294** of front undercarriage support assembly **206** is at least substantially aligned with, or collinear with, lateral axis **294** of rear undercarriage support assembly **208**. Similarly, front undercarriage support assembly **206** and rear undercarriage support assembly **208** are described herein as being “offset” from one another when lateral axis **294** of front undercarriage support assembly **206** is offset from, or not at least substantially aligned with, lateral axis **294** of rear undercarriage support assembly **208**.

(72) In some examples, connecting member **216** supports front undercarriage support assembly **206** and rear undercarriage support assembly **208** offset from one another such that front undercarriage support assembly **206** and rear undercarriage support assembly **208** are positioned to support a region of an undercarriage of an aircraft that are offset from one another along a transverse axis of the aircraft. Additionally or alternatively, in some examples, connecting member **216** supports front undercarriage support assembly **206** and rear undercarriage support assembly **208** laterally spaced apart from one another such that front undercarriage support assembly **206** and rear undercarriage support assembly **208** are positioned to support regions of an undercarriage of an aircraft that are spaced apart from one another along a longitudinal axis of the aircraft.

(73) In some examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** each comprise a connecting member receiver **258** that is configured to selectively interconnect with and disconnect from connecting member **216**. In some such examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** each comprise a respective plurality of connecting member receivers **258**. In some more specific examples, front undercarriage support assembly **206** includes a first connecting member receiver **258** positioned on a first side **260** of front undercarriage support assembly **206** and a second connecting member receiver **258** positioned on a second side **262** of front undercarriage support assembly **206** that is opposed to first side **260** of front undercarriage support assembly **206**. Likewise, in some examples, rear undercarriage support assembly **208** includes a first connecting member receiver **258** positioned on a first side **260** of rear undercarriage support assembly **208** and a second connecting member receiver **258** positioned on a second side **262** of rear undercarriage support assembly **208** that is opposed to first side **260** of rear undercarriage support assembly **208**.

(74) In some examples, connecting member **216** is configured to interconnect front undercarriage support assembly **206** with rear undercarriage support assembly **208** via first connecting member receiver **258** of the front undercarriage support assembly **206** and the first connecting member receiver **258** of the rear undercarriage support assembly **208**. Likewise, in some examples, connecting member **216** is configured to interconnect front undercarriage support assembly **206** with rear undercarriage support assembly **208** via second connecting member receiver **258** of front undercarriage support assembly **206** and the second connecting member receiver **258** of the rear undercarriage support assembly **208**.

(75) In some examples, aircraft handler **300** is configured to translate laterally beneath, and engage, the undercarriage of an aircraft from a first lateral side of the aircraft when connecting member **216** interconnects front undercarriage support assembly **206** and rear undercarriage support assembly **208** via the first connecting member receivers **258**. In some such examples, aircraft handler **300** is configured to translate laterally beneath, and engage, the undercarriage of the aircraft from a second lateral side of the aircraft that is opposed to the first lateral side of the aircraft when connecting member **216** interconnects front undercarriage support assembly **206** and rear undercarriage support assembly **208** via the second connecting member receivers **258**. In some examples, the first lateral side of the aircraft corresponds to a right side of the aircraft and the second lateral side of the aircraft corresponds to a left side of the aircraft. In this way, connecting member **216** and the plurality of connecting member receivers **258** collectively may be described being configured to permit aircraft handler **300** to translate laterally beneath, and engage, the undercarriage of an aircraft from either lateral side of the aircraft.

(76) In some examples, connecting member **216** is configured to permit adjustment of the lateral

spacing between front undercarriage support assembly **206** and rear undercarriage support assembly **208**. As shown in FIG. 2, in some examples, connecting member **216** is an articulated connecting member and comprises two or more sections **270** and a plurality of articulated connections **272** that interconnect adjacent sections **270** to one another and that interconnect connecting member **216** to each of front undercarriage support assembly **206** and rear undercarriage support assembly **208**. Articulated connections **272** are configured to selectively permit sections **270** to pivot relative to one another and relative to front undercarriage support assembly **206** and rear undercarriage support assembly **208** to adjust the lateral spacing therebetween. Stated differently, articulated connections **272** may be described as being configured to permit sections **270** to fold relative to one another to adjust the lateral spacing between front undercarriage support assembly **206** and rear undercarriage support assembly **208**. Similarly, in some examples, articulated connections **272** are configured to permit front undercarriage support assembly **206** and rear undercarriage support assembly **208** to move longitudinally relative to one another to permit adjustment of the offset or alignment between front undercarriage support assembly **206** and rear undercarriage support assembly **208**. In some examples, articulated connections **272** are configured to secure or retain each section **270** at a desired pivotal angle relative to one another and/or undercarriage support assemblies **200**, such that connecting member **216** supports and/or secures front undercarriage support assembly **206** and rear undercarriage support assembly **208** at a desired lateral spacing relative to one another and/or at a desired offset and/or in alignment.

(77) Additionally or alternatively, in some examples, connecting member **216** is disconnected from connecting member receivers **258** of front undercarriage support assembly **206** and rear undercarriage support assembly **208** to stow aircraft handler **300**. In such examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** are moved closer to one another once disconnected from connecting member **216**.

(78) As mentioned, in some examples, aircraft handler **300** comprises distal undercarriage support assembly **210** and at least one connecting arm **214** that operably couples distal undercarriage support assembly **210** to rear undercarriage support assembly **208**. In some such examples, aircraft handler **300** includes a first connecting arm **214** and a second connecting arm **214** that operably couple distal undercarriage support assembly **210** to rear undercarriage support assembly **208**. More specifically, in some examples, distal undercarriage support assembly **210** comprises a first portion **222** that is operably coupled to first connecting arm **214** and a second portion **224** of distal undercarriage support assembly **210**. For some examples in which rear undercarriage support assembly **208** comprises ground support chassis **236**, first connecting arm **214** is operably coupled to one projection **238** and second connecting arm **214** is connected to the other projection **238**.

(79) In some examples, first portion **222** and second portion **224** collectively comprise support member **202** and support region **20** of distal undercarriage support assembly **210**. In some examples, first portion **222** and second portion **224** of distal undercarriage support assembly **210** are configured to selectively couple to and uncouple from one another. More specifically, in some examples, distal undercarriage support assembly **210** comprises a distal coupling mechanism **226** that is configured to selectively couple first portion **222** and second portion **224** to one another. In some examples, first portion **222** and second portion **224** collectively are configured to support the second aft portion of the undercarriage of the aircraft when distal coupling mechanism **226** operably couples first portion **222** and second portion **224** to one another. In some examples, first portion **222** and second portion **224** are configured to translate relative to one another when distal coupling mechanism **226** disengages first portion **222** and second portion **224** from one another.

(80) In some examples, connecting arms **214** are configured to move first portion **222** and second portion **224** of distal undercarriage support assembly **210** relative to one another, such as when distal coupling mechanism **226** disengages first portion **222** and second portion **224** from one another. More specifically, as shown in FIG. 2, in some examples, aircraft handler **300** further

comprises a first connecting arm actuator **280** that operably couples the first connecting arm **214** to rear undercarriage support assembly **208**, and a second connecting arm actuator **280** that operably couples the second connecting arm **214** to rear undercarriage support assembly **208**. In some examples, the first connecting arm actuator **280** is configured to move the first connecting arm **214**, together with first portion **222**, along ground surface **16**, and the second connecting arm actuator **280** is configured to move second connecting arm **214**, together with second portion **224**, along ground surface **16**. In this way, first connecting arm actuator **280** and second connecting arm actuator **280** may be configured to adjust the spacing between first portion **222** and second portion **224** along ground surface **16**. In some examples, first connecting arm actuator **280** and/or second connecting arm actuator **280** is a rotary actuator that is configured to selectively pivot the respective connecting arm **214**, together with the respective portion of distal undercarriage support assembly **210** along ground surface **16**.

(81) As shown in FIG. 2, in some examples, distal undercarriage support assembly **210** comprises a supporting configuration **228** in which distal coupling mechanism **226** operably couples first portion **222** and second portion **224** to one another for operative support of the second aft portion of the undercarriage of the aircraft. As shown in FIG. 3, in some examples, distal undercarriage support assembly **210** also comprises a positioning configuration **230** in which distal coupling mechanism **226** disengages first portion **222** from second portion **224**, and first connecting arm actuator **280** and second connecting arm actuator **280** position first portion **222** and second portion **224** spaced apart from one another along ground surface **16**. In some examples, distal undercarriage support assembly **210** is configured to translate laterally beneath the second aft portion of the undercarriage of the aircraft in the positioning configuration **230**. In some such examples, distal undercarriage support assembly **210** also is configured to permit rear undercarriage support assembly **208** to translate laterally beneath the first aft portion of the undercarriage of the aircraft in positioning configuration **230**. More specifically, in some examples, the lateral spacing between first portion **222** and second portion **224** provides a path for distal undercarriage support assembly **210** to move around the first and second portions of the undercarriage of the aircraft during operative positioning of aircraft handler **300**.

(82) As discussed in more detail herein, in some examples, when distal undercarriage support assembly **210** is operably positioned relative to the second aft portion in the positioning configuration **230**, distal undercarriage support assembly **210** is configured to lift the second portion of the undercarriage of the aircraft from contact with ground surface **16** by transitioning from the positioning configuration **230** to supporting configuration **228**. In some such examples, distal coupling mechanism **226** is configured to pull first portion **222** and second portion **224** of distal undercarriage support assembly **210** towards one another to lift the second aft portion of the undercarriage of the aircraft from contact with ground surface **16**.

(83) When included, distal coupling mechanism **226** includes any suitable mechanism for selectively coupling first portion **222** and second portion **224** to one another. In some examples, distal coupling mechanism **226** comprises a pair of hydraulic cylinders that are configured to selectively couple first portion **222** and second portion **224** to one another and move first portion **222** and second portion **224** relative to one another to lift the second aft portion of the undercarriage of the aircraft from contact with ground surface **16**. In such examples, distal coupling mechanism **226** may be described as having a lift-spoon mechanism. In some examples, the hydraulic cylinders are operably coupled to and extend between first portion **222** and second portion **224** in supporting configuration **228** and disconnect first portion **222** from second portion **224** in positioning configuration **230**. More specifically, in such examples, each hydraulic cylinder is operably coupled to only one of first portion **222** and second portion **224** in positioning configuration **230**. In some such examples, one hydraulic cylinder extends from first portion **222** and the other hydraulic cylinder extends from second portion **224** in positioning configuration **230**. In some examples, each hydraulic cylinder comprises a coupling member that is configured to

selectively couple to and disconnect from one of first portion **222** and second portion **224**. In some examples, the hydraulic cylinders are positioned on either side of support region **20** of distal undercarriage support assembly **210**.

(84) As mentioned, in some examples, only one of front undercarriage support assembly **206** and rear undercarriage support assembly **208** comprise a respective drive assembly **212**. In other examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** each comprise a respective drive assembly **212**. As referred to herein, a drive assembly **212** that is included in front undercarriage support assembly **206** may be referred to as a front drive assembly **212**, and a drive assembly **212** that is included in rear undercarriage support assembly **208** may be referred to as a rear drive assembly **212**.

(85) For examples in which aircraft handler **300** only includes front drive assembly **212**, front drive assembly **212** is configured to move aircraft handler **300** and the aircraft along ground surface **16**. Likewise, for examples in which aircraft handler **300** only includes rear drive assembly **212**, rear drive assembly **212** is configured to move aircraft handler **300** along ground surface **16**. In particular, aircraft handler **300** may include only one of front drive assembly **212** and rear drive assembly **212** for examples in which front undercarriage support assembly **206** and rear undercarriage support assembly **208** are interconnected via connecting member **216**, and distal undercarriage support assembly **210** is interconnected with rear undercarriage support assembly **208** via connecting arm(s) **214**.

(86) In some examples, front undercarriage support assembly **206** and rear undercarriage support assembly **208** are mechanically disconnected from one another. In such examples, front undercarriage support assembly **206** comprises front drive assembly **212** and rear undercarriage support assembly **208** comprises rear drive assembly **212**. In some such examples, front drive assembly **212** and rear drive assembly **212** are configured to move front undercarriage support assembly **206** and rear undercarriage support assembly **208** independently of one another. In some such examples, rear drive assembly **212** is configured to move rear undercarriage support assembly **208** together with distal undercarriage support assembly **210**. In some examples, distal undercarriage support assembly **210** is mechanically disconnected from rear undercarriage support assembly **208** and comprises a respective drive mechanism **212** that is configured to move distal undercarriage support assembly **210** along ground surface **16** independently of front undercarriage support assembly **206** and rear undercarriage support assembly **208**.

(87) As shown in FIG. 2, in some examples, aircraft handlers **300** comprise one or more power sources **92** for supplying power to various actuators and/or powered components of **300**. In some examples, aircraft handler **300** further comprises controller **90** for regulating the various actuators and/or powered components such as by controlling the amount of power supplied to each component by power source(s) **92**. Examples of actuators and/or powered components of aircraft handler **300** that may be supplied power by power source(s) and/or regulated by controller **90** include lift mechanism(s) **26**, drive assembly(s) **212**, gates **242**, prong gate(s) **252**, connecting member **216**, articulated connection(s) **272**, connecting arm(s) **214**, connecting arm actuator(s) **280**, connecting arm articulated connection(s) **278**, and/or distal coupling mechanism **226**.

(88) FIG. 4 schematically represents examples of aircraft handlers **10** that are indicated at and referred to herein as aircraft handlers **400**. As shown in the examples of FIG. 4, aircraft handlers **400** include a front undercarriage support assembly **206** and a distal undercarriage support assembly **210** that are operably coupled to front undercarriage support assembly **206**. Support member **202** of front undercarriage support assembly **206** and support member **202** of distal undercarriage support assembly **210** collectively comprise a support arm **18** that is operably coupled to body **12** of front undercarriage support assembly **206** and body **12** of distal undercarriage support assembly **210**. Distal undercarriage support assembly **210** is configured to support support arm **18** on ground surface **16**.

(89) Aircraft handlers **400** further include a plurality of drive assemblies **212**, including a first drive

assembly **14** that includes translation assembly **204** of front undercarriage support assembly **206**, and a second drive assembly **24** that includes translation assembly **204** of distal undercarriage support assembly **210**. First drive assembly **14** and second drive assembly **24** are configured to move aircraft handler **400** along ground surface **16**. More specifically, first drive assembly **14** and second drive assembly **24** are configured to move aircraft handler **400** along ground surface **16** while support arm **18** operably supports at least the respective portion of the aircraft, such as to move aircraft handler **400**, together with aircraft, along ground surface **16**.

(90) Support arm **18** comprises at least one, and optionally a plurality of support regions **20**. Each support region **20** is positioned along any suitable region of support arm **18**. In some examples, support regions **20** are positioned along regions of support arm **18** that correspond to ground contacting portions of the undercarriage of the aircraft. As shown in FIG. **4**, in some examples, support arm **18** comprises a proximal end **28** that is operably coupled to body **12** of front undercarriage support assembly **206**, and a distal end **30** that is opposed to proximal end **28** and operably coupled to body **12** of distal undercarriage support assembly **210**. In some examples, second drive assembly **24** includes a chassis **64**, and distal end **30** of support arm **18** is at least partially overlapping with chassis **64**. In some such examples, support region **20** is positioned along support arm **18** proximate or abutting distal end **30**, such as to at least partially overlap with chassis **64** of second drive assembly **24**.

(91) As discussed herein, support arm **18** may be referred to as lift arm **18**, support prong **18**, support platform **18**, and/or support boom **18**. In some examples, support arm **18** is configured to support and/or suspend at least a portion of the undercarriage of the aircraft spaced apart from, spaced from direct contact with, and/or spaced above ground surface **16**. Support arm **18** also may be configured to support and/or suspend a substantial portion of, or the entirety of the undercarriage of the aircraft spaced above ground surface **16**.

(92) In some examples, support arm **18** is configured to selectively lift at least a portion of the undercarriage of the aircraft from direct contact with ground surface **16**. In some such examples, support arm **18** is configured to translate laterally beneath at least a portion of the undercarriage of the aircraft to lift at least a portion of the undercarriage from direct contact with ground surface **16**. As shown in FIG. **4**, in some examples, support arm **18** comprises one or more transition portions **86** that are configured to facilitate lifting and/or transferring of at least a portion of the undercarriage of the aircraft onto support arm **18**. For example, each transition portion **86** may be configured to facilitate lifting of at least a portion of the undercarriage of the aircraft onto a respective support region **20**. In some such examples, each transition portion **86** is positioned proximate, abutting, or overlapping with the respective support region **20** and is configured to lift at least a portion of the undercarriage of the aircraft onto the respective support region **20** when support arm **18** translates laterally beneath the undercarriage of the aircraft. Each transition portion **86** also may extend beyond a respective support region **20** to define an overhang, a peripheral portion, and/or an edge region of support arm **18**. As discussed herein, in some examples, support region **20** is positioned along support arm **18** to at least partially overlap with chassis **64** of second drive assembly **24**. In such examples, aircraft handler **400** may include a transition portion **86** that is positioned along chassis **64** proximate, or aligned with, support region **20** and is configured to facilitating lifting of at least a portion of the undercarriage of the aircraft onto the support region **20** that overlaps with chassis **64** of second drive assembly **24**.

(93) When included, transition portions **86** comprise any suitable structure or combination of structures for facilitating lifting and/or transitioning of at least a portion of the undercarriage of the aircraft from ground surface **16** onto support arm **18**. As examples, each transition portion **86** may include one or more of a ramp, a wedge, roller(s), a knuckle, and/or combinations thereof.

(94) In some examples, front undercarriage support assembly **206** comprises lift mechanism **26** that operably couples proximal end **28** of support arm **18** to body **12** of front undercarriage support assembly **206**. When included, lift mechanism **26** is configured to selectively raise and lower

proximal end **28** of support arm **18** relative to ground surface **16** to operably pivot support arm **18** relative to ground surface **16**. Stated another way, in some examples, lift mechanism **26** is configured to pivot support arm **18** among a level orientation and a plurality of tilted orientations relative to ground surface **16**. In some such examples, lift mechanism **26** also is configured to selectively retain and orient support arm **18** in each of the level orientation and the plurality of tilted orientations. Lift mechanism **26** also may be described as being configured to pivot support arm **18** among a plurality of tilt angles relative to the ground surface **16**, in which the plurality of tilt angles may be from -5° to 25° . In some examples, proximal end **28** of support arm **18** is pivotally or hingedly coupled to lift mechanism **26** to permit lift mechanism **26** to pivot support arm **18**. Similarly, in some examples, distal end **30** of support arm **18** is pivotally or hingedly coupled to body **12** of distal undercarriage support assembly **210** to permit lift mechanism **26** to pivot support arm **18**.

(95) In some examples, aircraft handlers **400** include one or more tilt sensors installed in support arm **18** that are configured to collect sensor data respective to the tilt angle of support arm **18**, and send the sensor data to controller **90**. Correspondingly, controller **90** may be configured to control lift mechanism **26** at least partially based on the sensor data received from the tilt sensor(s) installed in support arm **18**.

(96) As mentioned, in some examples, aircraft handlers **400** are configured to support an aircraft in the level aircraft orientation, and among a plurality of tilted aircraft orientations. In some such examples, aircraft handlers **400** are configured to selectively tilt or transition the aircraft between the level aircraft orientation and the plurality of tilted aircraft orientations. More specifically, for some examples in which aircraft handler **400** comprises lift mechanism **26**, lift mechanism **26** is configured to selectively raise and lower proximal end **28** of support arm **18** while support arm **18** supports at least a portion of the undercarriage of the aircraft. In such examples, when lift mechanism **26** pivots support arm **18**, support arm **18** correspondingly tilts the aircraft. As such, support arm **18** may be configured to tilt the aircraft among a level aircraft orientation and a plurality of tilted aircraft orientations, such as when lift mechanism **26** pivots support arm **18** among the plurality of tilted orientations. Similarly, support arm **18** also may be configured to operably support and selectively retain the aircraft in the level aircraft orientation and each of the plurality of tilted aircraft orientations.

(97) As discussed herein, in some examples, first drive assembly **14** and second drive assembly **24** are configured to operate independently, such as to steer and/or rotate aircraft handler **400**. Additionally or alternatively, in some examples, first drive assembly **14** and second drive assembly **24** are configured to operate in unison. As shown in FIG. 4, in some examples, first drive assembly **14** and/or second drive assembly **24** comprise one or more driven wheels **60**, which support first drive assembly **14** and/or second drive assembly **24** on ground surface **16**. Also as shown in FIG. 4, in some examples, first drive assembly **14** and/or second drive assembly **24** comprise one or more non-driven wheels **62**.

(98) In some examples, first drive assembly **14** and/or second drive assembly **24** are configured to facilitate lifting or transferring of at least a portion of the undercarriage of the aircraft from ground surface **16** onto support arm **18**. In particular, first drive assembly **14** and/or second drive assembly **24** may move aircraft handler **10** to translate support arm **18** laterally beneath at least a portion of the undercarriage of the aircraft to lift, such as via transition portions **86**, the portion of the undercarriage of the aircraft from contact with ground surface **16** and onto support region(s) **20**.

(99) As discussed in more detail herein with reference to FIG. 5, in some examples, support arm **18** has a deployed orientation **32**, in which support arm **18** is configured to support the undercarriage of the aircraft, and a stowed orientation **34**, in which support arm **18** is configured to occupy a smaller footprint or floor space than support arm **18** occupies in deployed orientation **32**. More specifically, in the deployed orientation, distal end **30** of support arm **18** is spaced further away from body **12** than in stowed orientation **34**. Stated differently, in stowed orientation **34**, distal end

30 and/or second drive assembly **24** are positioned closer to body **12** than in deployed orientation **32**.

(100) In some examples, support arm **18** is configured to be selectively and repeatedly transitioned between the deployed orientation and the stowed orientation. In some such examples, support arm **18** comprises a plurality of members that are configured to move relative to one another when support arm **18** is selectively transitioned between deployed orientation **32** and stowed orientation **34**. As examples, support arm **18** may include at least two, at least three, at least four, at least six, and/or at most ten members. As shown in FIG. 5, in some examples, support arm **18** comprises a first member **80** and a second member **82** that are configured to move relative to one another when support arm **18** is transitioned between deployed orientation **32** and stowed orientation **34**. In some examples, first member **80** is operably coupled to lift mechanism **26** of front undercarriage support assembly **206** and/or comprises proximal end **28** of support arm **18**. In some examples, second member **82** is operably coupled to distal undercarriage support assembly **210** and/or comprises distal end **30** of support arm **18**. In the deployed orientation, first member **80** and second member **82** may be in an overlapping, collinear, and/or in an extended, orientation. In the stowed orientation, first member **80** and second member **82** may be in a folded, tilted, nested and/or condensed orientation.

(101) First member **80** and second member **82** are configured to selectively move relative to one another in any suitable manner when support arm **18** is transitioned between the deployed orientation and the stowed orientation. As examples, first member **80** and second member **82** may be configured to selectively telescope, pivot, and/or slide relative to one another. As a more specific example, as shown in FIG. 4, support arm **18** may comprise a hinge mechanism **84** that hingedly or pivotally couples first member **80** to second member **82** and is configured to selectively permit first member **80** and second member **82** to pivot about hinge mechanism **84** when support arm **18** is transitioned between the deployed orientation and the stowed orientation.

(102) Aircraft handlers **400** may include and/or utilize any suitable mechanism for moving distal end **30** of support arm **18** relative to body **12** when support arm **18** is transitioned between deployed orientation **32** and stowed orientation **34**. In some examples, second drive assembly **24** is configured to move distal end **30** of support arm **18** relative to body **12** of front undercarriage support assembly **206** when support arm **18** is transitioned between deployed orientation **32** and stowed orientation **34**. In some such examples, second drive assembly **24** is configured to move distal end **30** of support arm **18** to cause first member **80** and second member **82** to pivot about hinge mechanism(s) **84** and transition support arm **18** between the deployed orientation and the stowed orientation.

(103) When included, each of first member **80** and second member **82** are positioned along support arm **18** with any suitable relationship relative to support region(s) **20**. As an example, when support arm **18** comprises first member **80**, second member **82**, and one support region **20**, support region **20** may be positioned along either or both first member **80** and second member **82**. For example, support region **20** may be positioned along both of first member **80** and second member **82** when first member **80** and second member **82** overlap in deployed orientation **32**. As another example, when support arm **18** comprises first member **80**, second member **82**, a first support region **20**, and a second support region **20**, the first support region **20** may be positioned along first member **80** and the second support region **20** may be positioned along second member **82**.

(104) With continued reference to FIG. 4, in some examples, aircraft handler **400** comprises a plurality of support arms **18**, such as at least two, at least three, at least four, and/or at most six support arms **18**. As shown in FIG. 4, each support arm **18** may include similar, or substantially similar, structures and/or elements and/or may be operably coupled to similar, or substantially similar, structures and/or elements, such as in a similar, or substantially similar, manner. Specifically, each support arm **18** may be operably coupled to body **12** of a respective undercarriage support assembly **200** in a similar, or at least substantially similar, manner, such as

via a lift mechanism **26**. Similarly, each support arm **18** may extend from body **12** of the respective undercarriage support assembly **200** in a similar direction as one or more other support arms **18** and/or may extend parallel to, or at least substantially parallel to one or more other support arms **18**. That said, each support arm **18** may include a unique number, quantity, and/or distribution of support regions **20** and optionally transition portions **86**. In this way, each support arm **18** may be configured to support, lift, suspend, and/or handle a different portion or region of the undercarriage of the aircraft. Similarly, each support arm **18** may comprise different placement of first member **80** and second member **82**.

(105) As shown in FIG. **4**, in some examples, distal undercarriage support assembly **210** is a first distal undercarriage support assembly **209**, support arm **18** is a first support arm **36**, and aircraft handler **400** further includes a rear undercarriage support assembly **208**, a pair of distal undercarriage support assemblies **210**, which includes the first distal undercarriage support assembly **209** and a second distal undercarriage support assembly **211**, and a pair of support arms **18**, which includes first support arm **36** and a second support arm **38**. In such examples, body **12** of front undercarriage support assembly **206** is referred to herein as a first body member **50**, and body **12** of rear undercarriage support assembly **208** is referred to herein as second body member **52**. As shown in FIG. **4**, in some examples, first body member **50** and second body member **52** are operably coupled to one another. In some examples, second distal undercarriage support assembly **211** possesses similar, or at least substantially, similar components, functions, attributes, etc. as that discussed herein for first distal undercarriage support assembly **209**.

(106) First support arm **36** operably interconnects front undercarriage support assembly **206** to first distal undercarriage support assembly **210** as discussed herein. Support member **202** of rear undercarriage support assembly **208** and support member **202** of second distal undercarriage support assembly **211** collectively comprise second support arm **38**. Second support arm **38** is operably coupled to second body member **52** of rear undercarriage support assembly **208** and to body **12** of second distal undercarriage support assembly **211** and interconnects rear undercarriage support assembly **208** with second distal undercarriage support assembly **211**. In some examples, second distal undercarriage support assembly **211** is operably coupled to distal end **30** of second support arm **38** and rear undercarriage support assembly **208** is operably coupled to proximal end **28** of second support arm **38**.

(107) As mentioned, aircraft handlers **400** comprise a plurality of drive assemblies **212**, which include first drive assembly **14** and second drive assembly **24**. In some examples, second distal undercarriage support assembly **211** further includes a drive assembly **212** that is indicated at and referred to herein as third drive assembly **40**. When included, third drive assembly **40** includes translation assembly **204** of second distal undercarriage support assembly **211** and is configured to move aircraft handler **400** along ground surface **16** together with the other drive assemblies **212**, which include first drive assembly **14**, second drive assembly **24**, and optionally a fourth drive assembly **44**.

(108) Additionally or alternatively, in some examples, rear undercarriage support assembly **208** further includes a drive assembly **212** that is indicated at and referred to herein as fourth drive assembly **44**. When included, fourth drive assembly **44** includes translation assembly **204** of rear undercarriage support assemblies **208** and is configured to move aircraft handler **400** along ground surface **16** together with the other drive assemblies **212**, which include first drive assembly **14**, second drive assembly **24**, and optionally third drive assembly **40**.

(109) When included, third drive assembly **40** may comprise similar, or substantially similar, features, functions, and/or components and/or may be operably coupled to similar, or substantially similar, elements and/or structures to those discussed in detail herein for second drive assembly **24**. Likewise, when included, fourth drive assembly **44** may comprise similar, or substantially similar, features, functions, and/or components, and/or may be operably coupled to similar, or substantially similar, elements and/or structures to those discussed in detail herein for first drive assembly **14**.

For example, as shown in FIG. 4, third drive assembly **40** and/or fourth drive assembly **44** each may comprise one or more driven wheels **60** and/or one or more non-driven wheels **62**. Additionally, third drive assembly **40** and/or fourth drive assembly **44** may comprise chassis **64**, and transition portion **86** may be operably coupled to chassis **64** of third drive assembly **40** and/or fourth drive assembly **44**.

(110) In some examples, first support arm **36** and second support arm **38** are configured to support different or discrete regions of an undercarriage of an aircraft. For example, first support arm **36** may be configured to support a forward portion of the undercarriage of the aircraft and second support arm **38** may be configured to support a rearward portion of the undercarriage of the aircraft or vice versa. Stated differently, in some examples, front undercarriage support assembly **206** and first distal undercarriage support assembly **209** collectively are configured to support the forward portion of the undercarriage of the aircraft, and rear undercarriage support assembly **208** together with second distal undercarriage support assembly **211** collectively are configured to support the rear portion of the undercarriage of the aircraft, or vice versa. In some examples, the undercarriage of the aircraft comprises a landing gear assembly having a front wheel assembly and one or more rear wheel assemblies. In such examples, first support arm **36** may be configured to support the front wheel assembly and second support arm **38** may be configured to support the one or more rear wheel assemblies or vice versa.

(111) First support arm **36** and second support arm **38** each include any suitable number of support regions **20** for supporting respective portions of the undercarriage of the aircraft. For example, in some examples, the landing gear assembly of the aircraft includes one set of front wheels and two sets of spatially separated rear wheels. In such examples, first support arm **36** may include a single support region **20** for supporting the single set of front wheels, and second support arm **38** may include two spatially separated support regions **20** that are configured to support the two sets of rear wheels or vice versa.

(112) In some examples, aircraft handler **10** includes a plurality of lift mechanisms **26**, in which each lift mechanism **26** operably couples proximal end **28** of a particular support arm **18** to body **12** of an undercarriage support assembly **200**. More specifically, in some examples, aircraft handler **400** includes a first lift mechanism **94** and a second lift mechanism **96**, in which first lift mechanism **94** operably couples proximal end **28** of first support arm **36** to first body member **50** and is configured to selectively pivot first support arm **36**, and second lift mechanism **96** operably couples proximal end **28** of second support arm **38** to second body member **52** and is configured to selectively pivot second support arm **38**, such as discussed herein. In some examples, first lift mechanism **94** and second lift mechanism **96** are configured to pivot first support arm **36** and second support arm **38** in unison, and in some examples, first lift mechanism **94** and second lift mechanism **96** are configured to pivot first support arm **36** and second support arm **38** independently of one another. Likewise, in some examples, first lift mechanism **94** and second lift mechanism **96** comprise the same type of lift mechanism and in other examples, first lift mechanism **94** and second lift mechanism **96** comprise different types of lift mechanisms.

(113) With continued reference to FIG. 4, in some examples, aircraft handler **400** comprises connecting member **216** that operably couples first body member **50** of front undercarriage support assembly **206** to second body member **52** of rear undercarriage support assembly **208**. In some examples, connecting member **216** is, or includes, a frame **54** that is configured to support first body member **50** and second body member **52** spaced apart along ground surface **16**. In some such examples, first body member **50** and second body member **52** support first support arm **36** and second support arm **38** spaced apart along ground surface **16**. In such examples, first support arm **36** and second support arm **38** extend from first body member **50** and second body member **52** with a desired lateral spacing being defined therebetween.

(114) In some examples, frame **54** includes a telescoping mechanism **58** that is configured to selectively and operably adjust the spacing between first body member **50** and second body

member **52**. Stated differently, in some examples, telescoping mechanism **58** is configured to selectively and operably adjust the spacing between first support arm **36** and second support arm **38**. When frame **54** includes telescoping mechanism **58**, frame **54** may be referred to as a telescoping frame **56**.

(115) In some examples, telescoping mechanism **58** is utilized to adjust the spacing between first support arm **36** and second support arm **38** to permit aircraft handler **400** to lift, move, and/or tilt aircraft having various undercarriage configurations. For example, telescoping mechanism **58** may be utilized to adjust the spacing between first support arm **36** and second support arm **38** to permit aircraft handler **400** to lift, move, and/or tilt aircraft having landing gear assemblies with various separations between the forward landing gear portion and the rear landing gear portion. In some examples, telescoping mechanism **58** is utilized to adjust the spacing between first support arm **36** and second support arm **38** when aircraft handler **400** lifts or transfers the undercarriage of the aircraft to and from ground surface **16**. For example, to lift the undercarriage of the aircraft from the ground surface, telescoping mechanism **58** may increase the spacing between first support arm **36** and second support arm **38**. Then, first support arm **36** may translate (such as via first drive assembly **14** and/or second drive assembly **24**) to a position on ground surface **16** that is fore of a forward portion of the undercarriage of the aircraft, while second support arm **38** translates (such as via third drive assembly **40** and/or fourth drive assembly **44**) to a position that is aft of a rearward portion of the undercarriage of the aircraft. Once positioned fore and aft of the undercarriage of the aircraft, telescoping mechanism **58** may be utilized to reduce the spacing between first support arm **36** and second support arm **38**, such that first support arm **36** and second support arm **38** translate laterally beneath and lift (such as via transition portions **86**) the forward and rear regions of the undercarriage from ground surface **16** onto respective support regions **20**.

(116) Telescoping mechanism **58** includes any suitable mechanism for adjusting the spacing between first body member **50** and second body member **52**, and/or the spacing between first support arm **36** and second support arm **38**. For example, telescoping frame **56** may include a first portion and a second portion that are configured to translate and/or telescope relative to one another, and telescoping mechanism **58** may be configured to facilitate translation and/or telescoping of the first portion and the second portion of telescoping frame **56**. As examples, telescoping mechanism **58** may include a hydraulic telescoping mechanism, a mechanical telescoping mechanism, an electromechanical telescoping mechanism, a screw jack telescoping mechanism, a chain driven telescoping mechanism and/or a rotary driven telescoping mechanism. As a more specific example, telescoping mechanism **58** includes one or more rails and one or more rollers disposed on the first portion and the second portion of telescoping frame **56** that interface with one another to facilitate translation. In some examples, telescoping mechanism **58** is an active telescoping mechanism and may power translation of first body member **50** and first support arm **36** relative to second body member **52** and second support arm **38**. As a more specific example, telescoping mechanism **58** may include powered rollers that power translation of the first and second portions of telescoping frame **56** relative to one another. In some such examples, telescoping mechanism **58** is configured to receive power from power source(s) **92** to adjust the spacing between first support arm **36** and second support arm **38**. Alternatively, in some examples, telescoping mechanism **58** is a passive telescoping mechanism and one or more of first drive assembly **14**, second drive assembly **24**, third drive assembly **40**, and/or fourth drive assembly **44** move first body member **50** and/or first support arm **36** relative to second body member **52** and/or second support arm **38**.

(117) As mentioned, in some examples, aircraft handler **400** has a deployed configuration and a stowed configuration, in which aircraft handler **400** is configured to occupy a smaller floor space or footprint in the stowed configuration relative to the deployed configuration. In some examples, support arm(s) **18** are in the deployed orientation when aircraft handler **400** is in the deployed configuration, and support arm(s) **18** are in the stowed orientation when aircraft handler **400** is in

the stowed configuration. In some examples, aircraft handler **400** is configured to selectively transition between the deployed configuration and the stowed configuration. In some such examples, telescoping mechanism **58** is configured to control the spacing between first body member **50** and second body member **52** and/or the spacing between first support arm **36** and second support arm **38**, to transition aircraft handler **400** between the deployed configuration and the stowed configuration.

(118) As shown in FIG. **4**, in some examples, aircraft handlers **400** comprise one or more power sources **92** for supplying power to various actuators and/or powered components of aircraft handler **400**. In some examples, aircraft handler **400** further comprises controller **90** for regulating the various actuators and/or powered components such as by controlling the amount of power supplied to each component by power source(s) **92**. Examples of actuators and/or powered components of aircraft handler **400** that may be supplied power by power source(s) **92** and/or regulated by controller **90** include lift mechanism(s) **26**, first drive assembly **14**, second drive assembly **24**, third drive assembly **40**, fourth drive assembly **44**, and/or telescoping mechanism **58**.

(119) Turning to FIG. **5**, illustrated therein is a schematic side view representing examples of aircraft handlers **400** with support arm **18** oriented in deployed orientation **32**, stowed orientation **34**, level orientation **100**, and among tilted orientations **102**. In particular, support arm **18** is illustrated in solid lines in level orientation **100**, support arm **18** is illustrated in dot-dash lines in tilted orientation **102**, and support arm **18** and second drive assembly **24** are illustrated in dot-dot-dash lines in stowed orientation **34**.

(120) Support arm **18** may be described as being in deployed orientation **32** while support arm **18** is in level orientation **100** and among tilted orientations **102**. In particular, as shown in FIG. **2**, support arm **18** may be configured to support at least a portion of an undercarriage **108** of the aircraft while oriented in level orientation **100** and among tilted orientations **102**. Aircraft handler **400** comprises first drive assembly **14** that supports body **12** of front undercarriage support assembly **206** on ground surface **16**. Aircraft handler **400** also includes distal undercarriage support assembly **210** that is operably coupled to distal end **30** of support arm **18** and supports support arm **18** on ground surface **16**. Distal undercarriage support assembly **210** comprises second drive assembly **24** that supports distal undercarriage support assembly **210** on ground surface **16**. As shown, first drive assembly **14** and second drive assembly **24** each may comprise one or more driven wheels **60**, and optionally one or more non-driven wheels **62**, that may support first drive assembly **14** and second drive assembly **24** on ground surface **16**.

(121) In level orientation **100**, support arm **18** extends between body **12** of front undercarriage support assembly **206** and body **12** of distal undercarriage support assembly **210** substantially parallel to ground surface **16**. Here, lift mechanism **26** supporting proximal end **28** of support arm **18** generally is aligned with, or at the same height as, distal end **30** of support arm **18**. When support arm **18** is in level orientation **100**, support region **20** may support undercarriage **108** of aircraft **22** and orient aircraft **22** in level aircraft orientation **104**. For example, when support arm **18** is oriented in level orientation **100**, support region **20** may provide a level surface, platform, or region for supporting undercarriage **108**, such as one or more wheels of a landing gear assembly, of aircraft **22**.

(122) When support arm **18** is among tilted orientations **102**, lift mechanism **26** supports proximal end **28** of support arm **18** spaced further above ground surface **16** than in level orientation **100**. In other words, lift mechanism **26** supports proximal end **28** spaced further above ground surface **16** than distal end **30** to orient support arm **18** among tilted orientations **102**. As shown in FIG. **5**, in some examples, lift mechanism **26** comprises a pillar **97** that extends upwardly and/or generally normal to ground surface **16** and lift mechanism **26** is configured to selectively move proximal end **28** of support arm **18** up and down pillar **97** to selectively transition support arm **18** among level orientation **100** and tilted orientations **102**.

(123) In some examples, each tilted orientation **102** corresponds to lift mechanism **26** supporting

proximal end **28** of support arm **18** and at a particular spacing above ground surface **16** and/or at a particular spacing above distal end **30**, such as to orient support arm **18** with a particular tilt angle. Further shown in FIG. 5, in some examples, support region **20** is configured to support at least a portion of the undercarriage **108** of the aircraft **22** while support arm **18** is oriented in any desirable tilted orientation **102**, such that support arm **18** supports or orients aircraft **22** in a corresponding tilted aircraft orientation **106**. In this way, lift mechanism **26** and support arm **18** are configured to selectively tilt aircraft **22** among level aircraft orientation **104** and tilted aircraft orientations **106**.

(124) In some examples, aircraft handler **400** and/or support arm **18** comprise one or more securing mechanisms for engaging undercarriage **108** of aircraft **22** with support arm **18**, such as while support arm **18** supports aircraft **22** among tilted aircraft orientations **106**. As discussed herein, in some examples, support region **20** comprises one or more portions that are configured to engage with undercarriage **108**, such as clamps and/or friction fits, while support arm **18** supports aircraft **22** among titled aircraft orientations **106**. Additionally or alternatively, in some examples, one or more tie-down mechanisms are utilized to secure aircraft **22** to support arm **18** and/or support region **20** while support arm **18** orients aircraft **22** among tilted aircraft orientations **106**.

(125) With continued reference to FIG. 5, illustrated in dot-dot-dash lines is an example of support arm **18** oriented in stowed orientation **34**. As shown, distal end **30** of support arm **18** is positioned closer to proximal end **28** and/or body **12** of front undercarriage support assembly **206** in stowed orientation **34** than in deployed orientation **32**. Similarly, second drive assembly **24** and/or body **12** of distal undercarriage support assembly **210** is positioned closer to first drive assembly **14**, body **12** of front undercarriage support assembly **206**, and/or proximal end **28** of support arm **18** when support arm **18** is in stowed orientation **34**. In this way, aircraft handler **400** occupies a smaller floor space or footprint when support arm **18** is oriented in stowed orientation **34**.

(126) In the specific example illustrated in FIG. 5, support arm **18** comprises first member **80**, second member **82**, and hinge mechanism **84** that hingedly couples first member **80** and second member **82** to one another. In some examples, when support arm **18** is in deployed orientation **32**, first member **80** and second member **82** are substantially collinear and may be overlapping. When support arm **18** is in stowed orientation **34**, first member **80** and second member **82** are tilted upwardly and pivoted relative to one another about hinge mechanism **84** to permit distal end **30**, second drive assembly **24**, and/or body **12** of distal undercarriage support assembly **210** to be positioned closer to proximal end **28**, body **12** of front undercarriage support assembly **206** and/or first drive assembly **14**.

(127) FIGS. 6-10 provide an illustrative, non-exclusive example of aircraft handlers **300** that is indicated at and referred to herein as aircraft handler **350**. In particular, FIGS. 6-7 illustrate examples of aircraft handler **350** supporting an aircraft **22**, and FIGS. 8-10 illustrate various configurations of example aircraft handler **350**. Where appropriate, the reference numerals from the schematic illustrations of FIGS. 1-3 are used to designate corresponding parts of the example aircraft handler **350** of FIGS. 6-10; however, aircraft handler **350** is non-exclusive and does not limit aircraft handlers **300** to the illustrated embodiments of FIGS. 6-10. That is, aircraft handlers **300** are not limited to the specific embodiment of FIGS. 6-10, and aircraft handlers **300** may incorporate any number of the various aspects, configurations, characteristics, properties, variants, options etc. of aircraft handlers **300** that are illustrated in and discussed with reference to the schematic representations of FIGS. 2-3 and/or the embodiment of FIGS. 6-10, as well as variations thereof, without requiring the inclusion of all such aspects, configurations, characteristics, properties, variants, options etc. Furthermore, any additional aspects, configurations, characteristics, properties, variants, options, etc. disclosed in connection with the example aircraft handler **300** of FIGS. 6-10 may be utilized with and/or otherwise included in other aircraft handlers **300**, including aircraft handlers **300** according to FIGS. 2-3. For the purpose of brevity, each previously discussed component, part, portion, aspect, region, etc. or variants thereof may not be discussed, illustrated, and/or labeled again with respect to the examples of FIGS. 6-10; however, it

is within the scope of the present disclosure that the previously discussed features, variants, etc. may be utilized with the examples of FIGS. 6-10.

(128) With initial reference to FIG. 6, aircraft handler 350 is illustrated supporting aircraft 22 in level aircraft orientation 104 and among tilted aircraft orientations 106. In particular, aircraft 22, an envelope 120 defined by aircraft 22, and support members 202 of front undercarriage support assembly 206 and rear undercarriage support assembly 208 are illustrated in dashed lines when aircraft handler 350 supports aircraft 22 in level aircraft orientation 104. Aircraft 22, envelope 120, and support members 202 of front undercarriage support assembly 206 and rear undercarriage support assembly 208 are illustrated in solid lines when aircraft handler 350 supports aircraft 22 among tilted aircraft orientations 106.

(129) As shown in FIG. 6, example aircraft handler 350 includes front undercarriage support assembly 206, rear undercarriage support assembly 208, and distal undercarriage support assembly 210. Distal undercarriage support assembly 210 is operably coupled to rear undercarriage support assembly 208 via connecting arms 214, and connecting member 216 operably couples front undercarriage support assembly 206 to rear undercarriage support assembly 208. Support member 202 of each undercarriage support assembly 200 operably engages and supports a respective portion of undercarriage 108 spaced above or apart from ground surface 16. More specifically, support member 202 of rear undercarriage support assembly 208 supports a first rear wheel assembly of undercarriage 108, and support member 202 supports a second rear wheel assembly of undercarriage 108 that is laterally spaced apart from the first wheel assembly. Support member 202 of front undercarriage support assembly 206 supports a front wheel assembly of undercarriage 108 that is spaced apart from the rear wheel assemblies along a longitudinal axis of aircraft 22, and positioned laterally between the rear wheel assemblies. Connecting member 216 supports front undercarriage support assembly 206 offset and laterally spaced apart from rear undercarriage support assembly 208 such that support member 202 of front undercarriage support assembly 206 is positioned to support the front wheel assembly.

(130) Thus, in these examples, aircraft handler 350 supports or suspends the entire undercarriage 108 of aircraft 22 spaced above or spaced apart from ground surface 16. Front undercarriage support assembly 206 comprises front drive assembly 212 that supports front undercarriage support assembly 206 on ground surface 16, rear undercarriage support assembly 208 comprises rear drive assembly 212 that supports rear undercarriage support assembly 208 on ground surface 16, and distal undercarriage support assembly 210 is supported on ground surface 16 by the respective translation assembly 204.

(131) As shown in FIG. 6, the envelopes 120 occupied or defined by aircraft 22 in level aircraft orientation 104 and in tilted aircraft orientations 106 may be described herein as rectangular prisms that encompass aircraft 22. More specifically, the bottom side or base of the rectangular prisms are defined by ground surface 16 and each other side of the rectangular prism has a point that contacts, abuts, or is defined by a maximal extent of aircraft 22 in that particular direction. In the specific example represented in FIG. 6, aircraft 22 is a fixed wing aircraft and the maximal extent of the wings 124 of aircraft 22, the fore-most point 126 of aircraft 22, and the aft-most point 128 of aircraft 22 abut, contact, or define the four vertical sides of the rectangular prisms of envelopes 120. The top side or upper side of the rectangular prisms of envelopes 120 may abut, contact, or be defined by the highest waterline or highest vertical extent of aircraft 22, such as the empennage 129 shown in the specific example of FIG. 6. The bottom side or base of the rectangular prisms of envelopes 120 may abut, contact, or be defined by the ground contact points of undercarriage 108 of aircraft 22 when aircraft 22 is supported directly on ground surface 16. When aircraft 22 is supported by aircraft handler 350, the base of the rectangular prism and abuts, contacts, or is defined by drive assembly 212 and translation assemblies 204 of aircraft handler 350.

(132) The width of each envelope 120 is defined by the distance between the opposing sides or surfaces of the rectangular prism that abut, contact, or are defined by the maximal extent of wings

124. In the specific example of FIG. 6, aircraft **22** includes wingtips that are configured to fold upwardly, such as to reduce the width of envelope **120**. However, this is not a requirement of the present disclosure. The height of envelopes **120** is defined by the distance between the top and bottom sides or surfaces of the rectangular prisms, and the length of envelopes **120** may be defined as the distance between the opposing sides or surfaces of the rectangular prisms that abut, contact, or are defined by the fore-most point **126** and aft-most point **128** of aircraft **22**.

(133) As shown in the example of FIG. 6, aircraft handler **350** is configured to fit completely within envelope **120** of aircraft **22** when aircraft handler **350** supports aircraft **22** in level aircraft orientation **104** and/or among tilted aircraft orientations **106**. Stated differently, in some examples, aircraft handler **350** is configured to support aircraft **22** without extending beyond, or expanding, the four sidewalls of the rectangular prism of envelope **120** and/or extending above the upper side or surface of the rectangular prism of envelope **120**. In this way, aircraft handler **10** may support aircraft **22** without increasing the width or length of envelope **120**.

(134) As shown in dashed lines in FIG. 6, envelope **120** of aircraft **22** in level aircraft orientation **104** comprises different dimensions than envelope **120** of aircraft **22** among tilted aircraft orientations **106**. In other words, aircraft handler **350** may be described as being configured to selectively alter the dimensions of envelope **120** by selectively tilting aircraft **22** between level aircraft orientation **104** and tilted aircraft orientations **106**. More specifically, as shown in the example of FIG. 6, when aircraft handler **350** supports aircraft **22** in level aircraft orientation **104**, or when aircraft **22** is supported on ground surface **16**, wings **124** of aircraft **22** extend generally horizontally or generally parallel to ground surface **16**. When aircraft handler **350** tilts aircraft **22** or supports aircraft **22** among tilted aircraft orientations **106**, wings **124** of aircraft **22** are tilted relative to ground surface **16**. In this way, aircraft handler **10** orients wings **124** to extend upwardly or partially traverse relative to ground surface **16** or level aircraft orientation **104**, such that the projection of wings **124** onto ground surface **16** is smaller when aircraft handler **10** tilts aircraft **22** among tilted aircraft orientations **106**. Consequently, as shown in the example of FIG. 6, envelope **120**, and/or the width thereof, of aircraft **22** in tilted aircraft orientations **106** is smaller or narrower than envelope **120**, and/or the width thereof, of aircraft **22** in level aircraft orientation **104**. Stated differently, when aircraft handler **350** tilts, supports, or orients aircraft **22** among tilted aircraft orientations **106**, the width of envelope **120** is smaller or narrower than the width of envelope **120** of aircraft **22** in level aircraft orientation **104** and/or when aircraft **22** is supported directly on ground surface **16**. Stated another way, in some examples, aircraft handler **350** is configured to reduce the width of envelope **120** of aircraft **22** by tilting aircraft **22** to among tilted aircraft orientations **106**. Stated yet another way, in some examples, aircraft handler **350** is configured to reduce the footprint of aircraft **22**, or floor space occupied by aircraft **22**, by tilting aircraft **22** to among tilted aircraft orientations **106**. As more specific examples, aircraft handler **350** may be configured to tilt aircraft **22** to among tilted aircraft orientations **106** to reduce the width of envelope **120** of aircraft **22** by at least 0.5%, at least 1%, at least 2%, at least 3%, at most 6%, and/or at most 10% the width of envelope **120** of aircraft **22** in level aircraft orientation **104** or supported directly on ground surface **16**.

(135) In some examples, aircraft handler **350** is configured to tilt aircraft **22** to fit aircraft **22** within a defined volume **130**, an external volume **130**, or a confined volume **130**, that is smaller or narrower than envelope **120** of aircraft **22** in level aircraft orientation **104** and/or envelope **120** of aircraft **22** supported directly on ground surface **16**. Stated differently, aircraft handler **350** may be configured to fit aircraft **22** to within defined volume **130** that aircraft **22**, or envelope **120** of aircraft **22**, cannot fit within defined volume **130** when aircraft **22** is oriented in level aircraft orientation **104** and/or when aircraft **22** is supported on ground surface **16**. More specifically, in some examples, aircraft handler **350** is configured to tilt aircraft **22** to among tilted aircraft orientations **106** to fit aircraft **22** within defined volume **130** having a width that is smaller than the width of envelope **120** of aircraft **22** in level aircraft orientation **104** and/or the envelope **120** of

aircraft **22** directly supported on ground surface **16**.

(136) As discussed herein, in some examples, aircraft handler **350** is configured to move aircraft **22** while supporting aircraft **22** among tilted aircraft orientations **106**. As such, in some examples, aircraft handler **350** is configured to move aircraft **22** to within, and from within, defined volume **130** that is smaller than envelope **120** of aircraft **22** in level aircraft orientation **106** and/or envelope **120** of aircraft **22** supported on ground surface **16**. As examples, aircraft handler **350** may be configured to fit aircraft **22** within, and/or move aircraft **22** to within, or from within, a defined volume **130** that is defined by a rectangular space that is surrounded by six walls, the internal space of a building, the internal space of a shipping container, the internal space of an elevator, and/or the internal space of an aircraft hangar.

(137) As another example, aircraft handler **350** may be configured to fit aircraft **22** within a parking space having an defined volume **130**, and/or a width thereof, that is smaller than envelope **120** of aircraft **22**, or width thereof, of aircraft **22** in level aircraft orientation **104** or supported on ground surface **16**. In some examples, defined volume **130** of the parking space is defined by adjacently parked aircraft and/or one or more access lanes surrounding the parking space. In some examples, aircraft handler **350** is configured to park aircraft **22** in the parking space while orienting aircraft **22** among tilted aircraft orientations **106** such that wings **124** of aircraft **22** overlap or extend above and below the wings of adjacently parked aircraft. In this way, aircraft handler **350** may be configured to park aircraft **22** with envelope **120** of aircraft **22** extending beyond defined volume **130** of the parking space and overlapping with the envelopes of adjacently parked aircraft, such as to permit more aircraft to be parked within a given area.

(138) In some examples, when aircraft handler **350** tilts or supports aircraft **22** among tilted aircraft orientations **106**, an aircraft tilt angle is formed between a line extending between the wing tips of, or the maximal extents of, wings **124** and the ground surface **16**. Stated another way, in some examples, aircraft handler **350** is configured to tilt aircraft **22** to among a plurality of aircraft tilt angles relative to level aircraft orientation **104**, in which the plurality of aircraft tilt angles may be from -5° to 25° . In some examples, the aircraft tilt angle corresponds to the tilt angle of support arms **18**.

(139) While FIG. **3** illustrates an example in which aircraft **22** is a fixed wing aircraft, aircraft handler **350** also may be configured to support, tilt, and alter the dimensions of the envelope of other types of aircraft, with examples including rotorcraft and/or helicopters. In some examples, the width of the envelope of a helicopter is defined by the diameter of, or a rectangle surrounding, the blades of the helicopter. In such examples, aircraft handler **350** may be configured to tilt the helicopter with the blades of the helicopter tilted relative to ground surface **16** and/or relative to the horizontal or level position of the helicopter to reduce the width or size of the envelope of the helicopter, such as discussed herein.

(140) Turning to FIG. **7**, illustrated therein is an example in which aircraft handler **350** is configured to rotate aircraft **22**. As discussed herein with reference to FIG. **3**, support member **202** of front undercarriage support assembly **206** supports the front wheel assembly of undercarriage **108**, support member **202** of rear undercarriage support assembly **208** supports the first rear wheel assembly of undercarriage **108**, and support member **202** of distal undercarriage support assembly **210** supports the second wheel assembly of undercarriage **108**. In this way, aircraft handler **350** supports or suspends the entire undercarriage of aircraft **22** spaced apart from ground surface **16**. More specifically, aircraft handler **350** supports or suspends aircraft **22** spaced above ground surface **16** while being positioned completely within the envelope of aircraft **22**.

(141) As illustrated in FIG. **7**, aircraft handler **350** is configured to rotate aircraft **22** about a yaw axis **132** of aircraft **22**. In other words, in some examples, aircraft handler **350** is configured to rotate aircraft **22** without moving yaw axis **132** of aircraft **22** along ground surface **16**. Stated another way, in some examples, aircraft handler **350** is configured to rotate aircraft **22** within a cylindrical volume **134**, in which the radius **136** of the cylindrical volume **134** is a threshold

fraction of a distance between yaw axis **132** and a maximum extent **122** of aircraft **22** from yaw axis **132**. As examples, radius **136** may be at least 100%, at most 110%, at most 120%, and/or at most 130% of the distance between yaw axis **132** and the maximum extent **122** of aircraft **22** from yaw axis **132**. As, in some examples, aircraft handler **350** is configured to fit completely within the envelope **120** of aircraft **22**, aircraft handler **350** also may be configured to rotate aircraft **22** while aircraft handler **350** remains completely within cylindrical volume **134**. In view of the above, aircraft handler **350** may be configured to rotate aircraft **22** with aircraft handler **350** and aircraft **22** requiring a minimal volume within which to rotate, which may be beneficial to relocating aircraft **22** within crowded or confined spaces. Additionally, while FIG. 7 illustrates an example in which aircraft handler **350** is configured to rotate aircraft **22** in level aircraft orientation **104**, aircraft handler **350** also may be configured to rotate aircraft **22** while aircraft handler **350** supports aircraft among tilted aircraft orientations **106**.

(142) Aircraft handler **350** may be configured to rotate aircraft **22** in any suitable direction of rotation, such as a clockwise rotation and/or a counter-clockwise rotation, and through any suitable angle of rotation, such as through 10°, through 30°, through 90°, through 180° and/or through 360°, such as while aircraft **22** and/or aircraft handler **350** remains within cylindrical volume **134**. In some examples, drive assembly **212** of front undercarriage support assembly **206** and/or drive assembly **212** of rear undercarriage support assembly **208** are utilized to rotate aircraft **22**. In some such examples, one or more driven wheels **60** of drive assembly **212** are configured to selectively pivot to a particular pivot angle and provide motive force along the pivot angle to rotate aircraft handler **350** and aircraft **22** about ground surface **16**.

(143) FIGS. 8-10 are isometric views of example aircraft handler **350** with FIGS. 8-10 each showing various components of aircraft handler **350** in different configurations. In particular, FIG. 8 illustrates an example of aircraft handler **350** in a configuration to support aircraft **22** in level aircraft orientation **104**, FIG. 9 illustrates an example of aircraft handler **350** in a configuration to translate laterally beneath aircraft **22** to be operably positioned to support aircraft **22**, and FIG. 10 illustrates an example of aircraft handler **350** in a configuration to support aircraft **22** among tilted aircraft orientations **106**.

(144) With initial reference to FIG. 8, front undercarriage support assembly **206** and rear undercarriage support assembly **208** are operably interconnected with one another via connecting member **216**. Connecting member **216** comprises a pair of sections **270** that are operably coupled to one another via articulated connection **272**. Connecting member **216** also includes two additional articulated connections **272** that are configured to hingedly couple sections **270** to front undercarriage support assembly **206** and rear undercarriage support assembly **208**. Front undercarriage support assembly **206** and rear undercarriage support assembly **208** are offset from one another such that connecting member **216** extends from articulated connections **272** at an off-normal angle relative to first side **260** of front undercarriage support assembly **206** and first side **260** of rear undercarriage support assembly **208**. In the example shown in FIG. 8, connecting member **216** selectively secures front undercarriage support assembly **206** and rear undercarriage support assembly **208** offset from one another by selectively retaining articulated connections **272** in the illustrated pivot angles. In some examples, each articulated connection **272** comprises a corresponding actuator that is configured to selectively permit and restrict pivotal movement about articulated connection **272**.

(145) In some examples, the articulated connection **272** that interconnects sections **270** is configured to selectively permit sections **270** to fold relative another, thereby allowing first side **260** of front undercarriage support assembly **206** to be positioned closer to first side **260** of rear undercarriage support assembly **208**. In some examples, articulated connection **272** that interconnects sections **270** is configured to selectively permit sections **270** to fold relative another to permit aircraft handler **350** to transition from the deployed configuration illustrated in FIG. 8 to the stowed configuration, in which aircraft handler **350** occupies a smaller footprint, as discussed

herein.

(146) With continued reference to FIG. 8, front undercarriage support assembly **206** and rear undercarriage support assembly **208** each include a respective lift mechanism **26**. Front undercarriage support assembly **206** and rear undercarriage support assembly **208** each also include a respective fork assembly **248** that is operably coupled to the respective body **12** via lift mechanism **26**. Each lift mechanism **26** is configured to move the respective fork assembly **248** relative to ground surface **16**, such as to tilt aircraft **22** as discussed herein with reference to FIG. 6. In FIG. 8, lift mechanism **26** positions the respective fork assembly **248** in lowered position **232**. Each fork assembly **248** comprises a pair of prongs **250** that at least partially define the respective support region **20**, and a respective prong gate **252**. In FIG. 8, both prong gates **252** are in prong gate closed configuration **254** and extend between and interconnect the distal ends of the respective prongs **250**.

(147) Front undercarriage support assembly **206** and rear undercarriage support assembly **208** each further include a respective ground support chassis **236**, which includes pair of projections **238** defining open region **240** therebetween. Both fork assemblies **248** are positioned laterally within the respective open regions **240**, such that projections **238** may enhance load distribution of the weight of the aircraft from fork assemblies **248** across ground surface **16**. Each projection **238** is supported on ground surface **16** by a portion of drive assembly **212** and/or translation assembly **204**, such as one or more driven wheels **60** and/or non-driven wheels **62**. Front undercarriage support assembly **206** further comprises gate **242** that is shown in FIG. 8 in closed configuration **244**, in which gate **242** is operably coupled to, and interconnects the distal ends of projections **238**. Similarly, rear undercarriage support assembly **208** further comprises gate **242** that is shown in FIG. 8 in closed configuration **244**.

(148) As mentioned, distal undercarriage support assembly **210** is operably coupled to rear undercarriage support assembly **208** via connecting arms **214**. In particular, each connecting arm **214** is coupled to a projection **238** of rear undercarriage support assembly **208** via a respective connecting arm actuator **280** such that connecting arms **214** extend from projections **238** laterally spaced apart from one another. Distal undercarriage support assembly **210** comprises first portion **222** and second portion **224** that are operably coupled to rear undercarriage support assembly **208** by separate connecting arms **214**. Distal undercarriage support assembly **210** further comprises distal coupling mechanism **226**, which in FIG. 8, operably interconnects first portion **222** and second portion **224** with one another. Stated differently, in FIG. 8, distal coupling mechanism **226** supports first portion **222** and second portion **224** in supporting configuration **228**.

(149) Also shown in FIG. 8, in some examples, each connecting arm **214** comprises two articulated portions **276** that are interconnected with one another via a connecting arm articulated connection **278**. In some examples, connecting arm articulated connections **278** are configured to permit articulated portions **276** to fold relative to one another to permit distal undercarriage support assembly **210** to move towards rear undercarriage support assembly **208**. In some examples, distal undercarriage support assembly **210** is moved from the position shown in FIG. 8 towards rear undercarriage support assembly **208**, and optionally at least partially within open region **240** to transition aircraft handler **350** from the deployed configuration shown in FIG. 8 to the stowed configuration discussed herein.

(150) FIG. 9 illustrates various components of aircraft handler **350** transitioned from the configurations shown in FIG. 8. More specifically, in FIG. 9, prong gates **252** are transitioned to the prong gate open configuration **256**, in which prong gate **252** is pivoted outwardly and is operably coupled to only one prong **250**. In prong gate open configuration **256**, prong gate **252** provides access to the lateral spacing between prongs **250**, such that each fork assembly **248** may be translated beneath the respective portion of the undercarriage of the aircraft. Similarly, each gate **242** is transitioned from the closed configuration of FIG. 8 to open configuration **246**. In open configuration **246**, gates **242** provide access to open regions **240**, such that front undercarriage

support assembly **206** and rear undercarriage support assembly **208** are configured to translate laterally beneath the respective portions of the undercarriage of the aircraft.

(151) Likewise, distal undercarriage support assembly **210** is transitioned from the supporting configuration of FIG. **8** to positioning configuration **230**. In positioning configuration **230**, distal coupling mechanism **226** disengages first portion **222** and second portion **224** from one another and connecting arm actuators **280** have pivoted connecting arms **214** together with the respective portions away from one another. In this way, distal undercarriage support assembly **210** is configured to provide access to open region **240** of rear undercarriage support assembly **208** and may be translated laterally beneath the first aft portion and the second aft portion of the undercarriage of the aircraft during operable positioning of aircraft handler **350**.

(152) In some examples, distal undercarriage support assembly **210** is configured to lift the second aft portion of the undercarriage of the aircraft from direct contact with ground surface **16** by transitioning from positioning configuration **230** of FIG. **9** to supporting configuration **228** of FIG. **8**. More specifically, in some examples, connecting arm actuators **280** are configured to pivot first portion **222** and second portion **224** towards one another once distal undercarriage support assembly **210** is positioned beneath or adjacent the second aft portion of the undercarriage of the aircraft in positioning configuration **230**. In particular, connecting arm actuators **280** may move first portion **222** and second portion **224** to be in contact with opposing sides of the ground-contacting region of the second aft portion, such as either side of the second rear wheel assembly in the examples of FIG. **6**. Once connecting arm actuators **280** bring first portion **222** and second portion **224** in proximity to one another, distal coupling mechanism **226** engages first portion **222** with second portion **224** with one another and pulls first portion **222** towards second portion **224** to reduce the lateral spacing therebetween. In this way, distal coupling mechanism **226** urges first portion **222** with second portion **224** beneath the ground-contacting region of the second aft portion of the undercarriage of the aircraft such that first portion **222**, with second portion **224**, lift the second aft portion of the undercarriage of the aircraft from contact with ground surface **16**. As shown in FIGS. **8** and **9**, in some examples, first portion **222** and second portion **224** each include a transition portion **86**, such as ramp, that is configured to facilitate lifting of the second aft portion of the undercarriage of the aircraft from contact with ground surface **16**.

(153) FIG. **10** illustrates an example of aircraft handler **350** configured to support aircraft **22** among tilted aircraft orientations **106**. As shown, each lift mechanism **26** comprises a pillar **97** along which lift mechanism **26** moves the respective fork assembly **248** relative to ground surface **16**. In FIG. **10**, lift mechanisms **26** have transitioned fork assemblies **248** from the lowered positions **232** of FIG. **8**, and support fork assemblies **248** spaced above ground surface **16** in raised positions **234**. In particular, fork assembly **248** of front undercarriage support assembly **206** and fork assembly **248** of rear undercarriage support assembly **208** each are positioned further above ground surface **16** than support region **20** of distal undercarriage support assembly **210**. Thus, when aircraft handler **350** supports the undercarriage **108** of aircraft **22** in this configuration, front undercarriage support assembly **206** and rear undercarriage support assembly **208** support the respective portions of the undercarriage **108** spaced further above ground surface **16** than does distal undercarriage support assembly **210** such that aircraft handler **350** tilts aircraft **22**. In some examples, lift mechanism **26** of rear undercarriage support assembly **208** is configured to support the respective fork assembly **248** spaced further above ground surface **16** than does lift mechanism **26** of front undercarriage support assembly **206**. In such examples, aircraft handler **350** is configured to support the second aft portion of undercarriage **108** aircraft **22** spaced further above ground surface **16** than the forward portion of undercarriage **108** when aircraft handler **350** tilts aircraft **22** such as shown in FIG. **6**.

(154) FIGS. **11-16** provide an illustrative, non-exclusive example of aircraft handlers **400** that is indicated at and referred to herein as aircraft handler **450**. In particular, FIGS. **11-12** illustrate examples of aircraft handler **450** supporting an aircraft **22**, and FIGS. **13-16** illustrate various

configurations of example aircraft handler **450**. Where appropriate, the reference numerals from the schematic illustrations of FIGS. **1** and **4-5** are used to designate corresponding parts of the example aircraft handler **450** of FIGS. **11-16**; however, aircraft handler **450** is non-exclusive and does not limit aircraft handlers **400** to the illustrated embodiments of FIGS. **11-16**. That is, aircraft handlers **400** are not limited to the specific embodiment of FIGS. **11-16**, and aircraft handlers **400** may incorporate any number of the various aspects, configurations, characteristics, properties, variants, options etc. of aircraft handlers **400** that are illustrated in and discussed with reference to the schematic representations of FIGS. **1** and **4-5** and/or the embodiment of FIGS. **11-16**, as well as variations thereof, without requiring the inclusion of all such aspects, configurations, characteristics, properties, variants, options etc. Furthermore, any additional aspects, configurations, characteristics, properties, variants, options, etc. disclosed in connection with the example aircraft handler **450** of FIGS. **11-16** may be utilized with and/or otherwise included in other aircraft handlers **400**, including aircraft handlers **400** according to FIGS. **4-5**. For the purpose of brevity, each previously discussed component, part, portion, aspect, region, etc. or variants thereof may not be discussed, illustrated, and/or labeled again with respect to the examples of FIGS. **11-16**; however, it is within the scope of the present disclosure that the previously discussed features, variants, etc. may be utilized with the examples of FIGS. **11-16**.

(155) With initial reference to FIG. **11**, aircraft handler **450** is illustrated as supporting aircraft **22** in level aircraft orientation **104** and among tilted aircraft orientations **106**. Similarly to FIG. **6**, aircraft **22**, envelope **120** defined by aircraft **22**, and support arms **18** of aircraft handler **450** are illustrated in dashed lines when aircraft handler **450** supports aircraft **22** in level aircraft orientation **104**. Support arms **18** of aircraft handler **450**, aircraft **22**, and envelope **120** defined by aircraft **22** are illustrated in solid lines when aircraft handler **450** supports aircraft **22** among tilted aircraft orientations **106**.

(156) In the specific example shown in FIG. **11**, aircraft handler **450** comprises first support arm **36** and second support arm **38** that are supporting discrete regions of undercarriage **108** of aircraft **22** spaced above or spaced apart from ground surface **16**. In particular, first support arm **36** comprises support region **20** that supports the front wheel assembly of undercarriage **108** and second support arm **38** comprises a pair of support regions **20** that support the rear wheel assembly of undercarriage **108**. As such, in this example, aircraft handler **450** supports or suspends the entirety of undercarriage **108** of aircraft **22** spaced above or spaced apart from ground surface **16**. Aircraft handler **450** further comprises first drive assembly **14**, second drive assembly **24**, third drive assembly **40**, and fourth drive assembly **44** that support aircraft handler **450** and aircraft **22** on ground surface **16**, such as discussed herein.

(157) Envelopes **120** illustrated in FIG. **11** may be defined similarly, or at least substantially similarly, to the envelopes **120** illustrated and discussed herein with reference to FIG. **6**. As shown in FIG. **11**, aircraft handler **450** is configured to fit completely within envelope **120** of aircraft **22** when aircraft handler **450** supports aircraft **22** in level aircraft orientation **104** and/or among tilted aircraft orientations **106**. Stated differently, in some examples, aircraft handler **10** is configured to support aircraft **22** without extending beyond, or expanding, the four sidewalls of the rectangular prism of envelope **120** and/or extending above the upper side or surface of the rectangular prism of envelope **120**. In this way, aircraft handler **450** may support aircraft **22** without increasing the width or length of envelope **120**.

(158) Similarly to that discussed herein with reference to FIG. **6**, aircraft handler **450** may be described as being configured to selectively alter the dimensions of envelope **120** by selectively tilting aircraft **22** between level aircraft orientation **104** and tilted aircraft orientations **106**. More specifically, as shown in the example of FIG. **11**, when aircraft handler **450** supports aircraft **22** in level aircraft orientation **104**, or when aircraft **22** is supported on ground surface **16**, wings **124** of aircraft **22** extend generally horizontally or generally parallel to ground surface **16**. When aircraft handler **450** tilts aircraft **22** or supports aircraft **22** among tilted aircraft orientations **106**, wings **124**

of aircraft **22** are tilted relative to ground surface **16**. In this way, aircraft handler **10** orients wings **124** to extend upwardly or partially traverse relative to ground surface **16** or level aircraft orientation **104**, such that the projection of wings **124** onto ground surface **16** is smaller when aircraft handler **450** tilts aircraft **22** among tilted aircraft orientations **106**. Consequently, as shown in the example of FIG. **11**, envelope **120**, and/or the width thereof, of aircraft **22** in tilted aircraft orientations **106** is smaller or narrower than envelope **120**, and/or the width thereof, of aircraft **22** in level aircraft orientation **104**. As more specific examples, aircraft handler **450** may be configured to tilt aircraft **22** to among tilted aircraft orientations **106** to reduce the width of envelope **120** of aircraft **22** by at least 0.5%, at least 1%, at least 2%, at least 3%, at most 6%, and/or at most 10% the width of envelope **120** of aircraft **22** in level aircraft orientation **104** or supported directly on ground surface **16**.

(159) As discussed herein with reference to FIG. **6**, in some examples, aircraft handler **450** is configured to tilt aircraft **22** to fit aircraft **22** within a defined volume **130**, an external volume **130**, or a confined volume **130**, that is smaller or narrower than envelope **120** of aircraft **22** in level aircraft orientation **104** and/or envelope **120** of aircraft **22** supported directly on ground surface **16**. Stated differently, aircraft handler **450** may be configured to fit aircraft **22** to within defined volume **130** that aircraft **22**, or envelope **120** of aircraft **22**, cannot fit within when aircraft **22** is oriented in level aircraft orientation **104** and/or when aircraft **22** is supported on ground surface **16**. In some examples, aircraft handler **10** is configured to move aircraft **22** while supporting aircraft **22** among tilted aircraft orientations **106**. As such, in some examples, aircraft handler **10** is configured to move aircraft **22** to within, and from within, defined volume **130** that is smaller than envelope **120** of aircraft **22** in level orientation and/or envelope **120** of aircraft **22** supported on ground surface **16**.

(160) In some examples, when aircraft handler **450** tilts or supports aircraft **22** among tilted aircraft orientations **106**, an aircraft tilt angle is formed between a line extending between the wing tips of, or the maximal extents of, wings **124** and the ground surface **16**. Stated another way, in some examples, aircraft handler **10** is configured to tilt aircraft **22** to among a plurality of aircraft tilt angles relative to level aircraft orientation **104**, in which the plurality of aircraft tilt angles may be from -5° to 25° . In some examples, the aircraft tilt angle corresponds to the tilt angle of support arms **18**.

(161) While FIG. **11** illustrates an example in which aircraft **22** is a fixed wing aircraft, aircraft handler **450** also may be configured to support, tilt, and alter the dimensions of the envelope of other types of aircraft such as discussed herein with reference to FIG. **6**.

(162) Turning now to FIG. **12**, illustrated therein is an example of aircraft handler **450** rotating aircraft **22** along ground surface **16**. As shown, aircraft handler **450** supports undercarriage **108** of aircraft **22** spaced above ground surface **16** while being positioned completely within the envelope of aircraft **22**. In some examples, aircraft handler **450** is configured to rotate aircraft **22** about yaw axis **132** of aircraft **22** without moving yaw axis **132** of aircraft **22** along ground surface **16**. Stated another way, in some examples, aircraft handler **450** is configured to rotate aircraft **22** within a cylindrical volume **134**, in which the radius **136** of the cylindrical volume **134** is a threshold fraction of a distance between yaw axis **132** and a maximum extent **122** of aircraft **22** from yaw axis **132**. As examples, radius **136** may be at least 100%, at most 110%, at most 120%, and/or at most 130% of the distance between yaw axis **132** and the maximum extent **122** of aircraft **22** from yaw axis **132**.

(163) In some examples, aircraft handler **450** also is configured to rotate aircraft **22** while aircraft handler **450** remains completely within cylindrical volume **134**. In view of the above, aircraft handler **450** may be configured to rotate aircraft **22** with aircraft handler **450** and aircraft **22** requiring a minimal volume within which to rotate, which may be beneficial to relocating aircraft **22** within crowded or confined spaces. Additionally, while FIG. **12** illustrates examples in which aircraft handler **450** is configured to rotate aircraft **22** in level aircraft orientation **104**, aircraft

handler **450** also may be configured to rotate aircraft **22** while aircraft handler **450** supports aircraft among tilted aircraft orientations **106**.

(164) Aircraft handler **450** may be configured to rotate aircraft **22** in any suitable direction of rotation and/or through any suitable angle, such as that discussed herein with reference to aircraft handler **350**. In some examples, first drive assembly **14**, second drive assembly **24**, third drive assembly **40**, and/or fourth drive assembly **44** are utilized to rotate aircraft **22**. In some such examples, one or more driven wheels **60** of first drive assembly **14**, second drive assembly **24**, third drive assembly **40**, and/or fourth drive assembly **44** are configured to selectively pivot to a particular pivot angle and provide motive force along the pivot angle to rotate aircraft handler **450** and aircraft **22** about ground surface **16**.

(165) Now with reference to FIG. **13**, aircraft handler **450** is illustrated in the deployed configuration with support arms **18** in deployed orientation **32**. As shown, aircraft handler **450** comprises first support arm **36** and first lift mechanism **94**, in which first lift mechanism **94** operably couples first support arm **36** to first body member **50** of front undercarriage support assembly **206** and orients first support arm **36** in level orientation **100**. Aircraft handler **450** also comprises second support arm **38** and second lift mechanism **96**, in which second lift mechanism **96** operably couples second support arm **38** to second body member **52** of rear undercarriage support assembly **208** and supports second support arm **38** in level orientation **100**. Aircraft handler **450** comprises frame **54** that extends between first body member **50** of front undercarriage support assembly **206** and second body member **52** of rear undercarriage support assembly **208**. Frame **54** supports first body member **50** and second body member **52** spaced apart along ground surface **16**, such that first support arm **36** and second support arm **38** are spaced apart along ground surface **16**. As shown, frame **54** includes telescoping mechanism **58** that is configured to telescope to adjust the spacing between first support arm **36** and second support arm **38**.

(166) With continued reference to FIG. **13**, first support arm **36** includes a single support region **20** while second support arm **38** includes a pair of support regions **20** that are spaced apart along the length of second support arm **38**. Support regions **20** form recessed platforms along support arms **18** and are positioned along support arms **18** to operably support discrete regions of the undercarriage of the aircraft. In particular, second support arm **38** includes a first support region **20** positioned proximate to, or that abuts, proximal end **28** and a second support region **20** that is positioned proximate to, or that abuts, distal end **30**. Chassis **64** of first distal undercarriage support assembly **209** overlaps with distal ends **30** of first support arm **36**, and chassis **64** of second distal undercarriage support assembly **211** overlaps with distal ends **30** of second support arm **38**. In this way, the second support region **20** of second support arm **38** overlaps with chassis **64** of third drive assembly **40**.

(167) Aircraft handler **450** further comprises transition portions **86** that are positioned proximate, or coupled to, each support region **20**, in which each transition portion **86** includes a ramp, or a wedge, that is inclined upwardly from ground surface **16** towards, or to, support region **20**. In particular, the transition portion **86** leading to the second support region **20** of second support arm **38** may be operably coupled to chassis **64** of second distal undercarriage support assembly **211**. Each other transition portion **86** is operably coupled to a support region **20**. As shown, transition portions **86** are positioned on the same side of first support arm **36** as second support arm **38**, such that transition portions **86** may facilitate lifting of the undercarriage of the aircraft onto first support arm **36** and second support arm **38** when first support arm **36** and second support arm **38** translate in the same direction laterally beneath the aircraft.

(168) With continued reference to FIG. **13**, first drive assembly **14**, second drive assembly **24**, third drive assembly **40**, and fourth drive assembly **44** each include a pair of driven wheels **60** and a pair of non-driven wheels **62** that are operably coupled to chassis **64**. In particular, non-driven wheels **62** are positioned inside of driven wheels **60**, such that driven wheels **60** define the outside-most ground-contacting points of aircraft handler **450**.

(169) Turning to FIGS. **14-16**, shown therein are side profile views of aircraft handler **450** showing examples in which support arms **18** have a deployed orientation **32** and a stowed orientation **34**. In particular, FIGS. **14-15** illustrate aircraft handler **450** with support arms **18** in deployed orientation **32**, while FIG. **16** illustrates aircraft handler **450** with support arms **18** in stowed orientation **34**. FIGS. **14-16** are side profile views taken, or viewed, perpendicular to the length of support arms **18**, such that only a single support arm **18**, such as first support arm **36**, is shown.

(170) With initial reference to FIG. **14**, aircraft handler **450** is illustrated with support arms **18** in level orientation **100**. As shown, proximal end **28** of support arm **18** is operably coupled to body **12** of front undercarriage support assembly **206** via lift mechanism **26**, and distal end **30** of support arm **18** is operably coupled to chassis **64** of distal undercarriage support assembly **210**. Support arm **18** is oriented in level orientation **100** and extends parallel, or substantially parallel, to ground surface **16**, with proximal end **28** and distal end **30** being level, or substantially level, with one another. Support arm **18** comprises a single support region **20** that defines a recessed region, or a depression, within the upper surface of support arm **18**, such that the upper surface of support region **20** is positioned closer to ground surface **16**. Support arm **18** also comprises first member **80**, second member **82**, and hinge mechanism **84** that hingedly couples first member **80** and second member **82** to one another. As shown, first member **80** extends beyond hinge mechanism **84** towards distal end **30** to overlap with at least a portion of second member **82**, such that a lower surface of first member **80** operably contacts at least a portion of an upper surface of second member **82** when support arm **18** is in deployed orientation **32** and/or level orientation **100**. In this way, first member **80** and second member **82** are prevented from pivoting downwardly, or towards ground surface **16**, about hinge mechanism **84**. In other words, first member **80** and second member **82** may remain collinear, fixedly positioned relative to one another, and/or are non-pivotal about hinge mechanism **84** while support arm **18** supports at least a portion of the undercarriage of the aircraft. As further shown in FIG. **14**, support region **20** is positioned along support arm **18** to be overlapping with first member **80** and second member **82**. In other words, in some examples, first member **80** and second member **82** both comprise portions of support region **20**.

(171) FIG. **15** is another side view of aircraft handler **450** with support arms **18** oriented among tilted orientations **102**. As shown, lift mechanism **26** supports proximal end **28** of support arm **18** higher above, or spaced further from, ground surface **16** relative to distal end **30** than that shown in FIG. **14**. In this way, lift mechanism **26** tilts support arm **18** relative to ground surface **16** and/or orients support arm **18** with a particular tilt angle relative to ground surface **16**. In other words, lift mechanism **26** moves proximal end **28** of support arm **18** further above ground surface **16** to transition support arm **18** from level orientation **100** shown in FIG. **14** to among tilted orientations **102** of FIG. **15**.

(172) In some examples, lift mechanism **26** comprises pillar **97** and lift mechanism **26** is configured to move proximal end **28** of support arm **18** up and down along pillar **97** to transition support arm **18** among level orientation **100** and tilted orientations **102**. In some examples, support arm **18** is pivotally coupled to lift mechanism **26** and distal undercarriage support assembly **210** to permit support arm **18** to be pivoted by lift mechanism **26**. As shown in FIG. **15**, in some examples, lift mechanism **26** comprises a lever arm **88** that is operably coupled to pillar **97** and proximal end **28** of support arm **18**, and lever arm **88** is configured to selectively move up and down pillar **97** to pivot support arm **18**.

(173) As further shown in FIG. **15**, first member **80** and second member **82** of support arm **18** engage with one another when support arm **18** is oriented among tilted orientations **102**, such that first member **80** and second member **82** are collinear, and/or do not pivot downwardly about hinge mechanism **84**, such as while supporting the undercarriage of the aircraft.

(174) FIG. **16** illustrates an example of aircraft handler **450** in the stowed configuration with support arms **18** oriented in stowed orientation **34**. As shown, first member **80** and second member **82** are pivoted upwardly about hinge mechanism **84**, such that first member **80** extends

downwardly from hinge mechanism **84** to operably couple to lift mechanism **26**, such as lever arm **88** thereof, and second member **82** extends downwardly from hinge mechanism **84** to operably couple to second drive assembly **24** (or third drive assembly **40** in the case of second support arm **38**). In particular, first member **80** is pivotally coupled to lift mechanism **26** and second member **82** is pivotally coupled to second drive assembly **24** to permit first member **80** and second member **82** to pivot about hinge mechanism **84** and when support arm **18** transitions between deployed orientation **32** and stowed orientation **34**. Further shown, lift mechanism **26** may support proximal end **28** higher above ground surface **16** than distal end **30** when support arm **18** is in stowed orientation **34**.

(175) With continued reference to FIG. **16**, distal end **30** of support arm **18** or first drive assembly **14** are positioned closer to body **12** of front undercarriage support assembly **206** or proximal end **28** of support arm **18** when support arm **18** is in stowed orientation **34** relative to when support arm **18** is in deployed orientation **32**. In this way, aircraft handler **450** occupies a smaller floor space, or possesses a smaller footprint, in the stowed configuration of FIG. **16** relative to the deployed configurations of FIGS. **6** and **7**. Thus, the stowed configuration of aircraft handler **450** and/or stowed orientation **34** of support arms **18** may permit aircraft handler **450** to be stored within a smaller stowage area. In particular, the area of the footprint of aircraft handler **450** when support arm(s) **18** are in stowed orientation **34** may be at most 30%, at most 40%, at most 50%, at most 55%, and/or at most 60% the area of the footprint of aircraft handler **450** when support arm(s) **18** are in deployed orientation **32**.

(176) In some examples, first drive assembly **14** and/or second drive assembly **24** (or third drive assembly **40** and/or fourth drive assembly **44** in the case of second support arm **38**) are configured to move relative to one another to transition support arm **18** between deployed orientation **32** and stowed orientation **34**. In some such examples, second drive assembly **24** (or third drive assembly **40** in the case of second support arm **38**) is configured to move towards and away from body **12** of front undercarriage support assembly **206** (or body **12** of rear undercarriage support assembly **208** in the case of second support arm **38**) to transition support arm **18** between deployed orientation **32** and stowed orientation **34**, and/or to selectively pivot first member **80** and second member **82** about hinge mechanism **84**.

(177) FIG. **17** is a flowchart that schematically represents illustrative, non-exclusive examples of methods **500**, according to the present disclosure. In FIG. **17**, some steps are illustrated in dashed boxes indicating that such steps may be optional or may correspond to an optional version of a method according to the present disclosure. That said, not all methods according to the present disclosure are required to include the steps illustrated in solid boxes. The methods and steps illustrated in FIG. **9** are not limiting and other methods and steps are within the scope of the present disclosure, including methods having greater than or fewer than the number of steps illustrated, as understood from the discussions herein. Furthermore, methods **500** are not limited to the order of steps illustrated in FIG. **17**, and each step of methods **500** may be performed with any suitable sequence or timing relative to any other step of methods **500**, including prior to, subsequent to, simultaneously with, and/or substantially simultaneously with any other step of methods **500**. Each step or portion of methods **500** may be performed with utilizing aircraft handlers **10**, portions thereof, and/or the example aircraft handlers thereof including example aircraft handlers **300**, example aircraft handler **350**, example aircraft handlers **400**, and/or example aircraft handler **450** discussed in detail herein with reference to FIGS. **1-16**. Likewise, any of the features, functions, and/or structures of aircraft handlers **10** discussed herein with reference to FIG. **17** may be included in and/or utilized with aircraft handlers **10** of FIGS. **1-16** without departing from the scope of the present disclosure.

(178) In some examples, controller **90**, as discussed above, is programmed to perform or otherwise facilitate or implement one or more of the methods and/or steps illustrated in FIG. **17** and/or discussed herein. In some examples, instructions for performing the various steps and/or methods

described herein are stored as computer-readable instructions on the non-transitory memory of controller **90**. The processing unit of controller **90** is configured to execute the computer-readable instructions to perform the one or more steps and methods illustrated in FIG. **17**. In other examples, instructions for performing the various steps and/or methods described herein are stored remotely from aircraft handler **10**, such as in the example of an aircraft handler **10** that is configured to be remotely controlled. In other examples, one or more steps or portions of methods **500** are executed responsive to control by an operator or one or more people that may control aircraft handler **10** remotely.

(179) As seen in FIG. **17**, methods **500** include supporting an undercarriage **108** of an aircraft **22** with an aircraft handler **10** at **502** and moving the aircraft **22** along a ground surface **16** with the aircraft handler **10** at **504**. In some examples, the moving the aircraft at **504** includes rotating the aircraft with the aircraft handler **10** at **506**. Methods **500** also may include tilting the aircraft with the aircraft handler **10** at **508**, stowing the aircraft **22** with the aircraft handler **10** at **510**, and/or transitioning the aircraft handler **10** between a deployed orientation and a stowed orientation at **512**.

(180) The supporting the undercarriage **108** of the aircraft **22** with the aircraft handler **10** at **502** includes supporting any suitable portion of the undercarriage **108** of the aircraft **22** with the aircraft handler **10**. In some examples, the supporting at **502** comprises supporting or suspending at least a portion of, some of, most of, or the entirety of the undercarriage **108** of the aircraft **22** spaced above, or apart from, contact with the ground surface **16**. When the supporting at **502** comprises supporting the entirety of the undercarriage **108** of the aircraft **22**, the supporting at **502** may include supporting the entire weight of the aircraft **22** with the aircraft handler **10**.

(181) The supporting at **502** comprises supporting the undercarriage of the aircraft with a plurality of support members **202** of aircraft handler **10**, as discussed herein. In some examples, the supporting at **502** comprises supporting the undercarriage of the aircraft with the support member **202** of a front undercarriage support assembly **206**, the support member **202** of a rear undercarriage support assembly **208**, and/or the support member of at least one distal undercarriage support assembly **210**, as discussed herein. In more specific examples, the supporting at **502** comprises supporting a forward portion of the undercarriage of the aircraft, such as a front wheel assembly, with the support member **202** of the front undercarriage support assembly **206**, optionally together with the support member **202** of a first distal undercarriage support assembly **209**. Additionally or alternatively, the supporting at **502** comprises supporting a first aft portion of the undercarriage **108** of the aircraft, such as a first rear wheel assembly, with the support member **202** of the rear undercarriage support assembly **208**, and/or supporting a second aft portion of the undercarriage **108** of the aircraft, such as a second rear wheel assembly, with the distal undercarriage support assembly **210**, which in some examples, is a second distal undercarriage support assembly **211**.

(182) For some examples in which the supporting at **502** is performed with aircraft handlers **300**, the supporting at **502** comprises supporting the forward portion of the undercarriage of the aircraft with a fork assembly **248** of the front undercarriage support assembly **206**, supporting the first aft portion of the undercarriage of the aircraft **22** with the fork assembly **248** of the rear undercarriage support assembly **208**, and supporting the second aft portion of the undercarriage of the aircraft with a support region **20** of the distal undercarriage support assembly **210**. In some such examples, the supporting is performed with prong gate(s) **252** in a prong gate closed configuration **254**, gates **242** in a closed configuration **244**, and/or with the distal undercarriage support assembly **210** in a supporting configuration **228**, as discussed herein.

(183) For some examples in which the supporting at **502** is performed with aircraft handlers **400**, the supporting at **502** comprises supporting at least a portion of the undercarriage of the aircraft with at least one support arm **18** of the aircraft handler **400**. As discussed herein, the at least one support arm **18** comprises one or more support regions **20**, each of which being configured to support at least a portion of the undercarriage of the aircraft. In some examples, the supporting at

502 comprises supporting one or more portions of the undercarriage **108** of the aircraft **22** with the one or more support regions **20**. In some examples, aircraft handlers **400** comprises a plurality of support arms **18**, and the supporting at **502** comprises supporting the undercarriage **108** of the aircraft at **502** with the plurality of support arms **18**. In some such examples, aircraft handlers **400** comprises a first support arm **36** and a second support arm **38**, each being configured to support a discrete region of the undercarriage of the aircraft, and the supporting at **502** comprises supporting at least two discrete regions of the undercarriage **108** of the aircraft with the first support arm **36** and the second support arm **38**.

(184) The supporting the aircraft at **502** comprises supporting the aircraft in any suitable orientation. In some examples, the supporting at **502** comprises supporting the aircraft in a level aircraft orientation **104**. Additionally or alternatively, the supporting the aircraft at **502** comprises supporting the aircraft among, and/or in any desirable tilted aircraft orientation **106**.

(185) The supporting the aircraft at **502** is performed with any suitable sequence or timing within methods **500**, such as subsequent to or substantially simultaneously with lifting at **503**. Additionally or alternatively, in some examples, the supporting at **502** is performed prior to, substantially simultaneously with, and/or subsequent to moving at **504** and/or tilting at **508**. As more examples, the supporting at **502** may be performed prior to, or substantially simultaneously with, the stowing at **510**. The supporting at **502** also may be performed prior to, or subsequent to, transitioning at **512**.

(186) In some examples, methods **500** comprise lifting the undercarriage of the aircraft from direct contact with the ground surface **16** at **503**. For some examples in which the lifting at **503** is performed with aircraft handlers **300**, the lifting at **503** comprises operably positioning each undercarriage support assembly **200** under the respective portion of the undercarriage of the aircraft such as with gates **242** in an open configuration **246**, prong gates **252** in a prong gate open configuration **256**, the fork assemblies **248** positioned in a translation position, and/or distal undercarriage support assembly **210** in a positioning configuration **230**, as discussed herein. In some examples, the lifting at **503** subsequently includes transitioning the fork assemblies **248** to a lowered position **232** and/or a raised position **234** with the lift mechanisms **26**, transitioning the gates **242** to the closed configuration **244**, transitioning prong gates **252** to the prong gate closed configuration **254**, and/or transitioning the distal undercarriage support assembly **210** to the supporting configuration **228**, as discussed herein.

(187) For some examples in which the lifting at **503** is performed with aircraft handlers **400**, the lifting at **503** comprises translating each support arm **18** laterally beneath the respective portion of the undercarriage of the aircraft with each support arm **18** positioned in a level orientation **100**, as discussed herein. In some such examples, the lifting at **503** comprises urging each respective portion of the undercarriage of the aircraft from the ground surface **16** onto a respective support region **20** utilizing transition portions **86**, as discussed herein. In some such examples, the lifting at **503** comprises utilizing one or more drive assemblies **212** of the aircraft handler **400** to facilitate lifting or transitioning the undercarriage of the aircraft from the ground surface **16** onto the support regions **20**. In particular, in some examples, the lifting at **503** comprises moving, with the one or more drive assemblies **212**, the one or more support regions **20** and/or the respective transition portions **86** laterally beneath the undercarriage of the aircraft and urging, such as utilizing the transition portions **86**, the undercarriage from contact with the ground surface **16** and onto the support regions **20**.

(188) The lifting at **503** is performed with any suitable sequence or timing within methods **500**, such as prior to or at least substantially simultaneously with the supporting at **502** and/or prior to any other step of methods **500**.

(189) With continued reference to FIG. **17**, methods **500** comprise moving the aircraft along the ground surface with the aircraft handler at **504**. In some examples, the moving at **504** comprises moving the aircraft while each support member **202** of the aircraft handler **10** supports the

respective portion of the undercarriage **108** of the aircraft **22**.

(190) For some examples in which the moving at **504** is performed with aircraft handlers **300**, the moving is performed while fork assembly **248** of front undercarriage support assembly **206**, fork assembly **248** of rear undercarriage support assembly **208**, and the support region **20** of the distal undercarriage support assembly **210**, each supporting the respective portion of the undercarriage of the aircraft, are spaced apart from the ground surface **16**. For some examples in which the moving at **504** is performed with aircraft handlers **400**, the moving is performed while the one or more support arms **18** of the aircraft handler **400** each support at least a portion of the undercarriage of the aircraft spaced apart from the ground surface **16**.

(191) The moving at **504** comprises moving the aircraft along any suitable or desirable ground surface, such as a level ground surface, a sloped ground surface, an even ground surface, and/or an uneven ground surface. The moving at **504** also may comprise moving the aircraft along a ground that is included in any suitable one or more environments, such as a runway, an aircraft hangar, a nautical ship, an aircraft carrier, an aircraft parking area, a shipping container, and/or an elevator. In some examples, the moving at **504** is performed to relocate the aircraft from an initial location to a final location. With this in mind, in some examples, the moving at **504** comprises moving the aircraft along a plurality of ground surfaces and/or within a plurality of environments.

(192) In some examples, the moving at **504** comprises moving the aircraft to within and/or from within a defined volume, as discussed herein, such as during relocation of the aircraft. In some examples, the moving at **504** may comprise moving the aircraft to within and/or from within a defined volume defined by a shipping container, a parking space, and/or an elevator, as discussed herein.

(193) The moving at **504** comprises moving the aircraft while the aircraft handler supports the aircraft in any suitable orientation, such as in a level aircraft orientation **104** and/or among a plurality of tilted aircraft orientations. In some examples, the moving the aircraft at **504** comprises moving the aircraft to within and/or **106** from within a defined volume **130** that is smaller than an envelope **120** of the aircraft in the level orientation and/or a defined volume that comprises a width that is narrower than a width of the envelope of the aircraft in the level aircraft orientation. In some such examples, the moving at **504** is performed while the aircraft handler supports the aircraft tilted among the plurality of the aircraft tilted orientations, such that the envelope of the aircraft is smaller than the defined volume and/or the width of the envelope of the aircraft is narrower than the width of the defined volume, while the aircraft handler moves the aircraft to within, and/or from within the defined volume.

(194) In some examples, the moving at **504** comprises moving the aircraft with one or more drive assemblies **212** of the aircraft handler **10**. In some examples, the moving at **504** comprises utilizing the one or more driven wheels **60** included at least one drive assembly **212**, as discussed herein. For some examples in which the moving at **504** is performed with aircraft handlers **400**, the moving at **504** comprises moving the aircraft **22** with one or more of a first drive assembly **14**, a second drive assembly **24**, a third drive assembly **40**, and/or a fourth drive assembly **44** of the aircraft handler **400**.

(195) In some examples, the moving at **504** comprises moving the aircraft along a plurality of directions of travel. In some examples, the moving at **504** comprises moving the aircraft along one or more straight paths. Additionally or alternatively, the moving at **504** comprises moving the aircraft **22** along one or more curved paths, such as between the one or more straight paths. In some examples, the moving at **504** comprises steering the aircraft **22**, such as along the one or more straight paths and/or along the one or more curved paths. In some such examples, the moving the aircraft at **504** comprises selectively pivoting the one or more driven wheels **60** to steer the aircraft along the one or more straight paths and/or the one or more curved paths.

(196) The moving at **504** is performed with any suitable sequence or timing within methods **500**. In some examples, the moving at **504** is performed substantially simultaneously with the supporting at

502 and/or subsequent to the lifting at **503**. As more examples, the moving at **504** may be performed prior to tilting at **508**, stowing at **510**, and/or transitioning at **512**. Additionally or alternatively, in some examples, the moving at **504** is performed substantially simultaneously with or, subsequent to, tilting at **508**, such as when the moving at **504** comprises moving the aircraft to within and/or from within the defined volume.

(197) With continued reference to FIG. **17**, in some examples, the moving at **504** comprises rotating the aircraft with the aircraft handler **10** at **506**. In some examples, the rotating at **506** is performed to reorient the aircraft and/or the aircraft handler **10** between a first direction of movement and a second direction of movement. In some examples, the rotating at **506** is performed to orient the aircraft and/or the envelope **120** of the aircraft to be aligned with and/or such as to fit within the defined volume **130**. With this in mind, the rotating at **506** may be performed while the aircraft handler **10** supports the aircraft in the level aircraft orientation **104** and/or while the aircraft handler **10** supports the aircraft **22** among the tilted aircraft orientations **106**.

(198) In some examples, the rotating at **506** comprises rotating the aircraft **22** about a yaw axis **132** of the aircraft **22**, as discussed herein. In some such examples, the rotating at **506** comprises rotating the aircraft **22** within a cylindrical volume **134** having a radius **136** that is at most a threshold fraction of a distance between the yaw axis **132** and a maximum extent **122** of the aircraft **22** from the yaw axis **132**. In some such examples, the threshold fraction of the radius **136** of the cylindrical volume **134** to the distance between the yaw axis **132** to the maximum extent **122** is at most 105%, at most 110%, at most 120%, and/or at most 130%. In this way, the rotating at **506** may be performed within a minimum volume and permit the aircraft and aircraft handler to be rotated, reoriented, and/or turned without extending beyond the cylindrical volume **134**.

(199) The rotating at **506** may comprise rotating the aircraft in a clockwise direction of rotation and/or in a counter-clockwise direction of rotation. The rotating at **506** also may include rotating the aircraft through any suitable degree angle of rotation, such as through 30°, through 90°, through 180°, and/or through 360°, such as while aircraft **22** and/or aircraft handler **10** remains within cylindrical volume **134**.

(200) In some examples, the rotating at **506** comprises rotating the aircraft and the aircraft handler **10** with one or more of the drive assemblies **212** discussed herein with reference to the moving at **504**. In some such examples, the rotating at **506** comprises selectively pivoting one or more driven wheels **60** to a particular pivot angle and providing motive force with the one or more driven wheels **60** along the pivot angle to rotate the aircraft **22** along the ground surface **16**.

(201) When included, the rotating at **506** is performed with any suitable sequence or timing within the moving at **504**. In some examples, the rotating at **506** is performed prior to and/or subsequent to moving the aircraft to within and/or from within the defined volume. Additionally or alternatively, in some examples, the rotating at **506** is performed between moving the aircraft along a first direction of travel and a second direction of travel. As more examples, the rotating at **506** may be performed subsequent to, or substantially simultaneously with, the supporting at **502**, and/or tilting at **508**. Additionally or alternatively, in some examples, the rotating at **506** is performed subsequent to tilting at **508**, and/or prior to stowing at **510**.

(202) With continued reference to FIG. **17**, in some examples, methods **500** comprise tilting the aircraft relative to the ground surface with the aircraft handler at **508**. In some examples, the tilting at **508** comprises tilting the aircraft between the level aircraft orientation **104** and the plurality of tilted aircraft orientations **106**. More specifically, in some examples, the tilting at **508** comprises tilting the aircraft from the level aircraft orientation **104** to among the plurality of tilted aircraft orientations **106** and/or a particular tilted aircraft orientation **106**. Additionally or alternatively, the tilting at **508** comprises tilting the aircraft from among the plurality of tilted aircraft orientations **106**, or from a particular tilted aircraft orientation **106**, to the level aircraft orientation **104**. In some examples, the tilting at **508** comprises tilting the aircraft to and/or from a particular aircraft tilt angle relative to the ground surface **16** or the level aircraft orientation **104**, such as in the range of

-5° to 25°.

(203) In some examples, the tilting at **508** comprises altering the dimensions of the envelope **120** of the aircraft **22**, as discussed herein. More specifically, for some examples in which the tilting at **508** comprises tilting the aircraft from the level aircraft orientation **104** to among the tilted aircraft orientations **106**, the tilting at **508** comprises reducing the width of the envelope **120** of the aircraft **22**, as discussed herein.

(204) When included, the tilting at **508** is performed while each support member **202** of aircraft handler **10** supports the respective portion of the undercarriage **108** of aircraft **22**. In some examples, the tilting at **508** comprises selectively raising and lowering at least one support member **202** relative to the ground surface **16**. For some examples in which the tilting at **508** is performed with aircraft handlers **300**, the tilting at **508** comprises selectively raising and/or lowering the fork assembly **248** of the front undercarriage support assembly **206** with the lift mechanism **26** of the front undercarriage support assembly **206** and/or selectively raising and/or lowering the fork assembly **248** of the rear undercarriage support assembly **208** with the lift mechanism **26** of the rear undercarriage support assembly **208**, as discussed herein. In some such examples, the tilting at **508** comprises raising the fork assembly **248** of the rear undercarriage support assembly **208** spaced further above the ground **16** than the fork assembly **248** of the front undercarriage support assembly **206**, as discussed herein.

(205) For examples in which the tilting at **508** is performed with aircraft handlers **400**, the tilting at **508** is performed while at least one support arm **18** of the aircraft handler **400** supports at least a portion of the undercarriage **108** of the aircraft **22** spaced apart from the ground surface **16**. For examples in which the tilting at **508** is performed with aircraft handlers **400**, the tilting at **508** comprises pivoting the at least one support arm **18** with the lift mechanism **26** while the at least one support arm **18** supports at least a portion of the undercarriage of the aircraft. More specifically, in some examples, the tilting at **508** comprises raising the proximal end **28** of the at least one support arm **18** to tilt or pivot the at least one support arm **18** from the level orientation **100** to among the tilted orientations **102** and/or to tilt the aircraft from the level aircraft orientation **104** to among the tilted aircraft orientations **106**. Additionally or alternatively, in some examples, the tilting at **508** comprises lowering the proximal end **28** of the at least one support arm **18** to tilt or pivot the at least one support arm **18** from among the tilted orientations **102** to the level orientation **100** and/or to tilt the aircraft **22** from among the tilted aircraft orientations **106** to the level aircraft orientation **104**. For some examples in which aircraft handlers **400** comprise the first support arm **36** and the second support arm **38**, the tilting at **508** comprises tilting the first support arm **36** and the second support arm **38** with the respective lift mechanisms **26**.

(206) When included, the tilting at **508** is performed with any suitable sequence or timing within methods **500**. In some examples, the tilting at **508** is performed substantially simultaneously with or subsequent to the supporting at **502** and/or the moving at **504**. Additionally or alternatively, in some examples, the tilting at **508** is performed prior to the moving at **504**.

(207) As shown in FIG. **17**, in some examples, methods **500** include stowing the aircraft with the aircraft handler at **510**. The stowing at **510** may include stowing the aircraft in any suitable stowage location and/or within any suitable environment. In some examples, the stowing at **510** includes stowing the aircraft in a parking space, within a defined volume **130**, within an aircraft hangar, on a ship deck, and/or within a shipping container. In some examples, the stowing at **510** comprises transferring the aircraft from operable support by the plurality of support members **202** of the aircraft handler **10** to the ground surface **16**. In some examples, the stowing at **510** comprises moving the aircraft handler **10** laterally from beneath the aircraft and/or from supporting the undercarriage of the aircraft to transfer the aircraft to the ground surface **16**.

(208) In some examples, the stowing at **510** comprises transferring the aircraft from the plurality of support members **202** to the ground surface **16** at the stowage location, which may be described as parking the aircraft with the aircraft handler. In some examples, the stowing at **510** comprises

orienting, such as rotating, and/or moving the aircraft with the aircraft handler **10** to fit the aircraft **22** within the stowage location. In other examples, the stowing at **510** comprises transferring the aircraft **22** from the plurality of support members **202** at a location other than the stowage location, and subsequently taxiing the aircraft **22** to the stowage location.

(209) When included, the stowing at **510** is performed with any suitable sequence or timing within methods **500**. As examples, the stowing at **510** may be performed substantially simultaneously with, or subsequent to the moving at **504**, the rotating at **506**, and/or the tilting at **508**. Additionally or alternatively, the stowing at **510** may be performed prior to transitioning at **512** and/or subsequent to the supporting at **502**.

(210) With continued reference to FIG. **17**, in some examples, methods **500** comprise transitioning the aircraft handler **10** between the deployed orientation and the stowed orientation at **512**. In some examples, the transitioning at **512** comprises transitioning the aircraft handler **10** from the deployed configuration to the stowed configuration, which is referred to herein as stowing the aircraft handler **10**. Additionally or alternatively, in some examples, the transitioning at **512** comprises transitioning the aircraft handler **10** from the stowed configuration to the deployed configuration, which is referred to herein as deploying the aircraft handler **10**. For some examples in which the transitioning at **512** comprises stowing the aircraft handler **10**, the transitioning at **512** comprises reducing the footprint of the aircraft handler **10**.

(211) For some examples in which the transiting at **512** is performed with aircraft handlers **400**, the transitioning at **512** comprises transitioning the one or more support arms **18** from the deployed orientation **32** to the stowed orientation **34**. Additionally or alternatively, in some examples, the transitioning at **512** comprises transitioning the one or more support arms **18** from the stowed orientation **34** to the deployed orientation **32**.

(212) For some examples in which the transitioning at **512** comprises stowing the aircraft handler **400**, the transitioning at **512** comprises reducing a length of the one or more support arms **18**, such as discussed herein. More specifically, in some such examples, the transitioning at **512** comprises moving the proximal end **28** and the distal end **30** of the one or more support arms **18**, closer to one another. In some such examples, the transitioning at **512** comprises moving the second drive assembly **24** towards the first drive assembly **14**, such as to bring the proximal end **28** and the distal end **30** of the first support arm **36** closer to one another and/or reduce the length of the first support arm **36**. Additionally or alternatively, in some such examples, the transitioning at **512** comprises moving the third drive assembly **40** towards the fourth drive assembly **44**, such as to move the proximal end **28** and the distal end **30** of the second support arm **38** closer to one another and/or reduce the length of the second support arm **38**.

(213) In some examples, the transitioning at **512** comprises pivoting a first member **80** and a second member **82** of each support arm **18** relative to one another about a hinge mechanism **84** to move the proximal end **28** and the distal end **30** of each support arm **18** relative to one another, as discussed herein. In particular, when the transitioning at **512** comprises transitioning the one or more support arms **18** from the deployed orientation **32** to the stowed orientation **34**, the transitioning at **512** may comprise pivoting the first member **80** and the second member **82** upwardly about the hinge mechanism **84**, such as to bring the proximal end **28** and the distal end **30** of the support arm **18** closer to one another.

(214) For some examples in which the transitioning at **512** is performed with aircraft handlers **300**, the transitioning at **512** comprises adjusting a spacing between the front undercarriage support assembly **206** and the rear undercarriage support assembly **208** along the ground surface **16**, and/or adjusting a spacing between the rear undercarriage support assembly **208** and the distal undercarriage support assembly **210**. More specifically, for some examples in which the transitioning at **512** comprises stowing the aircraft handler **300**, the transitioning at **512** comprises moving the body **12** of front undercarriage support assembly **206** closer to the body **12** of rear undercarriage support assembly **208**. In some such examples, the transitioning at **512** comprises

folding sections **270** of connecting member **216** about an articulated connection **272** to bring front undercarriage support assembly **206** and rear undercarriage support assembly **208** closer to one another, as discussed herein. Additionally or alternatively, for some examples in which the transitioning at **512** comprises stowing the aircraft handler **300**, the transitioning at **512** comprises moving the body **12** of distal undercarriage support assembly **210** closer to the body **12** of rear undercarriage support assembly **208**. In some such examples, the transitioning at **512** comprises folding articulated portions **276** of each connecting arm **214** relative to one another about the respective connecting arm articulated connection **278** to bring the body **12** of distal undercarriage support assembly **210** closer to the body **12** of rear undercarriage support assembly, as discussed herein.

(215) Illustrative, non-exclusive examples of inventive subject matter according to the present disclosure are described in the following enumerated paragraphs:

(216) A. An aircraft handler (**10**), comprising: a plurality of undercarriage support assemblies (**200**), each comprising: a body (**12**); a support member (**202**) operably coupled to and extending from the body (**12**), wherein the support member (**202**) comprises at least one support region (**20**) configured to engage and operably support at least a respective portion of an undercarriage (**108**) of an aircraft (**22**); and a translation assembly (**204**) operably coupled to the body (**12**) and configured to support a respective undercarriage support assembly (**200**) on a ground surface (**16**) and permit the respective undercarriage support assembly (**200**) to translate along the ground surface (**16**); and one or more drive assemblies (**212**) configured to move the aircraft handler (**10**) along the ground surface (**16**), wherein each drive assembly (**212**) comprises the translation assembly (**204**) of an undercarriage support assembly (**200**) of the plurality of undercarriage support assemblies (**200**).

(217) A1. The aircraft handler (**10**) of paragraph A, wherein the aircraft handler (**10**) is configured to support the aircraft (**22**) in a level aircraft orientation (**104**) and in a plurality of tilted aircraft orientations (**106**).

(218) A2. The aircraft handler (**10**) of any of paragraphs A-A1, wherein the aircraft handler (**10**) is configured to suspend the entire undercarriage (**108**) of the aircraft (**22**) spaced apart from the ground surface (**16**) when the support member (**202**) of each undercarriage support assembly (**200**) supports at least the respective portion of the undercarriage (**108**) of the aircraft (**22**).

(219) A3. The aircraft handler (**10**) of any of paragraphs A-A2, wherein the aircraft handler (**10**) is configured to rotate the aircraft (**22**) about a yaw axis (**132**) of the aircraft (**22**).

(220) A3.1. The aircraft handler (**10**) of any of paragraphs A-A3, wherein the aircraft handler (**10**) further is configured to rotate the aircraft (**22**) within a cylindrical volume (**134**), and wherein a radius (**136**) of the cylindrical volume (**134**) is 110% or less of a distance between a/the yaw axis (**132**) of the aircraft (**22**) and a maximum extent (**122**) of the aircraft (**22**), as measured radially from the yaw axis (**132**).

(221) A4. The aircraft handler (**10**) of any of paragraphs A-A3.1, wherein the support member (**202**) of each undercarriage support assembly (**200**) is configured to lift at least the respective portion of the undercarriage (**108**) of the aircraft (**22**) from direct contact with the ground surface (**16**).

(222) A5. The aircraft handler (**10**) of any of paragraphs A-A4, further comprising one or more lift mechanisms (**26**) each being included in a respective undercarriage support assembly (**200**) of the plurality of undercarriage support assemblies (**200**), wherein each lift mechanism (**26**) operably couples the support member (**202**) of the respective undercarriage support assembly (**200**) to the body (**12**) of the respective undercarriage support assembly (**200**), and wherein each lift mechanism (**26**) is configured to selectively raise and lower a respective support member (**202**) relative to the ground surface (**16**).

(223) A5.1 The aircraft handler (**10**) of paragraph A5, wherein the aircraft handler (**10**) is configured to tilt the aircraft (**22**) from a/the level aircraft orientation (**104**) to among a/the plurality of tilted aircraft orientations (**106**) when each lift mechanism (**26**) selectively raises the respective

support member (202) relative to the ground surface (16).

(224) A6. The aircraft handler (10) of any of paragraphs A-A5.1, wherein the one or more drive assemblies (212) are configured to move the aircraft handler (10) together with the aircraft (22) along the ground surface (16) when each support member (202) of the plurality of undercarriage support assemblies (200) each support at least the respective portion of the undercarriage (108) of the aircraft (22).

(225) A7. The aircraft handler (10) of any of paragraphs A-A6, wherein each drive assembly (212) comprises one or more driven wheels (60) that are configured to receive power from one or more power sources (92) and facilitate movement of the aircraft handler (10) along the ground surface (16).

(226) A7.1. The aircraft handler (10) of paragraph A7, wherein at least one of the one or more driven wheels (60) is configured to steer the aircraft handler (10).

(227) A7.2. The aircraft handler (10) of any of paragraphs A7-A7.1, wherein the at least one driven wheel (60) of the one or more driven wheels (60) is configured to selectively pivot to steer the aircraft handler (10).

(228) A7.3. The aircraft handler (10) of any of paragraphs A7-A7.2, wherein each of the one or more driven wheels (60) is configured to be independently operated.

(229) A8. The aircraft handler (10) of any of paragraphs A-A7.3, wherein the translation assembly (204) of at least one undercarriage support assembly (200) of the plurality of undercarriage support assemblies (200) comprises one or more non-driven wheels (62).

(230) A9. The aircraft handler (10) of any of paragraphs A-A8, wherein the aircraft handler (10) is configured to fit within an envelope (120) of the aircraft (22) when the plurality of undercarriage support assemblies (200) support the respective portions of the undercarriage (108) of the aircraft (22).

(231) A10. The aircraft handler (10) of any of paragraphs A-A9, wherein the aircraft handler (10) comprises a deployed configuration and a stowed configuration, wherein the aircraft handler (10) occupies a footprint on the ground surface (16), wherein the footprint of the aircraft handler (10) is smaller in the stowed configuration than the footprint of the aircraft handler (10) in the deployed configuration.

(232) B. The aircraft handler (10) of any of paragraphs A-A10, wherein the plurality of undercarriage support assemblies (200) comprises a front undercarriage support assembly (206) and a rear undercarriage support assembly (208); wherein the front undercarriage support assembly (206) is configured to support a forward portion of the undercarriage (108) of the aircraft (22), and wherein the rear undercarriage support assembly (208) is configured to support an aft portion of the undercarriage (108) of the aircraft (22); and wherein one or more of the front undercarriage support assembly (206) and the rear undercarriage support assembly (208) comprise the one or more drive assemblies (212).

(233) B1. The aircraft handler (10) of paragraph B, wherein the plurality of undercarriage support assemblies (200) further comprises a distal undercarriage support assembly (210), and wherein the aircraft handler (10) further comprises at least one connecting arm (214) that operably couples the distal undercarriage support assembly (210) to the rear undercarriage support assembly (208); and wherein the aft portion of the undercarriage (108) of the aircraft (22) is a first aft portion, and wherein the distal undercarriage support assembly (210) is configured to engage and support a second aft portion of the undercarriage (108) of the aircraft (22).

(234) B2. The aircraft handler (10) of paragraph B1, wherein the aircraft handler (10) comprises two connecting arms (214) including a first connecting arm (214) and a second connecting arm (214), and wherein the distal undercarriage support assembly (210) comprises a first portion (222) operably coupled to the first connecting arm (214), and a second portion (224) operably coupled to the second connecting arm (214).

(235) B2.1. The aircraft handler (10) of paragraph B2, wherein the first portion (222) and the

second portion (224) collectively comprise the support member (202) and the support region (20) of the distal undercarriage support assembly (210).

(236) B2.1.1. The aircraft handler (10) of paragraph B2.1, further comprising a first connecting arm actuator (280) that operably couples the first connecting arm (214) to the rear undercarriage support assembly (208) and a second connecting arm actuator (280) that operably couples the second connecting arm (214) to the rear undercarriage support assembly (208), wherein the first connecting arm actuator (280) is configured to move the first connecting arm (214) together with the first portion (222) along the ground surface (16) and the second connecting arm actuator (280) is configured to move the second connecting arm (214) together with the second portion (224) relative to the ground surface (16).

(237) B2.1.1.1. The aircraft handler (10) of paragraph B2.1.1, wherein the first connecting arm actuator (280) and the second connecting arm actuator (280) collectively are configured to adjust a spacing between the first portion (222) and the second portion (224) along the ground surface (16).

(238) B2.2. The aircraft handler (10) of any of paragraphs B2-62.1.1.1, wherein the distal undercarriage support assembly (210) comprises a distal coupling mechanism (226) configured to selectively couple the first portion (222) and the second portion (224) to one another.

(239) B2.2.1. The aircraft handler (10) of paragraph B2.2, wherein the first portion (222) and the second portion (224) are configured to collectively support the second aft portion of the undercarriage (108) of the aircraft (22) when the distal coupling mechanism (226) couples the first portion (222) and the second portion (224) to one another.

(240) B2.3. The aircraft handler (10) of any of paragraphs B2.2-B2.2.1, when depending from paragraph B2.1.1, wherein the distal undercarriage support assembly (210) comprises a supporting configuration (228) in which the distal coupling mechanism (226) operably couples the first portion (222) and the second portion (224) to one another for operative support of the second aft portion of the undercarriage (108), and a positioning configuration (230) in which the distal coupling mechanism (226) disengages the first portion (222) from the second portion (224) and the first connecting arm actuator (280) and the second connecting arm actuator (280) position the first portion (222) and the second portion (224) spaced apart from one another along the ground surface (16).

(241) B2.3.1. The aircraft handler (10) of paragraph B2.3, wherein the distal undercarriage support assembly (210) is configured to translate beneath the second aft portion of the undercarriage (108) of the aircraft (22) in the positioning configuration (230).

(242) B3. The aircraft handler (10) of any of paragraphs B-B2.3.1, when depending from paragraph A5, wherein the aircraft handler (10) comprises a plurality of lift mechanisms (26), wherein the front undercarriage support assembly (206) and the rear undercarriage support assembly (208) each comprise a respective lift mechanism (26) of the plurality of lift mechanisms (26).

(243) B3.1. The aircraft handler (10) of paragraph B3, wherein each lift mechanism (26) is configured to selectively position the respective support member (202) among a lowered position (232) and a plurality of raised positions (234).

(244) B3.1.1. The aircraft handler (10) of paragraph B3.1, when depending from paragraph A1, wherein the aircraft handler (10) is configured to support the aircraft (22) in the level aircraft orientation (104) when the support member (202) of the front undercarriage support assembly (206) and the support member (202) of the rear undercarriage support assembly (208) are positioned in the lowered position (232) by the respective lift mechanisms (26), and wherein the aircraft handler (10) is configured to support the aircraft (22) among the plurality of tilted aircraft orientations (106) when the support member (202) of the front undercarriage support assembly (206) and the support member (202) of the rear undercarriage support assembly (208) are among the plurality of raised positions (234).

(245) B4. The aircraft handler (10) of any of paragraphs B-B3.1.1, wherein the front undercarriage support assembly (206) comprises a front drive assembly (212) of the one or more drive assemblies

(212), and wherein the front drive assembly (212) is configured to move the front undercarriage support assembly (206) along the ground surface (16).

(246) B4.1. The aircraft handler (10) of paragraph B4.1, wherein the front drive assembly (212) is configured to move the aircraft handler (10) along the ground surface (16).

(247) B4.2. The aircraft handler (10) of any of paragraphs B4-B4.1, wherein the rear undercarriage support assembly (208) comprises a rear drive assembly (212) of the one or more drive assemblies (212), and wherein the rear drive assembly (212) is configured to move the rear undercarriage support assembly (208) along the ground surface (16).

(248) B4.2.1. The aircraft handler (10) of paragraph B4.2, when depending from paragraph B4.1, wherein the front drive assembly (212) and the rear drive assembly (212) are configured to move the aircraft handler (10) along the ground surface (16).

(249) B4.2. The aircraft handler (10) of any of paragraphs B4.2-B4.2.1, when depending from paragraph B1, wherein the rear drive assembly (212) is configured to move the distal undercarriage support assembly (210) together with the rear undercarriage support assembly (208).

(250) B5. The aircraft handler (10) of any of paragraphs B-B4.2, wherein the front undercarriage support assembly (206) further comprises a front ground support chassis (236) operably coupled to the body (12) of the front undercarriage support assembly (206), wherein the front ground support chassis (236) comprises a front pair of projections (238) that extend from the body (12) of the front undercarriage support assembly (206) laterally spaced apart from one another and that form a front open region (240) therebetween, and wherein a portion of the translation assembly (204) of the front undercarriage support assembly (206) supports the front pair of projections (238) above the ground surface (16).

(251) B5.1. The aircraft handler (10) of paragraph B5, wherein the support member (202) of the front undercarriage support assembly (206) is positioned laterally within the front open region (240).

(252) B5.2. The aircraft handler (10) of any of paragraphs B5-B5.2, wherein the front ground support chassis (236) further comprises a gate (242), wherein the gate (242) has a closed configuration (244) in which the gate (242) is operably coupled to and interconnects the front pair of projections (238) and an open configuration (246) in which the gate (242) is operably coupled to only one projection (238) of the front pair of projections (238) and provides access to the front open region (240).

(253) B6. The aircraft handler (10) of any of paragraphs B-B5.2, wherein the rear undercarriage support assembly (208) further comprises a rear ground support chassis (236) operably coupled to the body (12) of the rear undercarriage support assembly (208), wherein the rear ground support chassis (236) comprises a rear pair of projections (238) that extend from the body (12) of the rear undercarriage support assembly (208) laterally spaced apart from one another and that form a rear open region (240) therebetween, wherein a portion of the translation assembly (204) of the rear undercarriage support assembly (208) supports the rear pair of projections (238) above the ground surface (16).

(254) B6.1. The aircraft handler (10) of paragraph B6, wherein the rear ground support chassis (236) further comprises a/the gate (242), wherein the gate (242) has a closed configuration (244) in which the gate (242) is operably coupled to and interconnects the rear pair of projections (238) and an open configuration (246) in which the gate (242) is operably coupled to only one projection (238) of the rear pair of projections (238) and provides access to the rear open region (240).

(255) B7. The aircraft handler (10) of any of paragraphs B-B6.1, wherein the support member (202) of one or more of the front undercarriage support assembly (206) and the rear undercarriage support assembly (208) comprises a fork assembly (248), wherein the fork assembly (248) comprises a plurality of prongs (250) projecting outward from the respective body (12), wherein the plurality of prongs (250) are spaced laterally apart from one another, and wherein the plurality of prongs (250) include the support region (20) of the respective support member (202).

(256) B7.1. The aircraft handler (10) of paragraph B7, wherein the fork assembly (248) further comprises a prong gate (252) having a prong gate closed configuration (254), in which the prong gate (252) is operably coupled to and interconnects the plurality of prongs (250), and a prong gate open configuration (256) in which the prong gate (252) is operably coupled to only one prong (250) of the plurality of prongs (250).

(257) B7.1.1. The aircraft handler (10) of paragraph B7.1, wherein, the fork assembly (248) is configured to translate laterally beneath the respective portion of the undercarriage (108) of the aircraft (22) when the prong gate (252) is in the prong gate open configuration (256).

(258) B7.1.2. The aircraft handler (10) of any of paragraphs B7.1-B7.1.2, wherein the prong gate (252) is configured to reinforce the fork assembly (248) when the prong gate (252) is in the prong gate closed configuration (254).

(259) B7.2. The aircraft handler (10) of any of paragraphs B7-B7.1.2, wherein the fork assembly (248) is one of a plurality of fork assemblies (248), and wherein the support member (202) of the front undercarriage support assembly (206) and the support member (202) of the rear undercarriage support assembly (208) each comprise a respective fork assembly (248) of the plurality of fork assemblies (248).

(260) B8. The aircraft handler (10) of any of paragraphs B-B7.2, wherein the front undercarriage support assembly (206) and the rear undercarriage support assembly (208) are spaced apart from one another, and wherein the aircraft handler (10) further comprises a connecting member (216) configured to operably interconnect the front undercarriage support assembly (206) with the rear undercarriage support assembly (208).

(261) B8.1. The aircraft handler (10) of paragraph B8, wherein the front undercarriage support assembly (206) and the rear undercarriage support assembly (208) each comprise a connecting member receiver (258) configured to selectively interconnect with and disconnect from the connecting member (216).

(262) B8.1.1. The aircraft handler (10) of paragraph B8.1, wherein the front undercarriage support assembly (206) and the rear undercarriage support assembly (208) each comprise a respective plurality of connecting member receivers (258), wherein each respective plurality of connecting member receivers (258) comprises a first connecting member receiver (258) positioned on a first side (260) of the respective undercarriage support assembly (200), and a second connecting member receiver (258) positioned on a second side (262) of the respective undercarriage support assembly (200) that is opposed to the first side (260).

(263) B8.1.1.1. The aircraft handler (10) of paragraph B8.1.1, wherein the connecting member (216) is configured to interconnect the front undercarriage support assembly (206) with the rear undercarriage support assembly (208) via the first connecting member receiver (258) of the front undercarriage support assembly (206) and the first connecting member receiver (258) of the rear undercarriage support assembly (208), and wherein the connecting member (216) is configured to interconnect the front undercarriage support assembly (206) and the rear undercarriage support assembly (208) via the second connecting member receiver (258) of the front undercarriage support assembly (206) and the second connecting member receiver (258) of the rear undercarriage support assembly (208).

(264) B8.2. The aircraft handler (10) of any of paragraphs B8-B8.1.1.1, wherein the connecting member (216) is configured to permit adjustment of a spacing between the front undercarriage support assembly (206) and the rear undercarriage support assembly (208).

(265) B8.2.1. The aircraft handler (10) of paragraph B8.2, wherein the connecting member (216) is an articulated connecting member and comprises two or more sections (270) and a plurality of articulated connections (272) that interconnect adjacent sections (270) of the two or more sections (270) to one another and that interconnect the two or more sections (270) to front undercarriage support assembly (206) and rear undercarriage support assembly (208), wherein plurality of articulated connections (272) are configured to selectively permit the two or more sections (270) to

pivot relative one another and relative to front undercarriage support assembly (206) and rear undercarriage support assembly (208) to permit adjustment of the spacing of the front undercarriage support assembly (206) and the rear undercarriage support assembly (208).

(266) C. The aircraft handler (10) of any of paragraphs A-A10, wherein the plurality of undercarriage support assemblies (200) comprises a front undercarriage support assembly (206) and a distal undercarriage support assembly (210) operably coupled to the front undercarriage support assembly (206); wherein the one or more drive assemblies (212) comprises a first drive assembly (14) and a second drive assembly (24), wherein the first drive assembly (14) comprises the translation assembly (204) of the front undercarriage support assembly (206), and the second drive assembly (24) comprises the translation assembly (204) of the distal undercarriage support assembly (210), and wherein the first drive assembly (14) and the second drive assembly (24) are configured to move the aircraft handler (10) along the ground surface (16); and wherein the support member (202) of the front undercarriage support assembly (206) and the support member (202) of the distal undercarriage support assembly (210) collectively comprise a support arm (18), wherein the support arm (18) is operably coupled to the body (12) of the front undercarriage support assembly (206) and to the body (12) of the distal undercarriage support assembly (210), and wherein distal undercarriage support assembly (210) is configured to support support arm (18) on ground surface (16).

(267) C1. The aircraft handler (10) of paragraph C, when depending from paragraph A1, wherein the support arm (18) is configured to support the aircraft (22) in the level aircraft orientation (104) and in the plurality of tilted aircraft orientations (106).

(268) C2. The aircraft handler (10) of paragraph C1, wherein the first drive assembly (14) and the second drive assembly (24) are configured to move the aircraft handler (10) together with the aircraft (22) along the ground surface (16) when the support arm (18) supports the aircraft (22) among the plurality of tilted aircraft orientations (106).

(269) C3. The aircraft handler (10) of any of paragraphs C-C2, when depending from paragraph A5, wherein the front undercarriage support assembly (206) further comprises a lift mechanism (26) of the one or more lift mechanisms (26), wherein the support arm (18) has a proximal end (28) and a distal end (30), wherein the second drive assembly (24) of distal undercarriage support assembly (210) is operably coupled to the distal end (30) and the lift mechanism (26) operably couples the proximal end (28) to the body (12) and is configured to selectively raise and lower the proximal end (28) relative to the ground surface (16) to operably pivot the support arm (18) relative to the ground surface (16).

(270) C3.1. The aircraft handler (10) of paragraph C3, wherein the lift mechanism (26) is configured to pivot the support arm (18) among a level orientation (100) and a plurality of tilted orientations (102) relative to the ground surface (16).

(271) C3.1.1. The aircraft handler (10) of paragraph C3.1, wherein the support arm (18) is configured to support at least the portion of the undercarriage (108) of the aircraft (22) when the support arm (18) is pivoted among the plurality of tilted orientations (102).

(272) C3.1.2. The aircraft handler (10) of any of paragraphs C3.1—C3.1.1, wherein the support arm (18) is configured to tilt the aircraft (22) when the lift mechanism (26) pivots the support arm (18) from the level orientation (100) to among the plurality of tilted orientations (102).

(273) C3.2. The aircraft handler (10) of any of paragraphs C3-C3.1.2, wherein the lift mechanism (26) is configured to pivot the support arm (18) among a plurality of tilt angles relative to the ground surface (16), wherein the plurality of tilt angles are from -5° to 25° .

(274) C4. The aircraft handler (10) of any of paragraphs C1-C3.2, wherein the support arm (18) has a/the proximal end (28) and a/the distal end (30), wherein the second drive assembly (24) of distal undercarriage support assembly (210) is operably coupled to the distal end (30) and the body (12) and front undercarriage support assembly (206) is operably coupled to the proximal end (28), and wherein the support arm (18) has a deployed orientation (32) and a stowed orientation (34),

wherein the distal end (30) is spaced further away from the body (12) in the deployed orientation (32) than in the stowed orientation (34), and wherein the support arm (18) is configured to be transitioned between the deployed orientation (32) and the stowed orientation (34).

(275) C4.1. The aircraft handler (10) of paragraph C4, wherein the support arm (18) comprises a first member (80) and a second member (82) that are configured to move relative to one another when the support arm (18) transitions between the deployed orientation (32) and the stowed orientation (34).

(276) C4.1.1. The aircraft handler (10) of paragraph C4.1, wherein the first member (80) is hingedly coupled to the second member (82), wherein the first member (80) and the second member (82) pivot relative to one another about a hinge mechanism (84) when the support arm (18) transitions between the deployed orientation (32) and the stowed orientation (34).

(277) C4.1.2. The aircraft handler (10) of any of paragraphs C4.1-C4.1.1, wherein the second drive assembly (24) moves the distal end (30) of the support arm (18) relative to the body (12) when the support arm (18) transitions between the deployed orientation (32) and the stowed orientation (34).

(278) C5. The aircraft handler (10) of any of paragraphs C-C4.1.2, wherein the first drive assembly (14) and the second drive assembly (24) are configured to move the aircraft handler (10) along the ground surface (16) independently of one another.

(279) C5.1 The aircraft handler (10) of any of paragraphs C-C4.1.2, wherein the first drive assembly (14) together with the second drive assembly (24) are configured to move the aircraft handler (10) along the ground surface (16).

(280) C5.2. The aircraft handler (10) of any of paragraphs C-C5.1, when depending from paragraph A7, wherein the first drive assembly (14) and the second drive assembly (24) each comprise the one or more driven wheels (60) that are configured to receive power from one or more power sources (92) and facilitate movement of the aircraft handler (10) along the ground surface (16).

(281) C5.3. The aircraft handler (10) of any of paragraphs C-C5.2, wherein at least one of the first drive assembly (14) and the second drive assembly (24) further comprises the one or more non-driven wheels (62).

(282) C.6. The aircraft handler (10) of any of paragraphs C-C5.3, wherein the at least one support region (20) of the support arm (18) is configured to translate laterally beneath at least the portion of the undercarriage (108) of the aircraft (22) to move at least the portion of the undercarriage (108) onto the at least one support region (20) and lift at least the portion of the undercarriage (108) of the aircraft (22) from direct contact with the ground surface (16).

(283) C6.1. The aircraft handler (10) of paragraph C6, wherein the support arm (18) comprises one or more transition portions (86) positioned proximate each support region (20) that are configured to facilitate movement of at least the respective portion of the undercarriage (108) of the aircraft (22) from the ground surface (16) onto the at least one support region (20).

(284) C7. The aircraft handler (10) of any of paragraphs C-C6.1, wherein the support arm (18) is a first support arm (36), wherein the body (12) of the front undercarriage support assembly (206) comprises a first body member (50) and a second body member (52), wherein the first body member (50) is operably coupled to the first support arm (36) and the first drive assembly (14), wherein the distal undercarriage support assembly (210) is a first distal undercarriage support assembly (209), wherein the portion of the undercarriage (108) of the aircraft (22) is a first portion, and wherein the aircraft handler (10) further comprises: a rear undercarriage support assembly (208), wherein the body (12) of the rear undercarriage support assembly (208) is a second body member (52) that is operably coupled to the first body member (50); a second support arm (38) operably coupled to the second body member (52) of the body (12), wherein the second support arm (38) comprises at least one support region (20) configured to operably support at least a second portion of the undercarriage (108) of the aircraft (22); a second distal undercarriage support assembly (211) and a third drive assembly (40) operably coupled to the second support arm (38) and configured to support the second support arm (38) on the ground surface (16), wherein the

support member (202) of the rear undercarriage support assembly (208) and the support member (202) of the second distal undercarriage support assembly (211) collectively comprise the second support arm (38); and the third drive assembly (40), wherein the third drive assembly (40) comprises the translation assembly (204) of the second distal undercarriage support assembly (211), and wherein the third drive assembly (40) is configured to move the aircraft handler (10) along the ground surface (16), together with the first drive assembly (14) and the second drive assembly (24).

(285) C7.1. The aircraft handler (10) of paragraph C7, wherein the aircraft handler (10) further comprises a fourth drive assembly (44) that includes the translation assembly (204) of the rear undercarriage support assembly (208) and is operably coupled to the second body member (52) of the body (12).

(286) C7.1.1 The aircraft handler (10) of paragraph C7.1, wherein the first drive assembly (14) is configured to support the first body member (50) on the ground surface (16) and the fourth drive assembly (44) is configured to support the second body member (52) on the ground surface (16), and wherein the fourth drive assembly (44) is configured to move the aircraft handler (10) along the ground surface (16), together with the first drive assembly (14), the second drive assembly (24), and the third drive assembly (40).

(287) C7.2. The aircraft handler (10) of any of paragraphs C7-C7.1.1, wherein the body (12) further comprises a frame (54), wherein the first body member (50) of the body (12) and the second body member (52) of the body (12) are spaced apart and the frame (54) operably interconnects the first body member (50) and the second body member (52).

(288) C7.2.1. The aircraft handler (10) of paragraph C7.2, wherein the frame (54) is a telescoping frame (56) that is configured to adjust a spacing between the first body member (50) and the second body member (52).

(289) C7.2.1.1. The aircraft handler (10) of paragraph C7.2.1, wherein the first support arm (36) and the second support arm (38) are spaced apart by the telescoping frame (56), and wherein the telescoping frame (56) is configured to adjust a spacing between the first support arm (36) and the second support arm (38).

(290) D. A method (500), comprising: supporting (502) an undercarriage (108) of an aircraft (22) with an aircraft handler (10); and moving (504) the aircraft (22) along a ground surface (16) with the aircraft handler (10).

(291) D1. The method (500) of paragraph D, further comprising tilting (508) the aircraft (22) relative to the ground surface (16) with the aircraft handler (10).

(292) D1.1. The method (500) of any of paragraphs D-D1, wherein the tilting (508) the aircraft (22) comprises changing one or more dimensions of an envelope (120) of the aircraft (22).

(293) D1.2. The method (500) of any of paragraphs D1-D1.1, wherein the tilting (508) comprises tilting the aircraft (22) among a level aircraft orientation (104) and a plurality of tilted aircraft orientations (106).

(294) D1.2.1. The method (500) of paragraph D1.2, wherein the tilting (508) comprises tilting the aircraft from the level aircraft orientation (104) to one of the plurality of tilted aircraft orientations (106).

(295) D1.2.1.1. The method (500) of paragraph D1.2.1, wherein the tilting (508) comprises decreasing a width of a/the envelope (120) of the aircraft (22).

(296) D1.2.2. The method (500) of any of paragraphs D1.2-D1.2.1.1, wherein the moving (504) is performed while the aircraft (22) is in one of the plurality of tilted aircraft orientations (106).

(297) D1.2.3. The method (500) of any of paragraphs D1.2-D1.2.2, wherein the moving (504) comprises moving the aircraft (22) to within a defined volume (130), wherein the defined volume (130) comprises a width that is smaller than the width of a/the envelope (120) of the aircraft (22) when the aircraft (22) is oriented in the level aircraft orientation (104).

(298) D2. The method (500) of any of paragraphs D-D1.2.3, wherein the moving (504) comprises

rotating (506) the aircraft (22) within a cylindrical volume (134) having a radius (136) that is no longer than 110% of a distance between a yaw axis (132) of the aircraft (22) and a maximum extent (122) of the aircraft (22).

(299) D3. The method (500) of any of paragraphs D-D2, further comprising stowing (510) the aircraft (22) in a desired location.

(300) D4. The method (500) of any of paragraphs D-D3, further comprising transitioning (512) the aircraft handler (10) from a deployed configuration (264) to a stowed configuration (266).

(301) D4.1 The method (500) of paragraph D4, wherein the transitioning (512) the aircraft handler (10) from the deployed configuration (264) to the stowed configuration (266) comprises decreasing a footprint of the aircraft handler (10).

(302) D4.2 The method (500) of any of paragraphs D4-D4.1, wherein the transitioning (512) comprises decreasing a spacing along the ground surface (16) between a front undercarriage support assembly (206) of the aircraft handler (10) and a rear undercarriage support assembly (208) of the aircraft handler (10).

(303) D4.2.1 The method (500) of paragraph D4.2, wherein the decreasing the spacing comprises folding sections (270) of a connecting member (216) about an articulated connection (272) of the connecting member (216), and wherein the connecting member (216) operably interconnects the front undercarriage support assembly (206) and the rear undercarriage support assembly (208).

(304) D4.2.2 The method (500) of any of paragraphs D4.1-D4.2.1, wherein the transitioning (512) comprises decreasing a spacing between a rear undercarriage support assembly (208) of the aircraft handler (10) and a distal undercarriage support assembly (210) of the aircraft handler (10).

(305) D4.2.2.1 The method (500) of paragraph D4.2.2, wherein the decreasing the spacing between the rear undercarriage support assembly (208) and the distal undercarriage support assembly (210) comprises folding articulated portions (276) of a connecting arm (214) about a connecting arm articulated connection (278), wherein the connecting arm (214) operably interconnects the rear undercarriage support assembly (208) and the distal undercarriage support assembly (210).

(306) D4.3. The method (500) of any of paragraph D4-D4.2, wherein the transitioning (512) comprises transitioning at least one support arm (18) of the aircraft handler (10) from a deployed orientation (32) to a stowed orientation (34).

(307) D4.3.1 The method (500) of paragraph D4.3, wherein the transitioning the at least one support arm (18) comprises moving a proximal end (28) and a distal end (30) of the support arm (18) closer to one another.

(308) D4.3.1.1. The method (500) of paragraph D4.3.1, wherein the transitioning the at least one support arm (18) comprises moving a second drive assembly (24) that is operably attached to the distal end (30) of the support arm (18) towards the proximal end (28) of the support arm (18).

(309) D.4.3.2 The method (500) of any of paragraphs D4.3-D4.3.1.1, wherein the transitioning the at least one support arm (18) comprises pivoting a first member (80) and a second member (82) of the at least one support arm (18) relative to one another.

(310) D5. The method (500) of any of paragraphs D-D4.3.2, wherein the aircraft handler (10) comprises the aircraft handler (10) of any of paragraphs A-C7.2.1.1.

(311) E. The aircraft handler (10) of any of paragraphs A-C7.2.1.1, further comprising a controller (90), the controller (90) comprising: non-transitory memory comprising computer-readable instructions for executing the methods of any of paragraphs D-D5; and a processor for executing the computer-readable instructions to perform the methods of any of paragraphs D-D5.

(312) A controller may be any suitable device or devices that are configured to perform the functions of the controller discussed herein. For example, the controller may include one or more of an electronic controller, a dedicated controller, a special-purpose controller, a personal computer, a special-purpose computer, a display device, a logic device, a memory device, and/or a memory device having computer readable media suitable for storing computer-executable instructions for implementing aspects of systems and/or methods according to the present disclosure.

(313) As used herein, the phrase, “for example,” the phrase, “as an example,” and/or simply the term “example,” when used with reference to one or more components, features, details, structures, embodiments, and/or methods according to the present disclosure, are intended to convey that the described component, feature, detail, structure, embodiment, and/or method is an illustrative, non-exclusive example of components, features, details, structures, embodiments, and/or methods according to the present disclosure. Thus, the described component, feature, detail, structure, embodiment, and/or method is not intended to be limiting, required, or exclusive/exhaustive; and other components, features, details, structures, embodiments, and/or methods, including structurally and/or functionally similar and/or equivalent components, features, details, structures, embodiments, and/or methods, are also within the scope of the present disclosure.

(314) As used herein, the terms “selective” and “selectively,” when modifying an action, movement, configuration, or other activity of one or more components or characteristics of an apparatus, mean that the specific action, movement, configuration, or other activity is a direct or indirect result of one or more dynamic processes, as described herein. The terms “selective” and “selectively” thus may characterize an activity that is a direct or indirect result of user manipulation of an aspect of, or one or more components of, the apparatus, or may characterize a process that occurs automatically, such as via the mechanisms disclosed herein.

(315) As used herein, the terms “adapted” and “configured” mean that the element, component, or other subject matter is designed and/or intended to perform a given function. Thus, the use of the terms “adapted” and “configured” should not be construed to mean that a given element, component, or other subject matter is simply “capable of” performing a given function but that the element, component, and/or other subject matter is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the function. It is also within the scope of the present disclosure that elements, components, and/or other recited subject matter that is recited as being adapted to perform a particular function may additionally or alternatively be described as being configured to perform that function, and vice versa. Similarly, subject matter that is recited as being configured to perform a particular function may additionally or alternatively be described as being operative to perform that function.

(316) As used herein, the term “and/or” placed between a first entity and a second entity means one of (1) the first entity, (2) the second entity, and (3) the first entity and the second entity. Multiple entries listed with “and/or” should be construed in the same manner, i.e., “one or more” of the entities so conjoined. Other entities optionally may be present other than the entities specifically identified by the “and/or” clause, whether related or unrelated to those entities specifically identified. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising,” may refer, in one example, to A only (optionally including entities other than B); in another example, to B only (optionally including entities other than A); in yet another example, to both A and B (optionally including other entities). These entities may refer to elements, actions, structures, steps, operations, values, and the like.

(317) As used herein, the phrase “at least one,” in reference to a list of one or more entities should be understood to mean at least one entity selected from any one or more of the entities in the list of entities, but not necessarily including at least one of each and every entity specifically listed within the list of entities and not excluding any combinations of entities in the list of entities. This definition also allows that entities may optionally be present other than the entities specifically identified within the list of entities to which the phrase “at least one” refers, whether related or unrelated to those entities specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) may refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including entities other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including entities other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at

least one, optionally including more than one, B (and optionally including other entities). In other words, the phrases “at least one,” “one or more,” and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B, and C,” “at least one of A, B, or C,” “one or more of A, B, and C,” “one or more of A, B, or C,” and “A, B, and/or C” may mean A alone, B alone, C alone, A and B together, A and C together, B and C together, A, B, and C together, and optionally any of the above in combination with at least one other entity.

(318) As used herein, “at least substantially,” when modifying a degree or relationship, includes not only the recited “substantial” degree or relationship, but also the full extent of the recited degree or relationship. A substantial amount of a recited degree or relationship may include at least 75% of the recited degree or relationship. For example, an object that is at least substantially formed from a material includes an object for which at least 75% of the object is formed from the material and also includes an object that is completely formed from the material. As another example, a first direction that is at least substantially parallel to a second direction includes a first direction that forms an angle with respect to the second direction that is at most 22.5 degrees and also includes a first direction that is exactly parallel to the second direction. As another example, a first length that is substantially equal to a second length includes a first length that is at least 75% of the second length, a first length that is equal to the second length, and a first length that exceeds the second length such that the second length is at least 75% of the first length.

(319) In the present disclosure, several of the illustrative, non-exclusive examples have been discussed and/or presented in the context of flow diagrams, or flow charts, in which the methods are shown and described as a series of blocks, or steps. Unless specifically set forth in the accompanying description, it is within the scope of the present disclosure that the order of the blocks may vary from the illustrated order in the flow diagram, including with two or more of the blocks (or steps) occurring in a different order, concurrently, and/or repeatedly. It is also within the scope of the present disclosure that the blocks, or steps, may be implemented as logic, which also may be described as implementing the blocks, or steps, as logics. In some applications, the blocks, or steps, may represent expressions and/or actions to be performed by functionally equivalent circuits or other logic devices. The illustrated blocks may, but are not required to, represent executable instructions that cause a computer, processor, and/or other logic device to respond, to perform an action, to change states, to generate an output or display, and/or to make decisions.

(320) The various disclosed elements of apparatuses and steps of methods disclosed herein are not required to all apparatuses and methods according to the present disclosure, and the present disclosure includes all novel and non-obvious combinations and subcombinations of the various elements and steps disclosed herein. Moreover, one or more of the various elements and steps disclosed herein may define independent inventive subject matter that is separate and apart from the whole of a disclosed apparatus or method. Accordingly, such inventive subject matter is not required to be associated with the specific apparatuses and methods that are expressly disclosed herein, and such inventive subject matter may find utility in apparatuses and/or methods that are not expressly disclosed herein.

(321) It is believed that the disclosure set forth above encompasses multiple distinct inventions with independent utility. While each of these inventions has been disclosed in its preferred form, the specific embodiments thereof as disclosed and illustrated herein are not to be considered in a limiting sense as numerous variations are possible. The subject matter of the inventions includes all novel and non-obvious combinations and subcombinations of the various elements, features, functions and/or properties disclosed herein. Similarly, where the claims recite “a” or “a first” element or the equivalent thereof, such claims should be understood to include incorporation of one or more such elements, neither requiring nor excluding two or more such elements.

(322) It is believed that the following claims particularly point out certain combinations and subcombinations that are directed to one of the disclosed inventions and are novel and non-

obvious.

(323) Inventions embodied in other combinations and subcombinations of features, functions, elements, and/or properties may be claimed through amendment of the present claims or presentation of new claims in this or a related application. Such amended or new claims, whether they are directed to a different invention or directed to the same invention, whether different, broader, narrower, or equal in scope to the original claims, are also regarded as included within the subject matter of the inventions of the present disclosure.

Claims

1. An aircraft handler, comprising: a plurality of undercarriage support assemblies, each comprising: a body; a support member operably coupled to and extending from the body, wherein the support member comprises at least one support region configured to engage and operably support at least a respective portion of an undercarriage of an aircraft; and a translation assembly operably coupled to the body and configured to support a respective undercarriage support assembly on a ground surface and permit the respective undercarriage support assembly to translate along the ground surface; and one or more drive assemblies configured to move the aircraft handler along the ground surface, wherein each drive assembly comprises the translation assembly of an undercarriage support assembly of the plurality of undercarriage support assemblies; wherein the aircraft handler is configured to support the aircraft in a level aircraft orientation and in a plurality of tilted aircraft orientations; wherein the plurality of undercarriage support assemblies comprises a front undercarriage support assembly and a rear undercarriage support assembly; wherein the front undercarriage support assembly is configured to support a forward portion of the undercarriage of the aircraft, and wherein the rear undercarriage support assembly is configured to support an aft portion of the undercarriage of the aircraft; wherein one or more of the front undercarriage support assembly and the rear undercarriage support assembly comprise the one or more drive assemblies; wherein the plurality of undercarriage support assemblies further comprises a distal undercarriage support assembly, and wherein the aircraft handler further comprises at least one connecting arm that operably couples the distal undercarriage support assembly to the rear undercarriage support assembly; wherein the aft portion of the undercarriage of the aircraft is a first aft portion, and wherein the distal undercarriage support assembly is configured to engage and support a second aft portion of the undercarriage of the aircraft; and wherein the aircraft handler comprises two connecting arms including a first connecting arm and a second connecting arm, and wherein the distal undercarriage support assembly comprises a first portion operably coupled to the first connecting arm, and a second portion operably coupled to the second connecting arm.
2. The aircraft handler of claim 1, wherein the aircraft handler is configured to rotate the aircraft about a yaw axis of the aircraft.
3. The aircraft handler of claim 1, wherein the aircraft handler further is configured to rotate the aircraft within a cylindrical volume, and wherein a radius of the cylindrical volume is 110% or less of a distance between a yaw axis of the aircraft and a maximum extent of the aircraft, as measured radially from the yaw axis.
4. The aircraft handler of claim 1, further comprising one or more lift mechanisms each being included in a respective undercarriage support assembly of the plurality of undercarriage support assemblies, wherein each lift mechanism operably couples the support member of the respective undercarriage support assembly to the body of the respective undercarriage support assembly, and wherein each lift mechanism is configured to selectively raise and lower a respective support member relative to the ground surface.
5. The aircraft handler of claim 4, wherein the aircraft handler is configured to tilt the aircraft from the level aircraft orientation to among the plurality of tilted aircraft orientations when each lift mechanism selectively raises the respective support member relative to the ground surface.

6. The aircraft handler of claim 1, wherein each drive assembly comprises one or more driven wheels that are configured to receive power from one or more power sources and facilitate movement of the aircraft handler along the ground surface, wherein at least one of the one or more driven wheels is configured to steer the aircraft handler, wherein the at least one driven wheel of the one or more driven wheels is configured to selectively pivot to steer the aircraft handler, and wherein each of the one or more driven wheels is configured to be independently operated.
7. The aircraft handler of claim 6, wherein the translation assembly of at least one undercarriage support assembly of the plurality of undercarriage support assemblies comprises one or more non-driven wheels.
8. The aircraft handler of claim 1, wherein the aircraft handler is configured to fit within an envelope of the aircraft when the plurality of undercarriage support assemblies support the respective portions of the undercarriage of the aircraft.
9. The aircraft handler of claim 1, wherein the aircraft handler comprises a deployed configuration and a stowed configuration, wherein the aircraft handler occupies a footprint on the ground surface, wherein the footprint of the aircraft handler is smaller in the stowed configuration than the footprint of the aircraft handler in the deployed configuration.
10. The aircraft handler of claim 1, wherein the first portion and the second portion collectively comprise the support member and the support region of the distal undercarriage support assembly.
11. The aircraft handler of claim 10, further comprising a first connecting arm actuator that operably couples the first connecting arm to the rear undercarriage support assembly and a second connecting arm actuator that operably couples the second connecting arm to the rear undercarriage support assembly, wherein the first connecting arm actuator is configured to move the first connecting arm together with the first portion along the ground surface and the second connecting arm actuator is configured to move the second connecting arm together with the second portion relative to the ground surface.
12. The aircraft handler of claim 11, wherein the first connecting arm actuator and the second connecting arm actuator collectively are configured to adjust a spacing between the first portion and the second portion along the ground surface.
13. The aircraft handler of claim 11, wherein the distal undercarriage support assembly comprises a distal coupling mechanism configured to selectively couple the first portion and the second portion to one another; wherein the first portion and the second portion are configured to collectively support the second aft portion of the undercarriage of the aircraft when the distal coupling mechanism couples the first portion and the second portion to one another; wherein the distal undercarriage support assembly comprises a supporting configuration in which the distal coupling mechanism operably couples the first portion and the second portion to one another for operative support of the second aft portion of the undercarriage, and a positioning configuration in which the distal coupling mechanism disengages the first portion from the second portion and the first connecting arm actuator and the second connecting arm actuator position the first portion and the second portion spaced apart from one another along the ground surface.
14. The aircraft handler of claim 13, wherein the distal undercarriage support assembly is configured to translate beneath the second aft portion of the undercarriage of the aircraft in the positioning configuration.
15. The aircraft handler of claim 1, further comprising: a plurality of lift mechanisms each being included in a respective undercarriage support assembly of the plurality of undercarriage support assemblies; wherein each lift mechanism operably couples the support member of the respective undercarriage support assembly to the body of the respective undercarriage support assembly, and wherein each lift mechanism is configured to selectively raise and lower a respective support member relative to the ground surface; wherein the front undercarriage support assembly and the rear undercarriage support assembly each comprise a respective lift mechanism of the plurality of lift mechanisms; wherein each lift mechanism is configured to selectively position the respective

support member among a lowered position and a plurality of raised positions; wherein the aircraft handler is configured to support the aircraft in the level aircraft orientation when the support member of the front undercarriage support assembly and the support member of the rear undercarriage support assembly are positioned in the lowered position by the respective lift mechanisms; and wherein the aircraft handler is configured to support the aircraft among the plurality of tilted aircraft orientations when the support member of the front undercarriage support assembly and the support member of the rear undercarriage support assembly are among the plurality of raised positions.

16. The aircraft handler of claim 1, wherein the front undercarriage support assembly comprises a front drive assembly of the one or more drive assemblies, wherein the front drive assembly is configured to move the front undercarriage support assembly along the ground surface; wherein the rear undercarriage support assembly comprises a rear drive assembly of the one or more drive assemblies, and wherein the rear drive assembly is configured to move the rear undercarriage support assembly along the ground surface; and wherein the front drive assembly and the rear drive assembly are configured to move the aircraft handler along the ground surface.

17. The aircraft handler of claim 1, wherein the front undercarriage support assembly further comprises a front ground support chassis operably coupled to the body of the front undercarriage support assembly, wherein the front ground support chassis comprises a front pair of projections that extend from the body of the front undercarriage support assembly laterally spaced apart from one another and that form a front open region therebetween, and wherein a portion of the translation assembly of the front undercarriage support assembly supports the front pair of projections above the ground surface.

18. The aircraft handler of claim 1, wherein the rear undercarriage support assembly further comprises a rear ground support chassis operably coupled to the body of the rear undercarriage support assembly, wherein the rear ground support chassis comprises a rear pair of projections that extend from the body of the rear undercarriage support assembly laterally spaced apart from one another and that form a rear open region therebetween, wherein a portion of the translation assembly of the rear undercarriage support assembly supports the rear pair of projections above the ground surface, wherein the rear ground support chassis further comprises a gate, wherein the gate has a closed configuration in which the gate is operably coupled to and interconnects the rear pair of projections and an open configuration in which the gate is operably coupled to only one projection of the rear pair of projections and provides access to the rear open region.

19. The aircraft handler of claim 1, wherein the support member of one or more of the front undercarriage support assembly and the rear undercarriage support assembly comprises a fork assembly, wherein the fork assembly comprises a plurality of prongs projecting outward from the respective body, wherein the plurality of prongs are spaced laterally apart from one another, and wherein the plurality of prongs include the support region of the respective support member.

20. The aircraft handler of claim 1, wherein the front undercarriage support assembly and the rear undercarriage support assembly are spaced apart from one another, wherein the aircraft handler further comprises a connecting member configured to operably interconnect the front undercarriage support assembly with the rear undercarriage support assembly, and wherein the connecting member is configured to permit adjustment of a spacing between the front undercarriage support assembly and the rear undercarriage support assembly.

21. A method of using the aircraft handler of claim 1 to move the aircraft, the method comprising: supporting the undercarriage of the aircraft with the aircraft handler; tilting the aircraft relative to the ground surface with the aircraft handler; and moving the aircraft along the ground surface with the aircraft handler.

22. An aircraft handler, comprising: a plurality of undercarriage support assemblies, each comprising: a body; a support member operably coupled to and extending from the body, wherein the support member comprises at least one support region configured to engage and operably

support at least a respective portion of an undercarriage of an aircraft; and a translation assembly operably coupled to the body and configured to support a respective undercarriage support assembly on a ground surface and permit the respective undercarriage support assembly to translate along the ground surface; and one or more drive assemblies configured to move the aircraft handler along the ground surface, wherein each drive assembly comprises the translation assembly of an undercarriage support assembly of the plurality of undercarriage support assemblies; wherein the aircraft handler is configured to support the aircraft in a level aircraft orientation and in a plurality of tilted aircraft orientations; wherein the plurality of undercarriage support assemblies comprises a front undercarriage support assembly and a rear undercarriage support assembly; wherein the front undercarriage support assembly is configured to support a forward portion of the undercarriage of the aircraft, and wherein the rear undercarriage support assembly is configured to support an aft portion of the undercarriage of the aircraft; and wherein one or more of the front undercarriage support assembly and the rear undercarriage support assembly comprise the one or more drive assemblies; and wherein the rear undercarriage support assembly further comprises a rear ground support chassis operably coupled to the body of the rear undercarriage support assembly, wherein the rear ground support chassis comprises a rear pair of projections that extend from the body of the rear undercarriage support assembly laterally spaced apart from one another and that form a rear open region therebetween, wherein a portion of the translation assembly of the rear undercarriage support assembly supports the rear pair of projections above the ground surface, wherein the rear ground support chassis further comprises a gate, wherein the gate has a closed configuration in which the gate is operably coupled to and interconnects the rear pair of projections and an open configuration in which the gate is operably coupled to only one projection of the rear pair of projections and provides access to the rear open region.

23. The aircraft handler of claim 22, wherein the aircraft handler is configured to fit within an envelope of the aircraft when the plurality of undercarriage support assemblies support the respective portions of the undercarriage of the aircraft.

24. The aircraft handler of claim 22, wherein the aircraft handler comprises a deployed configuration and a stowed configuration, wherein the aircraft handler occupies a footprint on the ground surface, wherein the footprint of the aircraft handler is smaller in the stowed configuration than the footprint of the aircraft handler in the deployed configuration.

25. The aircraft handler of claim 22, wherein the front undercarriage support assembly and the rear undercarriage support assembly are spaced apart from one another, wherein the aircraft handler further comprises a connecting member configured to operably interconnect the front undercarriage support assembly with the rear undercarriage support assembly, and wherein the connecting member is configured to permit adjustment of a spacing between the front undercarriage support assembly and the rear undercarriage support assembly.

26. A method of using the aircraft handler of claim 22 to move the aircraft, the method comprising: supporting the undercarriage of the aircraft with the aircraft handler; tilting the aircraft relative to the ground surface with the aircraft handler; and moving the aircraft along the ground surface with the aircraft handler.
